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**Using MRI technique in clinical applications to
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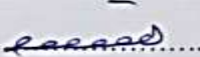
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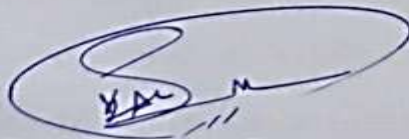
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DEDICATION

To my dear parents who illuminated my path of science and supported me and never forgot me in their prayers, the greatest credit goes to them after Allah Almighty.

And to my great husband Mr. Bassam Mohammed who shared with me and made the difficulty of searching easier for me.

And to my dear brother Dr. Mohsen who advise, help and provide me with knowledge.

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Finally, I thank everyone who support me with word and advice and wished me success.

List of Abbreviations

Shortcut	Meaning
FA	Flip Angle
BBB	Blood-Brain Barrier
CSF	Cerebrospinal Fluid
DIP	Digital Image Processing
FID	Free Induction Decay
FLAIR	Fluid Attenuated Inversion Recovery
FS	Field Strength
FSE	Fast Spin Echo
FT	Fourier Transform
FT⁻¹	Inverse Fourier Transform
Gd	Gadolinium
Gd-DTPA	Gadolinium diethylenetriamine penta-acetic acid
IR-FSE	Inversion Recovery Fast Spin Echo
MRI	Magnetic Resonance Imaging
MSSE	Multi Slice Spin Echo
NMR	Nuclear Magnetic Resonance
RF	Radio Frequency
ST	Slice Thickness
T1WI	T1-Weighted Image
T2WI	T2-Weighted Image
TE	Time of Echo
TI	Time of Inversion
TR	Time of Reptation

List of Symbols

Symbol	Meaning
J	Angular Momentum
I	Magnetic Quantum Number (Spin)
$\hbar$	Reduced Planck Constant
M	Magnetic Moment of the Nucleus
γ	Gyromagnetic Ratio
g_n	Land Coefficient
β	Bohr Magneton for the Nucleus
e	Electron Charge
m_p	Proton Mass
τ	Torque
B_o	Primary Magnetic Field
ν_o	Lamor Frequency
E	Energy
ω_o	Larmor Angular Frequency
N	Population of Protons
k	Boltzmann Constant
T	Absolute Temperature
B_1	Secondary Magnetic Field
M_z	Longitudinal Magnetization Component
M_0	Original Magnetization
M_{xy}	Transverse Magnetization Component
T_1	Longitudinal Relaxation Time Constant

T_2	Transverse Relaxation Time Constant
B_s	Magnetic Field Added by the Gradients
r	Position Vector
G_s	Values of the Added Gradients
$B(r)$	Resultant Magnetic Field
ω_{zc}	precession frequency of the slice center
G_{fe}	frequency encoding gradient
$s(t)$	Signal from Voxels
G_p	Phase Encoding Gradient
T_p	Time Period
C	Contrast
I	Intensity
E_n	Number of Generated Echoes
n^{th}	Number of Transmitted Pulses
α	Flip Angle
$\rho(r)$	Location of Object
$s(\mathcal{K})$	Signal in K Space
FOV	Field of View
T_A	Time of Acquisition
n	Number of Steps
m_j	Received Signal by Coil j
S_j	Sensitivity of Coil j

ABSTRACT

Nuclear magnetic resonance (NMR) is a physical phenomenon used to investigate the magnetic properties of atomic nuclei. It is the absorption and emission of energy by nuclei in a magnetic field, which can provide wide-ranging information about the structure, dynamics, reaction state and chemical environment of molecules. Magnetic resonance imaging (MRI) is the safest imaging method. It gives a visual representation of human tissues without invasive intervention for clinical diagnosis. In this research, Fast Spin Echo (FSE), Multi-Slice Spin Echo (MSSE) and Fluid Attenuated Inversion Recovery (FLAIR) sequences were used. The cases under study are placed in a permanent i-open type MRI system in order to image brain lesions. The obtained signals are encoded to fill the k-space through three stages; slice selection, phase encoding and frequency encoding. The obtained analog signals are processed to discrete signal using Fourier transform, then digitized and reconstructed using inverse Fourier transform to obtain an image of the brain. T1-weighted before and after gadolinium-based contrast agent injection, T2-weighted, and FLAIR tests were performed to obtain image types and compare the resulting images with normal images in each type of sequence applied. In the FLAIR image, the first case, the signal was enhanced at the target site, turned bright, and appeared as bound water, indicating that it was a tumor, while in the second case, the fluid signal was suppressed specifically at the target site, turned dark, and appeared as free water, like cerebrospinal fluid, indicating that it was fluid and not tumor. All of this in turn helps to distinguish between tumor and fluid in the T2-weighted image. Also, the contrast agent (Gadolinium) in T1WI shortened the T1 relaxation time and accurately revealed the edges of the tumor and helped to differentiate it from the fluid. Thus, the lesions were detected. Our study establishes a basis for using MRI technology with contrast sequences such as FLAIR and other sequences to distinguish between different disease signs such as fluid

cysts and tumors when other techniques fail to do the same. This thesis is written in five chapters; the same are abstracted briefly as follows:

1. **Introduction:** This section introduces the concept of NMR and its applications in the medical field. It also covers the history of NMR and MRI. Moreover, it contains significance and justification, and objectives of research as well as a literature review of previous studies in MRI technology.
2. **Theoretical Part:** This chapter discusses the physical principles behind MRI technology, the hardware components that work together to generate sectional images, and the computer systems related to MRI.
3. **Practical Part:** This section provides an overview of the equipment, agents, and cases used, as well as the methodology employed.
4. **Results & Discussion:** This chapter presents the results of two case studies that underwent MRI, along with a discussion and analysis of the findings.
5. **Conclusion & Future Scope:** This final section summarizes the conclusions drawn from the research and offers recommendations for future developments in the field.

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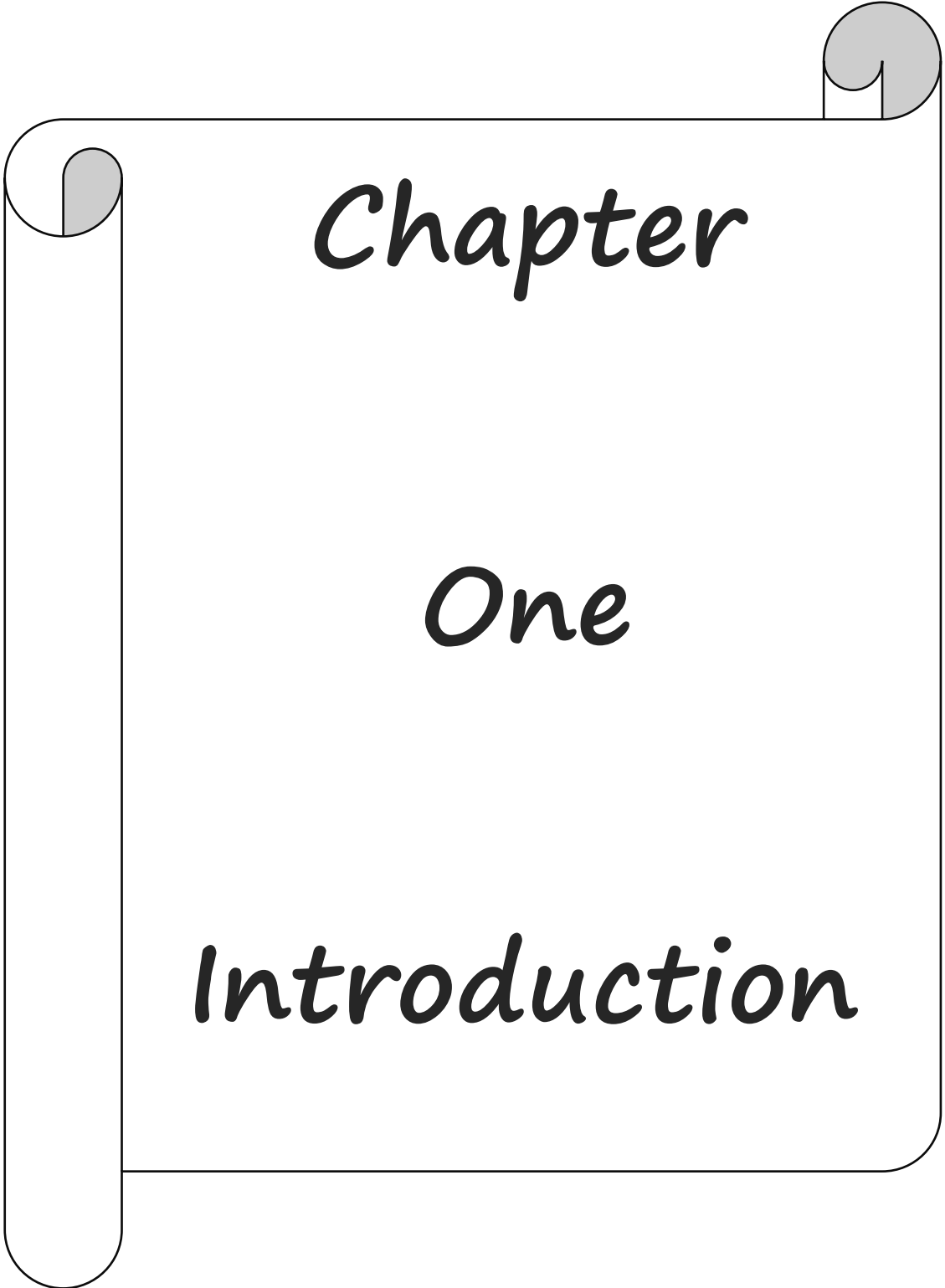
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Chapter

One

Introduction

Introduction

Nuclear Magnetic Resonance (NMR) is a spectroscopic technique that exploits the magnetic properties of atomic nuclei. It involves the study of the interaction between nuclear spins and an applied magnetic field. Protons within a nucleus possess unique spin and charge distributions, which endow them with magnetic characteristics. When subjected to a strong external magnetic field, these nuclei absorb specific frequencies of electromagnetic energy, a phenomenon known as resonance. This energy absorption is characteristic of the nucleus and its molecular environment, providing valuable insights into molecular structure and composition [1], [2].

Our knowledge of the atomic world and the quantum theory have led to the development of NMR. Actually, a thorough examination of the key principles of NMR would be impossible without taking quantum physics into account. In particular, we must comprehend spin in quantum physics in order to explain the ideas underlying this technology. One of the most fascinating periods in the history of contemporary research is the discovery of quantum spin. This relates to our deeper comprehension of matter, which has been enlightened by the identification of atoms' constituents [3].

Understanding the Zeeman effect, which shows how the interaction of nuclear spins with external magnetic fields was essential to the development of quantum physics, forms the basis of NMR. Both the dynamics of the NMR system and the dynamics of spin relaxation in a quantum spin system are described by the Bloch equation. In order to convert the time domain NMR signal to the frequency domain signal, the Fourier transform (FT) method—which corresponds to the free induction decay (FID) signal—later describes the idea of the NMR signal. It is crucial to note that MRI is the second generation of this phenomenon, while NMR was the first generation of Rabi's finding [3], [4].

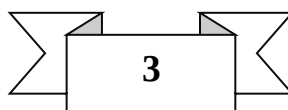
Magnetic resonance imaging (MRI) is a non-invasive diagnostic technique used for imaging the internal tissues of the body. The patient is placed inside a huge homogenous magnetic field in clinical routine whose intensity ranges in conventional MRI up to 3.0 Tesla (closed-type designs), and in modern low-field MRI around 0.5 T (open-type designs) to produce a net magnetic field inside the human body [5], [6], [7], [8], and then specific radio frequency pulses are sent to flip and measure the net magnetic field that produces images of high resolution for the human body tissues in order to examine them and detect the diseases [9], [10].

1.1. The importance of MRI technique

Magnetic resonance imaging provides more contrast to soft tissues than computed tomography (CT), which is better for imaging bones [10]. There is a special type of MRI called functional magnetic resonance imaging (fMRI) of the brain, which gives a visual representation of blood flow to specific areas of the brain and identifies locations of critical functions of the brain [11]. Magnetic resonance imaging is the clinically optimal method for a more detailed view of soft internal organs, ligaments, and cartilages. It is safe for pregnant women and children because it does not use ionizing radiation. Through this technique, pathological signs are highlighted, thus helping doctors evaluate a variety of problems such as disorders of the eye and inner ear, tumors, infections, stroke, Alzheimer's disease, and multiple sclerosis [10].

1.2. Stages of development of MRI technique

Researches and theoretical ideas of physicists about the spin of nuclei and their movement formed the scientific background behind the MRI technique. In 1924, the Austrian physicist Pauli assumed that the nucleus of the atom rotates around its axis and that this leads to the generation of a magnetic field around the nucleus. In the 1930s, the



American physicist Rabi innovated the term nuclear magnetic resonance, which explains how atomic nuclei absorb energy at different frequencies and then re-release it when placed in a magnetic field. The resulting spectrum is evidence of the identity and nature of this nucleus [12]. Rabi was able to observe the resonance experimentally and published in 1938 in the Journal of Physics, “A New Method for Measuring Nuclear Magnetic Moment.” Rabi won the Nobel Prize in 1944 [13].

The basics of nuclear magnetic resonance were developed in 1946 by the Swiss-American physicist Bloch and the American physicist Purcell, who independently studied the nuclear magnetic properties of solids and liquids and won the Nobel Prize in Physics in 1952. Bloch developed a series of equations to calculate nuclear magnetization as a function of time given relaxation times [10], [12], [14]. The American physicist Erwin Hahn discovered spin echoes and free induction decay in 1950 for the first time. Then obtained a one-dimensional NMR spectrum by the American physicist Herman Carr in 1952 [15], followed by a study by the physicist Jay Singer in 1959 in humans about blood flow by measuring NMR relaxation times. Although a patent for measuring blood flow with NMR was registered in 1967 by Alexander Ganssen, it did not enter into medical practice until the 1980s. In the same year, 1967, Ligon published his report about water relaxation in NMR of living human arms. After that, Jackson & Langham published the first magnetic resonance signals of a live animal, an anesthetized rat, in 1968 [16], [17], [18], [19].

A Damadian doctor realized that it was possible to differentiate between cancer cells and normal cells through differences in magnetic resonance signals. He assumed that cancer cells contain more water. He conducted his experiment on rats with cancer and healthy ones and discovered that there were clear differences in the relaxation times of the released signals by different tissues as numerical information. His study was published in

1971 under the title "Tumor Detection by Nuclear Magnetic Resonance" [15]. A patent application for "Apparatus and Method for Detecting Cancer in Tissues" was submitted and obtained in 1974. There were early research groups, most notably Damadian, Hazelwood, and Chang, who began cataloging relaxation times for a wide range of biological tissues [12], [14].

From spectra to imaging, Vladislav Ivanov in the Soviet Union proposed the idea of using gradients with frequency-selective excitation for spatial encoding in 1960. His proposal was rejected as improbable, and it was finally approved in 1984 [15], [20]. The technique was developed in 1973 by a British physicist Mansfield, and the American chemist Lauterbur [12], who won the Nobel Prize in Medicine and Physiology in 2003 [20]. Lauterbur proposed using gradients in the magnetic field as a secondary magnetic field to form a map of the internal body, thus obtaining a two-dimensional anatomical image. As for Mansfield, He studied NMR diffraction in solid materials and then used mathematical methods to quickly scan and obtain higher-resolution images [12].

In the late seventies, specifically in 1977, the first human image was obtained using NMR imaging. It was taken by Damadian doctor of the chest of his assistant Minkoff. It required five hours to produce. At that time, radiology developed and in addition to X-rays, it became possible to image in several other ways, such as CT scans and ultrasound (US) and NMR [12], [14], [17], [20]. In the early 1980s, magnetic resonance systems were manufactured commercially, and the term was changed from NMR to MRI technique, and the word "nuclear" was canceled because people feared it and linked it to nuclear weapons, and its name was made to indicate that it is an imaging machine, regardless of its physical basis. Later, successive developments were made in this technology aims to produce high-quality images and reduce scanning time [12], [18].

1.3. Components of the MRI machine

The MRI technique consists of the following main parts: magnets, gradient coils, radiofrequency (RF) coils, computer system and patient table (Figure 1.1).

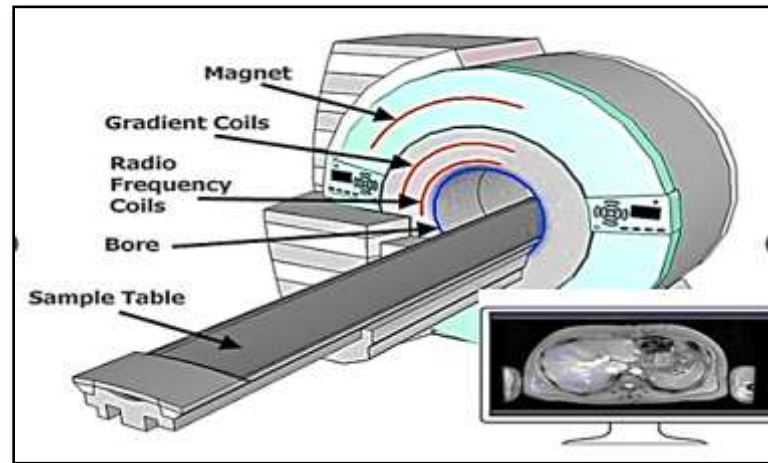


Figure 1.1: The component of the MRI machine.

- Magnet: The basic component in the MRI machine, and its function is to generate a uniform primary magnetic field (B_0) in order to place the patient in it [21], [22].
- Gradient coils: Responsible for the secondary magnetic field that overlaps the primary magnetic field in order to obtain spatial encoding [21], [22].
- RF coils: An antenna for transmitting and receiving radiofrequency signals to and from tissues [21], [22].
- Computer system: It is the control device for producing MRI images by giving commands (type of protocol) after which parameters are related, including the time of repetition (TR) of imaging cycles, the number of imaging cycles, the series of RF pulses, activation of gradients, etc., and so on. Then the signals that carry information about the examined tissue are obtained during the image acquisition process, and they are converted from analog signals to digital and then stored temporarily in K space, from which the image is reconstructed through a mathematical process (FT) to appear

on the screen, and finally the print command is given via a laser camera for producing MRI films [22].

- Patient table: Its main function is to bear the weight and position of the patient who will be examined. As for the laser positioning device to determine the examination location, it consists of two parts, the control and display unit and the moving and braking unit [23].

1.4. Definition of Research Problem

During the development of medical diagnosis, which uses several methods and techniques to detect and identify the lesions. In the field of medical imaging, reliable and exact identification of diseases such as tumors and fluid cysts remain a significant issue. While traditional MRI methods give useful information, their sensitivity and specificity in distinguishing between lesion types may be limited. To address this, researchers have been exploring the potential of advanced MRI sequences like FLAIR and others to enhance lesion detection and characterization. These sequences use the fundamental principles of magnetic resonance to produce distinct contrast mechanisms that may distinguish tissues based on their water content, relaxation durations, and other physical characteristics. However, refining these sequences for specific lesion types and comprehending the underlying physics of picture creation are critical for improving diagnostic accuracy and repeatability.

Despite significant advancements in MRI technology, the optimal combination of sequences and parameters for different lesion types are still being researched. Image quality and diagnostic accuracy are influenced by factors such as lesion size, location, and composition, as well as patient-specific characteristics. Furthermore, interpreting

MRI data frequently necessitates knowledge and experience, resulting in possible inter-observer variability.

1.5. Significance and Justification

The development of robust and reliable MRI protocols for lesion detection is of paramount importance for improving patient care. Accurate and early diagnosis enables timely intervention, leading to better treatment outcomes and potentially increased survival rates. By optimizing MRI sequences and analysis techniques, researchers aim to enhance diagnostic confidence, reduce the need for invasive procedures, and facilitate personalized treatment planning. Furthermore, the ability to differentiate between various lesion types using MRI can aid in determining the most appropriate management strategy. For example, distinguishing between benign and malignant tumors is crucial for treatment decisions. Similarly, accurate characterization of fluid cysts can help avoid unnecessary interventions.

This study aims to contribute to the field by investigating the physics underlying FLAIR and other MRI sequences in the context of lesion detection. By elucidating physical mechanisms responsible for image contrast and exploring the relationship between image features and lesion characteristics. This research seeks to contribute to the ongoing efforts in optimizing MRI techniques for lesion detection by investigating the effectiveness of FLAIR and other sequences in differentiating various lesion types. The findings of this research expected the potential to improve diagnostic accuracy, streamline clinical workflows, and ultimately enhance patient outcomes.

1.6. Objectives of Research

The main objectives of this research can be summarized as follows:

- 1) To investigate the use of various MRI sequences such as Fast Spin Echo (FSE), Multi-Slice Spine Echo (MSSE), and FLAIR for differentiating between types of brain lesions, such as tumors and fluid cysts.
- 2) To examine the effectiveness of FLAIR in enhancing tumor detection by contrasting water-bound and free water signals.
- 3) To compare T1-weighted, T2-weighted, and FLAIR imaging techniques in distinguishing between tumor and fluid, particularly using the contrast agent gadolinium in T1-weighted images to reveal tumor edges.
- 4) To evaluate the role of MRI technology with contrast sequences in providing accurate, non-invasive diagnosis when other diagnostic techniques are less effective.

1.7. Literature Survey

Some previous theoretical and practical and survey studies on magnetic resonance imaging machines with different types and magnetic strengths, how to identify lesions in different sequences according to their physical and chemical properties, and improving the quality of acquired images by processing them with MATLAB are reviewed as follows:

Benson, et al. [24], studied the cervical spine by using MRI machine. They used a Picker International 0.15 Tesla resistive MRI machine and performed with a neck surface coil at a university hospital in England; the slice thickness was 10 mm and the pixel format was 256x256 views. They found excellent tissue contrast & sensitivity in the detection of diseases.

Bhargav, et al. [25], conducted the utility of contrast-enhanced fluid-attenuated inversion recovery (FLAIR) in MRI and compared results with contrast-enhanced T1-

weighted images in different intracranial lesions. They used Forty-nine patients with known intracranial lesions who underwent a gadolinium agent. Their results were that FLAIR images are useful in intra-axial lesions and not useful in extra-axial lesions.

Aprile, et al. [26], studied intracranial lesions performed with conventional techniques and FLAIR. They used forty-five cases of different cystic intracranial lesions and applied T1-weighted, T2-weighted, FLAIR, and PD-weighted sequences. They concluded that the FLAIR sequence was better for imaging intracranial lesions.

Annisa, et al. [27], researched the comparison between T2WI FLAIR and T2WI TSE MRI sequences in brain tumor. The research was conducted with an MRI 3Tesla in the Radiology Installation. The data was in the form of 20 MRI brain images from the T2WI TSE and T2WI FLAIR sequences with cases of intracranial tumors. Then determined the most optimal results of image information generated from the two sequences. The results were that the T2WI TSE sequence showed good intensity and tumor boundaries, while the FLAIR T2WI sequence can show edema around the tumor.

Sahu, et al. [28], searched on the comparison between spin echo T2-weighted and FLAIR for characterizing brain pathology. They used 0.2 Tesla open resistive MRI on twenty cases of different brain pathologies. They found the FLAIR sequence is more beneficial than the T2 sequence.

Zheng, et al. [29], studied the features of recurrent gliomas postoperative intracranial dissemination by using MRI sequences. Ten cases with gliomas underwent MRI, including T1WI, T2WI, fat-suppressed T1WI, and FLAIR. They concluded that MRI with different sequences can diagnose glioma metastasis.

Le, et al. [30], searched about the mismatch between T2 and fluid-attenuated inversion recovery in tumefactive multiple sclerosis. They tested one case of a 46-year-old male with tumefactive multiple sclerosis. They found the T2-FLAIR mismatch sign wasn't a differential feature between glioma and tumefactive demyelination.

Gurwawa, et al. [31], studied the comparison between FLAIR and post-contrast T1-weighted MRI sequences for determining the volume of gliomas in the brain. In this study, volumetric analysis was conducted on the post-contrast T1-weighted and FLAIR MR image sequences of 38 patients with brain gliomas. The results are FLAIR image sequences provide a more complete characterization of gliomas and are more useful when observing tumor response after treatment.

Almutairi, et al. [32], studied multiple sclerosis (MS) in the brain using MRI and applied DIR (Double Inversion Recovery), FLAIR, and T2WI sequences. The images from the DIR, FLAIR and T2WI sequences were compared on axial planes. The patients were randomly selected from the major hospital in Saudi Arabia. All images were obtained using a 3T scanner (Siemens Skyra). A total of 97 MS patients were included. They concluded DIR was a valuable MRI sequence for better delineation, greater contrast measurements, and the increasing total number of MS lesions in MRI, compared with FLAIR and T2WI. In MS patients, it was recommended to add DIR sequence to daily routine imaging sequences.

Moharamzad, et al. [33], researched on diagnosis the grade of brainstem glioma using conventional MR imaging sequences. 62 males and 34 females with brain stem glioma underwent conventional 1.5T MR imaging, which included T1WI, T2WI and post-contrast T1WI sequences. MRI features such as necrosis and inhomogeneous contrast

enhancement in adults and heterogeneous contrast enhancement in children indicated high grade brainstem glioma.

Yamamoto, et al. [34], conducted research on the effect of different static magnetic fields of 1.5T and 3.0T MRI on the tumor diameter measurements in breast cancer. In this study, the maximum diameters of the tumor obtained by DWI and Gd-T1WI on 1.5-T and 3.0-T MRI systems were divided by the maximum diameter from surgical pathology diagnosis to arrive at the tumor ratio index. In the results, there was no significant difference in the ratios of tumor length diameter measured by 1.5-T Gd-T1WI and DWI compared to pathology. But there was a significant difference in the ratios of tumor length diameter measured by 3.0-T DWI and Gd-T1WI compared to pathology.

Leandrou [35], conducted research on the comparison between magnetic resonance imaging and computed tomography in Evaluation of Stroke. The methods and materials that support his project were based on previous publications of the literature, indexed on Pubmed-Medline, Medscape, AMED, CINAHL, and SCIRUS from 2000-2010. A computerized database search was performed using the Oxford's University OxLIP+ interface. He found MRI more effective and accurate than CT in the detection of stroke. MRI can be used for the emergency imaging of patients with suspected acute stroke.

Gossuin, et al. [36], researched the physics principles of MRI from spin to pixel and applications. The study aimed to give a basic description of the physical principles of this imaging technique as well as some examples of its clinical applications. They concluded that MRI is continuously evolving, is used in clinical routine, and will be widely used in the future, and maybe with new sequences, higher magnetic fields, and stronger gradients to improve resolution and reduce imaging times.

Cheng, et al. [37], researched contrast agents in MRI to diagnose the tumors. Through the literature survey. They showed both T1 and T2 MRI contrast agents employed to improve the accuracy, sensitivity, and specificity of tumor diagnosis with MRI, but more efforts in developing MRI contrast agents responsive to particular metabolites or biomarkers were needed.

Caspani, et al. [38], searched on nanomaterial contrast agents that are used in magnetic resonance imaging. This review discussed the principles associated with the use of contrast agents in MRI, as well as the most recent reports focused on nanostructured contrast agents. They concluded that there was a variety of paramagnetic and superparamagnetic nanoparticles with control over their composition, size, and shape. The development of different magnetic effects was important in the pursuit of novel contrast agents in MRI.

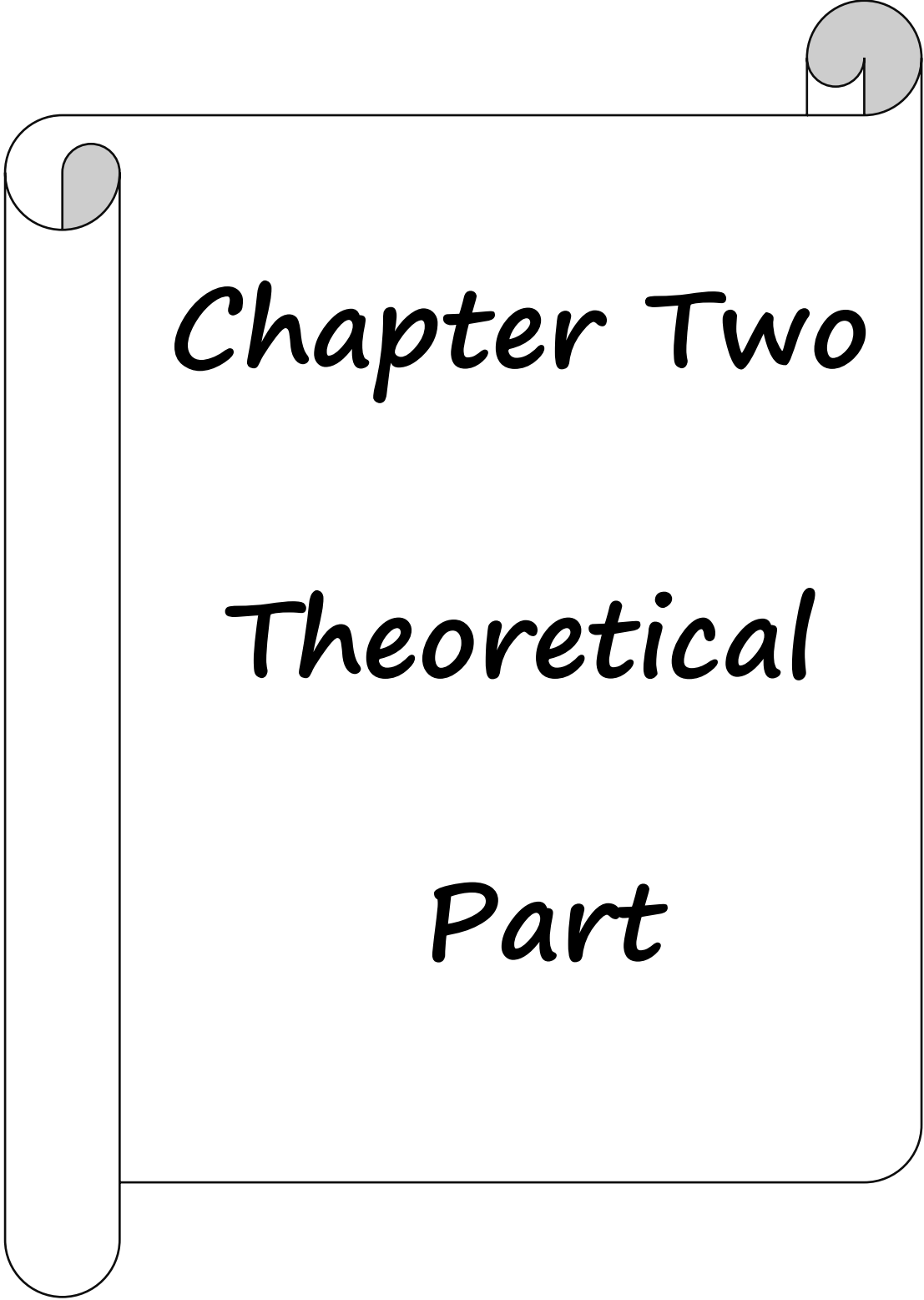
Andriyani & Hasibuan [39], studied the applications of Gadolinium contrast agents in MRI. The method was a literature search on databases (PubMed, Web of Science, EMBASE, and Cochrane Libraries). The studies were conducted within the time frame 2013-2023. The results were that gadolinium had magnetic properties that increased the contrast in the MRI image, thus making it possible to identify structures in the body and early detection of lesions such as tumors, infections, etc. Gadolinium is safe and relatively non-toxic, and it was useful for accurate diagnosis, but there was a risk of its accumulation in patients with impaired kidney function.

Gamage [40], studied the detection of brain tumors by using image processing. The following image processing steps were used: input MRI image, preprocessing, image segmentation, feature extraction, and classification. He concluded tumors can be segmented using various image segmentation techniques. Thresholding was the best and

easiest approach among image segmentation techniques when the contrast between foreground object and background object is comparatively high.

Gaonkar, et al. [41], conducted a study on edge detection by the MATLAB program for brain tumor diagnosis. The methodology was brain tumor, and system software will be written and modelled in the Matlab image processing tool. They took MRI image data as an input source and went to math works in MATLAB processing, such as changing to RGB to Gray and increase or decrease of gray levels then gray was transformed to black and white images that showed desired information in the images. The MATLAB simulation was performed on brain images, and tumor detection was performed for image segmentation and optimal global thresholding.

AL-Najjar [42], conducted a search about the processing for the MRI image MATLAB program to detect brain tumors. The proposed study utilized MATLAB software to detect tumors in multi-echo MRI brain images. Steps followed in the experiment: input the image, preprocessing, enhancing the image, segmenting the image, extracting the tumor, surrounding the tumor with a box, specifying the region of interest, and specifying tumor properties. The study utilized image processing techniques to enhance the MRI images, facilitating the segmentation of tumors. The utilization of computer software for diagnosing medical conditions was on the rise. .



Chapter Two

Theoretical

Part

Theoretical Part

This part provides an in-depth explanation of the physical principles underlying the NMR phenomenon, detailing how atomic nuclei respond to magnetic fields and radiofrequency pulses. It explores how these principles are harnessed in MRI technology, which is widely used in medical imaging to obtain detailed images of the internal structures of the body. Additionally, this section illustrates the role of computer technology in processing and analyzing the data obtained from MRI scans, highlighting how advanced algorithms and computational tools are crucial for generating high-quality, accurate images that assist in diagnosis and treatment planning.

2.1. Physics Background

The physical section refers to primary physical principle behind NMR are based on the behavior of hydrogen nuclei in a magnetic field, which are manipulated using radiofrequency pulses to create detailed images and the hardware components that work together to produce high-resolution images of the body's internal structures. The magnetic field aligns the hydrogen atoms in the body, and radiofrequency pulses induce them to emit signals, which are then detected and analyzed to produce images. Understanding the physical components and their functions is essential for grasping how MRI technology works and its applications in medical diagnostics. These components include the powerful magnetic field, RF coils, gradient coils.

2.1.1. Spin and Magnetic properties

The angular momentum of the nucleus J due to the motion of the spin I , is explained by the following relationship [36]:

$$J = \hbar I \quad (2.1)$$

Where I represent spin, J denotes angular momentum, and $\hbar$ is reduced Planck constant (represent Planck constant h divided by 2π). The magnetic moment of the nucleus M is proportional to the angular momentum J , and the proportionality factor is the gyromagnetic ratio γ , according to the equation below [36], [43], [44]:

$$M = \gamma J \quad (2.2)$$

Where $\gamma = g_n \beta / \hbar$, where g_n is the Land coefficient for the studied nucleus and β is the Bohr magneton for the nucleus $\beta = e\hbar/2m_p$. Where torque τ is given by the following relationship [44], [45]:

$$\tau = M \times B_o \quad (2.3)$$

Where B_0 represents the applied external magnetic field. Torque is also the rate of change in the angular momentum of the nucleus with respect to time [44]:

$$\tau = \frac{dJ}{dt} \quad (2.4)$$

So, the following equation can be formulated [44], [45]:

$$\frac{dM}{dt} = \gamma M \times B_o \quad (2.5)$$

The above equation shows the rate of precession of the magnetic moment of the nucleus around the external magnetic field at Lamor frequency, which is equal to [44], [46]:

$$\nu_o = \gamma B_o / 2\pi \quad (2.6)$$

When there is no external magnetic field, there is no split of energy, meaning that there is degeneracy, but in the event of the presence of an external magnetic field, the hydrogen atoms present in abundance in the human body are affected, especially the isotope H_1^1 , which contains a single proton. This leads to a splitting in the energy level, which is called the Zeeman effect, which results from the interaction between the nuclear magnetic moment and the external magnetic field and leads to the hydrogen

atoms having two types of alignment in the external magnetic field, and is described by the Hamiltonian in the relationship below [36], [43]:

$$H = -M \cdot B_o \quad (2.7)$$

The value of energy in terms of the magnetic quantum number I , according to the following relationship [36]:

$$E = -\gamma \hbar I B_o \quad (2.8)$$

The Hamiltonian has two eigenvalues that correspond to the energies of protons with magnetic moments aligned and misaligned with the external magnetic field, where protons spin up where $I = 1/2$ have a lower energy level, as shown in the equation below [36], [43]:

$$E \uparrow = -\gamma \hbar B_o / 2 \quad (2.9)$$

The protons spin down, where $I = -1/2$, they have a high energy level, as shown in the relationship below [32], [45] :

$$E \downarrow = \hbar B_o / 2 \quad (2.10)$$

The energy difference ΔE between the upper and lower levels is represented as follows [36], [43], [47]:

$$\Delta E = \gamma \hbar B_o = \hbar \omega_o \quad (2.11)$$

That is [44], [48],

$$\omega_o = \gamma B_o \quad (2.12)$$

Where ω_o is called the Larmor angular frequency. According to the Boltzmann distribution, the number of atoms in the upper level $N \downarrow$ (which have a spin-down motion) is less than the number of atoms in the lower level $N \uparrow$ (which have a spin-up motion) [36], [43], [47]:

$$\frac{N \uparrow}{N \downarrow} = \exp\left(-\frac{\Delta E}{KT}\right) = \exp\left(-\frac{\hbar \omega_o}{KT}\right) \quad (2.13)$$

Where k is the Boltzmann constant, and T is the absolute temperature. The protons system act as a paramagnetic magnet in the direction of the effect of the applied magnetic field. The magnetic moment can be calculated as described in the equation [36]:

$$M_z = (N \uparrow)(M_z \uparrow) + (N \downarrow)(M_z \downarrow) = N \frac{(\gamma \hbar)^2}{4KT} B_0 \quad (2.14)$$

Where N is the population of protons.

2.1.2. Excitation

When an RF pulse is transmitted at an appropriate frequency of $\nu_0 = \gamma B_0 / 2\pi$, excitation occurs. This is explained by the fact that this happens, from a quantum standpoint, the population of the upper Zeeman level increases, and this opposes the state of equilibrium and leads to a disturbance of the stable state. As for the classical side, the secondary magnetic field $B_1(t)$, suppose in the direction X , perpendicular to the primary magnetic field B_0 , in the direction Z , and the result is the rate of change of the magnetic moment of the nucleus, which is described by this equation [36]:

$$\frac{dM}{dt} = \gamma M \times (B_0 + B_1) \quad (2.15)$$

In the rotating frame, it simplifies to the following form [32]:

$$\left(\frac{dM}{dt}\right)_{\text{rot}} = \gamma M \times B_1 \widehat{e}_x \quad (2.16)$$

By comparing equation (3.16) with equation (3.5), it is clear that the angular frequency of the torque around the secondary magnetic field B_1 is shown in the equation below [32], [50], [51]:

$$\omega_1 = \gamma B_1 \quad (2.17)$$

The flip angle of the magnetic moment changes by changing the duration or amplitude of the application of the secondary field B_1 , and this is explained by the equation below [36]:

$$\alpha = \omega_1 t = \gamma B_1 t \quad (2.18)$$

In short, when a radio frequency pulse (secondary magnetic field) is sent, the low-energy protons are excited and disturbed, flipping from top to bottom [50], [51], moving synchronously (in the same phase), and their transverse magnetization grows to its maximum value, while the longitudinal magnetization in the z direction decays to zero. Transverse magnetization represents the magnetic moment vector perpendicular to the direction of the external magnetic field, i.e., in the XY plane [10], [52].

2.1.3. Resonance

In general, resonance is a phenomenon that occurs when the frequency of the applied force is equal to or close to the natural frequency of the system. In MRI, resonance occurs when the frequency of the radio frequency pulse is equal to or close to the precession frequency (Larmor frequency) of the nucleus of the hydrogen atom in the water molecule in the human body [50]. The exchange or transfer of energy from the radio frequency pulse to the tissues of the body (excitation mode) does not occur except when the resonance condition occurs (the radio frequency of the transmitted pulse matches the precession frequency of the proton (Lamor frequency)). The precession frequency depends on the strength of the magnetic field and the type of tissue [48].

2.1.4. Relaxation

After stopping the radio frequency pulse, the protons return to normal through a process of time-dependent interactions between the protons and their environment, called relaxation. There are two independent processes explained as follows:

- *T₁-Relaxation*

Longitudinal relaxation describes the process of transferring the longitudinal projection of the net magnetization vector to the initial state before excitation occurs. A transfer of

energy to the surroundings (lattice) occurs, often in the form of thermal energy and Bloch described this with the following equation [36]:

$$\frac{dM_z}{dt} = - \frac{[M_z(t) - M_0]}{T_1} \quad (2.19)$$

The amount of the longitudinal magnetization is recovered according to the time constant T_1 of the particular tissue, as shown by the following equation [48], [49], [50], [52]:

$$|M_z| = |M_0| (1 - e^{-t/T_1}) \quad (2.20)$$

where M_z represents the longitudinal magnetization component, M_0 represents the original magnetization, and t denotes the time. This process is called spin-lattice relaxation and the longitudinal relaxation time constant T_1 , which is the time required for 63% recovery $(1 - 1/e)$ of the original magnetization [50].

- *T_2 -Relaxation*

Transverse relaxation is the decay of the transverse component in the transverse plane $X'Y'$ in the rotating frame during rebalancing, and is described according to the Bloch equation as follows [32]:

$$\frac{dM_{X'Y'}}{dt} = - \frac{[M_{X'Y'}(t)]}{T_2} \quad (2.21)$$

The amount of transverse magnetization decays according to the time constant T_2 of the given tissue, and this is shown by the following equation [36], [46], [47], [48], [53]:

$$|M_{x'y'}| = |M_0| e^{-t/T_2} \quad (2.22)$$

Where $M_{x'y'}$ represents the transverse magnetization component. Decay of transverse magnetization due to phase differences. This process is called spin-spin relaxation, and the relaxation time T_2 is the time required for 37% $(1/e)$ of the transverse magnetization to decay [50]. In practice, the decay time of transverse magnetization is faster than expected and is indicated by the symbol T_2^* [51]. This is due to defects of field

homogeneity due to magnetic susceptibility that depends on the type of tissue and the make of the magnet [52]. T_2^* is shorter than T_2 [46].

In MRI, the tissues (excited protons) emit the energy they absorbed in the form of radio frequency signals, from which magnetic resonance images are formed. The time for the tissues to emit these signals varies according to the physical characteristics of these tissues (materials), and from here the contrast appears in the image [48], [50]. During the process of relaxation and return to equilibrium, the changes resulting from the magnetic field generate an induced electrical current, meaning that the change in tissue magnetization induces an electrical voltage in the receiving radio frequency coils, and this induced current is in the form of a sine wave with the same frequencies as the nuclei precession (Lamor frequency) in the tissue voxel [36].

2.1.5. Bloch Equations

The interaction between weak and strong magnetic fields $M(t)$, $B(t)$ respectively at an angle α produces a torque of the weak magnetic field around the strong magnetic field described by the scientist Bloch in the fixed (laboratory) frame of reference, with the following equations in a simplified manner [54]:

$$\frac{dM(t)}{dt} = M(t) \times B(t) \quad (2.23)$$

$$\frac{dM(t)}{dt} = \gamma M(t) \times B(t) \quad (2.24)$$

By putting [45]:

$$M(t) = M_x(t)\hat{i} + M_y(t)\hat{j} + M_z(t)\hat{k} \quad (2.25)$$

And [54]:

$$B(t) = B_x(t)\hat{i} + B_y(t)\hat{j} + B_z(t)\hat{k} \quad (2.26)$$

With $B_x(0)=0$, $B_y(0)=0$, $B_z(0)=B_0$, we get [47]:

$$\Rightarrow M(t) \times B(t) = \begin{bmatrix} i & j & k \\ M_x(t) & M_y(t) & M_z(t) \\ 0 & 0 & B_0 \end{bmatrix} \quad (2.27)$$

more clarification [54]:

$$\frac{dM_x(t)}{dt} = \gamma M_y(t) B_0 \quad (2.28)$$

$$\frac{dM_y(t)}{dt} = -\gamma M_x(t) B_0 \quad (2.29)$$

$$\frac{dM_z(t)}{dt} = 0 \quad (2.30)$$

The strong magnetic field is explained according to the following equation [45]:

$$B(t) = B_z(t) \hat{k} = B_0 \hat{k} \quad (2.31)$$

The projection of the nuclear magnetic moment onto the XY plane is as shown below [45]:

$$M_{xy}(0) = M_0 \sin \alpha \quad (2.32)$$

The initial value M(t) projection in three directions as shown below [45]:

$$M_x(0) = M_{xy}(0) \cos(\emptyset) = M_0 \sin(\alpha) \cos(\emptyset) \quad (2.33)$$

$$M_y(0) = M_{xy}(0) \sin(\emptyset) = M_0 \sin(\alpha) \sin(\emptyset) \quad (2.34)$$

$$M_z(0) = M_0 \cos(\alpha) \quad (2.35)$$

To solve equation (3.23), we assume that [45]:

$$M_{xy}(t) = M_x + M_y j, j = \sqrt{-1} \quad (2.36)$$

The reformulation of equations (2.28) and (2.29) is as follows [45]:

$$\frac{dM_x(t)}{dt} + j \frac{dM_y(t)}{dt} = \gamma M_y(t) B_z - j \gamma M_x(t) B_z(t) \quad (2.37)$$

$$\Rightarrow \frac{dM_{xy}}{dt} = -j \gamma B_z(t) M_{xy}(t) \quad (2.38)$$

$$\Rightarrow M_{xy}(t) = K e^{-j \gamma B_0 t} \quad (2.39)$$

Where Bz(t)=B₀

$$M_{xy}(0) = M_0 \sin(\alpha) \cos(\emptyset) + j M_0 \sin(\alpha) \sin(\emptyset) \quad (2.40)$$

Applying the initial conditions, we get [45]:

$$\Rightarrow M_{xy}(t) = [M_0 \sin(\alpha) \cos(\emptyset) + jM_0 \sin(\alpha) \sin(\emptyset)] e^{-j\gamma B_0 t} \quad (2.41)$$

$$\Rightarrow M_{xy} = M_0 \sin(\alpha) e^{j\emptyset} e^{-j\gamma B_0 t} \quad (2.42)$$

$$\Rightarrow M_x(t) = M_0 \sin(\alpha) \cos(\emptyset - \gamma B_0 t) \quad (2.43)$$

$$M_y(t) = M_0 \sin(\alpha) \sin(\emptyset - \gamma B_0 t) \quad (2.44)$$

$$M_z = M_0 \cos(\alpha) \quad (2.45)$$

The precession causes changes in the magnetic flux which, according to Faraday, will be detected as an induced voltage in a coil tuned to the Lamor frequency, and the signal will be proportional to the magnetic moment and the speed of the flux change.

2.1.6. Main components of MRI

The main components of the MRI technique, including the powerful magnetic field, radiofrequency pulses, and gradient coils, work in unison to generate the physical phenomenon known as NMR within the human body. When a patient is placed within the MRI scanner, the strong magnetic field aligns the hydrogen nuclei in the body, which primarily reside in water molecules. The radiofrequency pulses are then applied, causing these hydrogen nuclei to absorb energy and enter an excited state. Once the pulse is turned off, the hydrogen nuclei relax back to their original state, releasing energy in the form of emitted signals. These signals are detected by the MRI system, and through complex processing, they are used to generate detailed images of the internal structures of the body, allowing for non-invasive medical imaging.

2.1.6.1. Magnet

The magnet is the main component in the MRI machine [48], [55]. The magnet works permanently and is called the static primary magnetic field. There are different types of magnets used in MRI technology, including permanent magnet, resistive magnet, and superconducting magnet [22], [46], [56], [57].

2.1.6.1.1. *Permanent Magnet*

In this type of magnet, magnetic iron is used that is coiled in the shape of the letter C. The two poles of the magnet are parallel to ensure homogeneity of the field in the gap between them and to obtain a vertical magnetic field between the two poles [63]. The advantages of this type are that it is less expensive in terms of maintenance, and it does not need an electricity source for operation or a cooling source. The disadvantage is that it needs a large amount of iron, which leads to an increase in its weight. To reduce its weight, it is doped with some elements, such as neodymium, and this leads to an increase in its cost. When transporting it, we must be careful. We cannot turn it off, as well as its effect on the devices and people around it [62], [63], [65].

2.1.6.1.2. *Resistive Magnet*

Its name is due to the important property of the materials (they are coils made of copper, aluminum, or other metallic materials), which is resistance (loss of energy in the form of heat) [55]. When an electric current is passed, a magnetic field is produced in this type of magnet. One of its advantages is controlling its operation and stopping, which makes it safe to move from one place to another. Its disadvantages are a large loss of energy in the form of heat, which impedes the production of a strong magnetic field [56], [58].

2.1.6.1.3. *Superconducting Magnet*

It uses coils consisting of special superconducting composite materials, and they are cooled to a very low temperature below the critical temperature (is the temperature below which a material becomes superconducting, exhibiting zero electrical resistance, and this transition depends on the material's specific properties in order of 3-10K) [55], [59], [60], so that the materials are transformed from ordinary materials that have resistance when an electric current passes to superconducting materials that do not have any resistance to

an electric current, and thus a very large current passes through them, generating a strong magnetic field that exceeds the field generated by other types of magnets . However, it is necessary to maintain cooling to below the critical temperature, usually using liquid helium ($-269\text{ }^{\circ}\text{C}$) in order to ensure that it remains in the superconducting state [48], [61]. However, if its temperature rises, this leads to the loss of its superconductivity, and the energy is dissipated in the form of heat and a small current pass, resulting in a weak magnetic field.

2.1.6.2. *RF Coils*

RF refers to the electromagnetic wave frequencies that fall within the range typically between 500 kHz and 1000 GHz [62]. RF coils produce a magnetic field perpendicular to the main magnetic field. They act as transmitting and receiving antennas for radio frequency signals [55]. In other words, they send radio frequency pulses that are received by tissues in resonance state and receive signals in the form of an electrical current, and then send them to the computer system to process these signals to obtain an image of the tissues to be imaged [59]. These coils are usually designed into several parts depending on the place to be examined: a special part for the head, a part for the body, and another part for the limbs. These coils are placed near the area to be examined to capture the highest intensity signals emitting from the tissue [57].

2.1.6.3. *Gradient Coils*

Gradient coils are operated by electric current and work to change the strength of the basic regular magnetic field from one point to another in a gradual manner along the patient in the X, Y, and Z directions [59], [63]. Gradient coils work to change the magnetic field [48], [55], and this leads to a difference in precession frequencies and is useful in spatial encoding of tissue voxels (represents the smallest unit of the volume of the scanned

area that derived from the words "volume" and "pixel.") and slices, i.e., knowing the location of each signal later helps in reconstruction the image. The unit for measuring the change in the magnetic field per unit of distance is milliTesla/meter, where the X-gradient direction produces sagittal images, the Y-gradient direction produces coronal images, and the Z-gradient direction produces axial images. Turning the gradient coils on and off at a speed of a millisecond or less during imaging several times produces a loud noise, which causes discomfort to the patient [10], [57], [64], [65].

2.1.7. Interactions of Magnetic Fields with Tissues

The human body consists of about 80% water, and the water molecule is composed of two hydrogen atoms and one oxygen atom. The hydrogen isotope H_1^1 , which is abundant in the human body, contains a single proton with a positive charge and spinning about its own axis that gives it a magnetic moment [10], [46]. In other words,

- The magnetic moments of the protons align, and a magnetic field is created in the human body in the direction of the external field [50].
- Protons precess around the axis of the external magnetic field, and this precession occurs at a radio frequency called the Larmor frequency. The rate of their precession depends on the strength of the external magnetic field and the type of tissue atoms [10], [53], [57].

Hydrogen nuclei have magnetic properties, and the human body is a rich source of hydrogen isotope H_1^1 [10]. NMR imaging relies on the interaction between the magnetic properties of a hydrogen proton with both a magnetic field and radio frequency waves to produce cross-sectional images of the human body. There are other nuclei, such as C13, F19, Na23, and P31, that can be used in magnetic resonance imaging, but they are less abundant and sensitive than hydrogen (Table 2.1) [50]. These nuclei have an odd mass number; they have magnetic properties and therefore possess magnetic moments. As for

nuclei that have an even mass number, they do not have any magnetic properties, and so they do not have magnetic moments [10], [57], [66].

Table (2.1): Relative sensitivity of some magnetic nuclides.

Nuclide	sensitivity	Reffrence
H-1	1.0	[50]
F-19	0.83	[22]
Na-23	0.093	[67]
P-31	0.066	[68]

This is due to the magnetic properties of the nucleus of the hydrogen atom resulting from spin (a moving charge that generates a magnetic field). Each proton works like a tiny magnet. Some of the magnetic moments of the protons (one part in a million) are aligned, and they are useful in providing magnetic resonance images [10], [53], [66].

2.1.8. Spatial Encoding

The basic idea of spatial encoding is to apply gradients to the primary magnetic field in three directions in order to change the precession frequencies of the magnetized voxels, which are three shapes to form three directions:

2.1.8.1. Slice Selection

When selecting a two-dimensional slice, this is done through magnetic field gradients, according to the following equation [47]:

$$B(r) = B_0 + B_s(r) \quad (2.46)$$

If the direction of the initial magnetic field is in the Z direction, then the gradients of the magnetic field for selecting the slice will be in the Z direction, and this is described in the following equation [47], [69]:

$$B(r) = (B_0 + G_s r) \hat{e}_z \quad (2.47)$$

Where $B(r)$ represents the resultant magnetic field, B_0 represents the initial external magnetic field, $B_s(r)$ denotes the magnetic field added by the gradients, r represents the position vector, and G_s represents the values of the added gradients. This affects the Larmor frequency and thus becomes [47]:

$$\omega_r = \gamma |B(r)| \quad (2.48)$$

That is, less or more than the uniform initial precession frequency ω_0 , and it is explained according to the equation [47], [70]:

$$\omega_r = \omega_0 + \gamma G_s r \quad (2.49)$$

The thickness of the slice is determined by the bandwidth of the radio frequency (the range of frequencies within the RF spectrum), which is essentially the difference between the highest and lowest frequencies used in a pulse, and the strength of gradients of the magnetic field. If we assume that the Z axis is the slice selection axis, then the precession frequency of the slice center (ω_{zc}) is [47], [53], [69]:

$$\omega_{zc} = \gamma (B_0 + G_z Z_c) \quad (2.50)$$

In other words, changing the location of the slice up or down is done by increasing or decreasing the applied radio frequency because the gradient of the magnetic field causes differences in the resonance frequencies gradually along the line (z-direction) [48], [53].

2.1.8.2. Frequency Encoding

During the acquisition process, the signals superposition and using frequency encoding. assume that frequency encoding is performed in the x direction, then the precession frequency of the proton is added to the frequency encoding gradient G_{fe} according to the equation [47], [70]:

$$\omega_r = \omega_0 + \gamma G_{fe} r \quad (2.51)$$

Where ω_r is the added precession frequency. The signal from the entire sample, voxels, and demodulated are given by the following relationships [47]:

The signal from sample

$$S(t) = \int dS(r, t) \quad (2.52)$$

$$S(t) = \int_{-\infty}^{\infty} \rho(r) e^{-i\gamma(B_0 + G_{fe}r)t} dr \quad (2.53)$$

where $s(t)$ is the signal from the voxel, more simply

$$S(t) = \left[\int_{-\infty}^{\infty} \rho(r) e^{-i\gamma G_f r t} dr \right] e^{-i\omega_0 t} \quad (2.54)$$

Then the carrier signal can be separated and demodulated

$$\Rightarrow \check{s}(t) = \int \rho(r) e^{-i\gamma G_f r t} dr \quad (2.55)$$

The frequency encoding gradient is added in order to produce information about the location of voxels in the frequency encoding direction (X-direction) [63]. Frequency encoding is an encoding method by applying a linear gradient field, which causes a distortion in the primary magnetic field in the direction of the frequency encoding to produce different precession frequencies for the voxels in order to distinguish the locations of the voxels on this basis.

2.1.8.3. Phase Encoding

The phase is encoded through the pulse gradient, which gives an additional phase to the tissue voxels, described in the equation below [47]:

$$\phi(r) = -\gamma G_p r T_p \quad (2.56)$$

The signal from the entire sample is given by [47]:

$$S(t) = \left[\int \rho(r) e^{-i\gamma G_p r T_p} dr \right] e^{-i\omega_0 t} \quad (2.57)$$

By removing the carrier wave and demodulating it, we get [47]:

$$\Rightarrow \check{s}(t) = \int \rho(r) e^{-i\gamma G_p r T_p} dr \quad (2.58)$$

The phase encoding gradient is activated on the y-axis, causing changes in frequencies for each part of the column. Then it is stopped, so the column regains its uniform frequency but is out of phase, and the cycle is repeated, but with a different gradient strength to obtain different phases in each cycle [48]. The phase encoding cycle is the time repetition time TR. The data echo signals obtained in this way will eventually begin to fill the K space from the negative pole to the phase sequentially. The phase encoding steps determine the size of the imaging matrix. Time of Acquisition (TA) = phase

encoding steps $\times$ TR $\times$ NEX, where TR is the repetition time, NEX is the number of excitations, and phase encoding steps is the number of row data scan lines [71].

when the signals are collected during the acquisition process [22], and then via Fourier transform, the signals are separated and arranged in their locations in the image pixels in the image reconstruction process [63]. The NMR signal after frequency encoding is a superimposed signal of phase and frequency, and a two-dimensional FT is able to resolve phase differences and frequency differences [48].

2.1.9. K-Space

The signals are collected temporarily and then later reconstructed according to the effect of frequency encoding, as shown below [47]:

$$= \begin{cases} \frac{\gamma}{2\pi} G_{fe} t & \text{FID signals} \\ \frac{\gamma}{2\pi} G_{fe} (t - TE) & \text{echo signals} \end{cases} \quad (2.59)$$

The effect of phase encoding is as follows [47]:

$$\mathcal{K} = \frac{\gamma}{2\pi} G_p T_p \quad (2.60)$$

Briefly in MRI, during the process of tissue relaxation, many waves of different spatial frequencies and directions are emitted, which are captured by radio frequency coils installed near the patient's body and sent to the computer. This stage is called acquisition, and the data is stored in the computer in a K space temporarily [57], [63]. K space, also known as Fourier space, reciprocal space, or momentum space, is a symmetric mathematical space in which the wavelength signals obtained during the acquisition process are stored in a coordinate system whose vector components are K_x , K_y , and K_z and whose units are the reciprocals of distance [53]. The points of this space are complex values and spatial frequency matching; each column in this space represents data acquired from a single frequency encoding step, and each row contains data acquired from repeated phase encoding steps [57]. The data matrix obtained in K

space is later used to reconstruct the image according to both the effect of frequency and phase encoding [63]. This is accomplished through a fast mathematical process called the Fourier transform, during which the overlapping signals are decoded from the voxels and placed in the appropriate pixels [48]. Each pixel is a weighted sum of the individual points of k space, so any defect in a particular point gives distortion in the image. In other words, every pixel is derived from the entire k space, and every point in the k space contributes to the entire image [57].

2.1.10. Image Reconstruction

The measured signal is acquired in the frequency domain through a K-space, the Fourier transform between the object $\rho(r)$ and the signal $S(\mathcal{K})$, described by the following relationship [47], [53], [69]:

$$S(\mathcal{K}) = \int \rho(r) e^{-2\pi i \mathcal{K} r} dr \quad (2.61)$$

To find the location of the body $\rho(r)$ we use the inverse FT and obtain the image, which is given by [47]:

$$\rho(r) = \int_{-\infty}^{\infty} s(\mathcal{K}) e^{2\pi i \mathcal{K} r} d\mathcal{K} \quad (2.62)$$

In other form, if we assume that the two-dimensional frequency field in K space is represented by (K_x, K_y) , and the location in image space is represented by $I(x, y)$, then this can be formulated [36], [69]:

$$S(K_x, K_y) = \text{FT}[I(x, y)] \ \& \ I(x, y) = \text{FT}^{-1}[S(K_x, K_y)] \quad (2.63)$$

In a gradient echo sequence, a k-space along a k_x -direction is filled with the signal resulting from the frequency encoding, and the signal is digitized for successive steps (Δt), the number of steps $n = T_A / \Delta t$, where T_A the time of acquisition, moving to fill a new line by repeating the sequence with a k_y new value, by changing the gradient of the

phase encoding. All k-space is sampled by repeating the sequence to a n different value of (G_y), thus obtaining an $n \times n$ matrix of k-space, and then applying a Fourier transform to produce an $n \times n$ matrix in the image space. The field of view (FOV) of an image can be represented simply as $FOV_x = 2x_{MAX}$, $FOV_y = 2y_{max}$, and the resolution is given by Δy and Δx . Through the properties of the Fourier transform and sampling theory, we give the following equations [36]:

$$\begin{aligned} FOV_x &= \frac{1}{\Delta k_x} = \frac{2\pi}{\gamma G_x \Delta t}, & FOV_y &= \frac{1}{\Delta k_y} = \frac{2\pi}{\gamma G_y T_p} \\ \Delta x &= \frac{FOV_x}{n} = \frac{2\pi}{\gamma G_x T_A}, & \Delta y &= \frac{FOV_y}{n} = \frac{2\pi}{2\gamma G_y MAX T_p} \end{aligned} \quad (2.64)$$

where G_x is the frequency encoding gradient, occupied for a time period T_p .

2.1.11. Contrast

Contrast in general is the relative difference in intensity between adjacent pixels. The contrast (C) between two adjacent pixels, which have two different signals, the intensity of each is I_1 and I_2 , respectively, MR images have specific contrast between soft tissues. The T1-weighted (T1W) image contrast is mostly due to differences in T1, while in the T2- and PD-weighted images (T2W and PDW), the contrast is mostly due to differences in T2 or PD, and described in the following equation [36], [72]:

$$C = \frac{I_1 - I_2}{I_1 + I_2} \quad (2.65)$$

The main reason for contrast in MRI images is the difference in relaxation times and proton densities from one tissue to another, so MRI usually has sufficient contrast for diagnosis [53], except that in some cases we need factors that increase contrast [73], as in detecting small tumors, their sizes, and precisely defining their edges [36]. Contrast materials, or called dyes, are chemical contents used to improve the internal view of the

body during magnetic resonance imaging and increase the quality of images by shortening the relaxation time of some protons for specific tissues. Contrast media work to cause an imbalance in the relaxation times of water molecules adjacent to it, which leads to an increase the contrast in the image, thus highlighting the pathological signs in the tissues more clearly [73], [74].

One of the most widely used and common compounds is gadolinium ion (Gd^{3+}), which is a paramagnetic agent. The compound name in detail is gadolinium diethylenetriamine acetic acid (Gd-DTPA) [73], but due to its toxicity, it is chelation in a light molecule. Gadolinium causes a shortening of the T_1 and T_2 relaxation times, thus increasing the T_1 signal and reduces the T_2 signal. It is considered a contrast agent for T1-weighted images, any organs that received this agent appear bright and it is called a positive contrast agent [36]. The contrast agent is injected into a vein in the arm, and it is necessary to ensure that the patient does not have a history of kidney or liver disease, and it is used in detection of brain tumors [10]. There are negative contrast agents used in detecting tumors of the liver and spleen [73]. The organs that take this agent appear dark on T2-weighted images. Among these agents are iron oxide nanoparticles, which are considered superparamagnetic and work to shorten T_2 and T_2^* relaxation times due to the effect of relaxation of water protons with magnetic field inhomogeneity in the presence of magnetic particles [36], [75].

2.1.12. Sequence

Every sequence has a way of filling K-space. At each repetition of the sequence, a row is acquired in K-space. A pulse of 180 works to shift the location relative to the center of K-space. The following equation explains this [36]:

$$\mathcal{S}(k_x, k_y) \xrightarrow{180^\circ} \mathcal{S}^*(k_x + k_y) = [\iint_{slide} I(x, y) e^{-2i\pi[k_x x + k_y y]} dx dy]^*$$

$$\begin{aligned}
 &= \iint_{slide} I(x, y) e^{-2i\pi[-k_x x + -k_y y]} dx dy \\
 &= \mathcal{S}(-k_x, -k_y) \quad \dots \quad (2.66)
 \end{aligned}$$

The flip angle (α) controls the resulting amount of transverse magnetization, as shown in the equation below [53]:

$$|M_{xy}| = |M_0| \sin \alpha \quad (2.67)$$

The residual longitudinal magnetization is as follows [52]:

$$|M_z| = |M_0| \cos \alpha \quad (2.68)$$

Where M_0 is the original magnetization.

It is a successive repetition of a specific set of RF pulses and gradients to fill K-space. The sequence includes many parameters, such as flip angle, repetition time, and echo time. K-space modeling is also useful in building the image later [52]. The sequence is repeated after a specified time, known as Time of Repetition, and symbolized by the symbol (TR). The number of times the sequence is repeated is equal to the number of phase encoding steps required and depends on the size of the matrix (resolution). Scan time = TR x number of phase encoding steps x number of excitations (NEX). This Sequence includes [57], [63]:

- 1- RF pulse
- 2- Select the slice
- 3- Phase encoding
- 4- Frequency encoding
- 5- Repeat the sequence

Different sequences used in MRI have the main purpose of increasing contrast between the studied tissues. We will present the following:

2.1.12.1. Spin Echo (SE) Sequence

This sequence aims to preserve the distinct T2 effect between tissues and eliminate the effects of field inhomogeneity T2*, as shown in Figures 2.23 [51], [53], [76], [77]. This type of sequence is characterized by a high signal-to-noise ratio (SNR), T2 weighting of distinct features between tissues, and reduced sensitivity to field heterogeneity. The disadvantages of this sequence are that it takes a long time to scan, and a 180 pulse to rephase requires additional RF energy, in addition to activating more than one gradient [48]. Types of spin echo:

2.1.12.1.1. Fast Spin Echo (FSE)

A FSE pulse sequence is a fast pulse sequence in which more than one line of k-space is filled during a single TR, unlike a standard spin-echo (SE) pulse sequence that fills one line of k-space per pulse repetition, which requires a long imaging time [78]. It affects the hospital's financial resources, so the so-called turbo spin echo (TSE) or (FSE) was developed, through which we obtain a train of echoes and this reduces the scanning time while maintaining the diagnostic quality of the image [76]. It enables us to reduce the imaging time by filling multiple lines of K-space while maintaining a relatively long TR, as short TRs will not allow the longitudinal magnetization to be completely recovered and therefore the signal will be weaker. The TR determines the amount of magnetization that flips to the transverse plane at the next pulse. The imaging time is equal to the number of phase steps multiplied by the TR and the NEX, meaning that the imaging time is inversely proportional to the length of the echo train [78].

2.1.12.1.2. HASTE Sequence

There is another fast pulse sequence in the SE pulse sequence family, called Half-Fourier Acquisition Single-shot Turbo-spin Echo (HASTE), and this sequence is very fast, and this sequence uses the same principle as FSE but requires a very long train of

echoes, as it can produce multiple echoes of up to more than 128. In other words, image data is acquired in one shot using long echo trains. We will not need to repeat the cycle, thus imaging an entire slice of the body in less than half a second. This sequence shortens the time for a 180 pulse and shortens the spacing between the internal echo times. This sequence has other names depending on the device manufacturers. For example, Ge Company uses the name Single Shot Fast Spin Echo (SS-FSE), and Philips uses the name Single Shot Half-Fourier Turbo Spin Echo and its abbreviation (SSH-TSE) [79], [80], [81].

This sequence uses a single-shot technique to acquire data for the entire image within a single TR, i.e., a single 90-degree excitation pulse, in order to accelerate image acquisition. It differs from a FSE or TSE sequence, which uses a multi-shot technique with multiple excitation pulses through which it is possible to acquire the entire K-space. For example, if the K-space consists of 128 lines and the length of the echo train is 16, the number of shots equals $128/16=8$, i.e., eight shots [82].

There is a certain symmetry in the Fourier space. K space, which represents the spatial frequency data of the reconstructed MR image, where its center determines the contrast of the image. It can be used to obtain more than half of the required number of echoes and synthesize the rest of the echoes numerically; that is, only approximately half of the K space is collected. The rest is deduced using the inherent symmetry of K space, and thus the entire image is reconstructed. It usually gives transverse relaxation time-weighted images (T2WI), where fluids such as cerebrospinal fluid (CSF) and edema appear bright in the image, despite the advantages of this sequence in scanning speed [82]. This is useful for patients who cannot lie down for a long time and reduces the effects resulting from movement, but it does not provide high-quality diagnostic

images, as it may hide important subtleties such as the size of the ventricle in the brain and limited contrast between tissues [80].

2.1.12.2. Gradient Echo (GE) Sequence

This sequence is known as the fast sequence because the acquisition occurs faster compared to the SE sequence. This is due to the flip angle being less than 90, i.e., small, which leads to the decay of the transverse magnetization in less time and therefore the TR is short, and there is no 180-pulse to re-phase. Gradient is used instead, and this leads to its sensitivity to field inhomogeneity and T_2^* weighting [51], [53], [76], [77]. Types of gradient echo:

2.1.12.2.1. Echo Planar Imaging (EPI)

Echo planar imaging is from the family of GE and its abbreviation is (EPI). Through this sequence, we quickly fill the entire K-space in one shot, and within a time period ranging from 20 to 100 milliseconds, that is, capturing a complete image of an entire slice inside the patient. We obtain multiple echoes of different phase steps acquired by rapidly inverting the frequency encoding gradients [51], [76].

The advantages of the aforementioned sequence are reducing the effects resulting from motion due to the speed of imaging and the ability to image physiological processes inside the human body and the heart. However, one of its disadvantages is that the signal at the edges of the K space is of very small amplitude. This leads to a loss in spatial resolution, so we reduce the size of the matrix, meaning that reduces the resolution of the image to allow rapid filling of the K space, and is also sensitive to susceptibility effects and inhomogeneities of the main magnetic field, and the length of the echo train causes a weighting of the free decay time T_2^* . The limitations of this sequence are how fast the devices are able to apply alternating frequency encoding gradients and also how much the speed of the devices in converting analog signals to

digital to put it in K space. In a single-shot echo planar sequence, more than 128 lines are often acquired in a single TR. However, in a multi-shot echo planar sequence, the range of phase steps is divided into several multiple shots or TRs [83], [84].

2.1.12.2.2. Stimulated Echo (SE)

Stimulated echo, also from the family of GE, and its abbreviation is (STE), arises from three or more radio pulses with flip angles of 90 or less. It produces a stimulated echo whose amplitude depends on the T_1 and T_2 . This sequence gives an image that contains a mixture of contrast. Here contrast is a combination of T_2 weighting from the stimulated echo and T_1 weighting from the FID or GE [85]. The number of generated echoes (E_n), depends on the number of transmitted pulses (n^{th}), and this equation explains that as follows [86]:

$$E_n = [3^{(n-1)} - 1]/2 \quad (2.69)$$

The amount of magnetization generated the signal intensity by three pulses at flip angles $\alpha_1, \alpha_2, \alpha_3$ with time intervals t and T is explained by the following equation [86]:

$$M = \frac{1}{2}M_0 \sin \alpha_1 \sin \alpha_2 \sin \alpha_3 e^{-[T/T_1 + 2t/T_2]} \quad (2.70)$$

2.1.12.2.3. Coherent Gradient Echo

This sequence is also from the GE family and is characterized by the fact that at each TR the spins are completely identical in phase, which is why it is called a coherent echo. The phase encoding gradient is applied to remove the phase before the echo time and the rephase gradient after the echo time, that is, before the radio frequency pulse. In this sequence, TE and TR are short, and the sample signal is made up of the stimulated echo and the gradient echo, so the resulting image contains a mixture of T_1 and T_2 weighting.

This depends on TR, TE, flip angle, and the T_1 and T_2 relaxation times of the tissues [48], [85].

2.1.12.2.4. *Incoherent Gradient Echo*

One of the MRI techniques that relies on GE, where residual transverse magnetization is destroyed at each end of a cycle, so it is called the spoiled gradient [51], where the phase encoding gradient is activated to remove the phase before the next pulse - after TE and before TR. Therefore, there is a loss of transverse magnetization in the next pulse, and the generated signal will be the result of the longitudinal magnetization vector, which will remain for a specific period of time before the free induction decays [48], [85], [87], [88], [89]. The signal is the free excitation decay, and because TE is very short, there are very small differences in the free excitation decay between tissues, so what we get is a T1WI. The typical time for TR is (30-500) milliseconds [48]. TE is less than 15 milliseconds. The weight of T_1 increases with increasing flip angle (α), and the weight of T_2^* increases with increasing TE [90].

2.1.12.2.5. *Steady State Free Precession (SSFP) sequence*

This sequence is a type of GE sequence, as it maintains a stable residual transverse magnetization between successive radio pulses. The transverse magnetization decays in the middle between RF pulses, and using Rewinder Gradient to rephase in the timing of the next RF pulse. The repetition time is much less than the transverse relaxation time ($TR < T_2$), as the free induction decay tails and echoes combine. This sequence gives us a predominantly T2WI [48], [76], [91].

2.1.12.3. *Inversion Recovery (IR) Sequence*

The inversion recovery (IR) sequence is one of the types of sequences used in MRI. The goal of this sequence is to increase the contrast between tissues by zeroing out a

specific tissue signal [28]. This sequence is similar to the spin echo sequence by adding a 180 pulse at the beginning, where the spin echo sequence consists of two pulses (Pulse 90, Pulse 180) and their echo [51]. The inversion recovery sequence contains three pulses (Pulse 180, Pulse 90, Pulse 180) and echo [48]. Timing parameters for the IR pulse sequence (Figure 2.39) are explained as follows [51]:

- Time of Inversion (TI): The time between 180 RF pulse and 90 RF pulse.
- Time of Repetition (TR): The time between the 180 first inversion pulse and the 180-inversion pulse of the next scan.
- Time of Echo (TE): It is the time between the 90 RF pulse and the appearance of the echo signal.

We can know the TI by knowing the relaxation time (T_1) of the tissue whose signal is to be zeroed, using the equations below [48] :

$$\text{Signal intensity} = 0 = 1 - 2e^{\frac{-TI}{T_1}} \quad (2.71)$$

$$TI (\text{null}) = 0.693 T_1 \quad (2.72)$$

Where T_1 is the value of the tissue whose signal is to be zeroed.

2.1.12.3.1. Short Time Inversion Recovery (STIR)

Cancel the bright fat signal in STIR by making the TI short (120-130 ms) [57]. The signal from the fat is canceled by adjusting the inversion time (TI) so that there is no transverse magnetization of the fat at equilibrium. Tissues with a T_1 appear dark and have a lower signal intensity [51]. This reflects the longitudinal relaxation time-weighted image (T1WI), and for the tissues that pass through the bounce point, its signal is cancelled according to the choice of the inversion time (TI). This makes pathological areas that are filled with fluids, such as edemas, infections, tumors, and lesions, brighter and more prominent, and this is useful in imaging muscles and bones. The fat signal is

suppressed, and tissue pathological signals that may be hidden by fat in other types of images are highlighted [57], [92].

2.1.12.3.2. Fluid Attenuated Inversion Recovery (FLAIR)

Canceling the water signal in FLAIR by making the TI long (2500 milliseconds), where the signal from the liquid is canceled by adjusting TI so that there is no transverse magnetization of the liquid at equilibrium [57]. The image appears similar to T₂ except that free flowing water such as CSF appears in black, as for non-free flowing water, it appears white and fat appears black. It is characterized by the fact that the repetition time and TE are very long, and it is usually used for the brain [93]. Giving a low signal to the CSF enables us to distinguish between lesions and CSF, and precisely determine the edges of the CSF [94]. Delineation of lesions near ventricles, and differentiation between tumors and tumors surrounding edema [57].

2.1.12.4. Parallel Imaging (PI)

The main goal of parallel imaging is to reduce the image acquisition time [53], [95], by using several coils to capture the MRI signals [58], and thus each coil receives a signal m_j , and the following equation explains this [36]:

$$m_j(k) = \int I(x, y) s_j(x, y) e^{-ik \cdot r} dr \quad (2.73)$$

Where S_j is the sensitivity of coil j , and then the image is reconstructed by processing signals from different coils. The repetition steps decrease by a factor proportional to the number of coils, but increasing the number of coils increases the complexity of signal processing and reduces the signal-to-noise ratio [96]. There are two ways to reconstruct the image:

- 1- The K-space filling method, abbreviated as (SMASH), works on different linear combinations of the different coil signals to fill the K-space, and then using the FT to produce the image [97].
- 2- The method of filling the image space, the abbreviation for this method is (SENSE). This method is very complex, as one voxel of the image is measured in several coils [53], [98].

2.1.13. Characteristics of Magnetic Resonance Images

MRI images simulate lifelike anatomy and appear in gray scales. Normal tissues and pathological changes are accurately displayed because they reflect different intensities of MRI signals due to differences in relaxation times of different tissues, while CT images appear in different gray levels because they reflect different tissue densities. Magnetic properties are what control the contrast in MRI images, not the attenuation of the rays in the imaged structures. It is possible to obtain many different contrasts by applying different sequences with different weights.

2.1.13.1. Spin-Lattice Relaxation Time Weighted Image (T_1 WI)

These images depend on the longitudinal relaxation time (T_1) of the tissue. The time for recovery of longitudinal magnetization varies from one tissue to another depending on its characteristics [53]. The images reflect differences in the T_1 parameter between tissues, so fat appears bright in the image due to the shortness of T_1 , meaning that it recovers longitudinal magnetization faster [73]. The signal is strong and appears white. On the other hand, water is characterized as CSF, cyst and edema along T_1 , that is, the longitudinal magnetization slowly recovers after the RF pulse, giving a faint signal and appearing in the image in a dark black color, while the rest of the tissues take on gray levels. As for bone and air contain little hydrogen, so the signal is weak and appears black. The TR must be short, that is, shorter than the time to return to equilibrium and fully

recover the original magnetized tissue to contrast appear between the tissues. The TE must also be short to reduce the contribution of the T_2 signal, as it is very useful in identifying anatomical structures and details [48].

2.1.13.2. *Spin -Spin Relaxation Time Weighted Image (T2WI)*

These images depend on the transverse relaxation of the tissue, in which dephasing of the magnetic spinning vectors occurs in the transverse plane. The decay time of the transverse magnetization varies from one tissue to another according to its characteristics [53], and the images reflect the differences in the T_2 parameter between tissues [73], so T2WIs differ from T1WIs in that T_2 is short in fat and appears dark in the image as black, and long T_2 in water and appears bright in the image as white. Prolonging the TR in order to reduce the contribution of T_1 contrast, lengthen the TE and to avoid the effect of field inhomogeneity T_2^* a refocusing pulse is sent. It is very useful in identifying pathological changes (lesions), but one of its disadvantages is the difficulty in differentiating between lesions such as edema and CSF because both give a bright white color in the image [48].

2.1.13.3. *FLAIR*

In FLAIR the signal from the fluid is canceled by adjusting TI so that there is no transverse magnetization of the fluid at equilibrium, and the image shows a similar to T_2 except that the free-flowing water, like the cerebrospinal fluid, appears black, while non-free flowing water appears white. It is characterized by the fact that the repetition time and echo time are very long, and is usually used for the brain [99]. Giving a low signal to the CSF enables us to distinguish between lesions and CSF, and precisely determine the edges of the CSF, delineation of lesions near cavities, differentiation between tumors, and tumors surrounding edema [100]. TI varies based on the material from which we want to zero the signal [57].

2.1.13.4. Diffusion Weighted Image (DWI)

Diffusion-weighted images are one of the types of magnetic resonance imaging in which the contrast in the image is shown from the property of water diffusion in tissues (measuring Brownian motion within tissues). In pure water and gel, diffusion is free and unhindered, but in soft tissues, water diffusion is restricted by cell membranes and organelles. Restricted water appears bright and is very useful in distinguishing tumors, pus, and cerebral ischemia [58].

2.1.14. Magnetic Resonance Images Quality Evaluation Indexes

There are several indexes that evaluate the image quality in MRI, which are explained in detail below:

2.1.14.1. Noise

Magnetic resonance image signals include noise signals. Noise is a statistical parameter of a non-imaging signal intensity, meaning that noise is unwanted signals [50]. The signal-to-noise ratio is one of the most important indexes of evaluating image quality and measuring contrast enhancement, abbreviated as SNR, where the detected magnetic resonance signals from voxels are unavoidably mixed with noise. Magnetic Resonance Image = signal + noise. Signal-to-Noise Ratio (SNR) = signal / noise. The signal-to-noise ratio is affected by several factors such as [81]:

- Magnetic field strength.
- Radio frequency coils.
- Radio frequency bandwidth.
- Characteristics of the imaged tissues.
- Repetition time and echo time.
- Flip angle.
- Thickness and space of the slice.

- Matrix size and field of view.
- Number of excitations.

2.1.14.2. Contrast

Image contrast is one of the indexes of the quality of MRI images. Image contrast comes from the difference in intensity of NMR signals resulting from tissue voxels. The quality of contrast between tissues, especially between normal and pathological tissues, plays a major role in clinical diagnosis. Factors influencing contrast are [48], [53]:

1. Pulse sequences: spin echo sequence, gradient echo sequence, inverse recovery sequence.
2. Pulse parameters: repetition time (TR), echo time (TE), inversion time (TI), relaxation times, flip angle.
3. Contrast enhancing agent: gadolinium.

2.1.14.3. Resolution

Resolution is one of the important indexes for evaluating the quality of magnetic resonance images. Resolution is affected by the size of the voxel, which is determined by the field of view, the size of the scanning matrix, and the thickness of the layer. The smaller the voxel, the higher the resolution. Increasing the resolution in MRI images enables us to see small details and thus diagnose diseases better early [57].

Resolution is expressed in the number of pixels. The pixel value is equal to the ratio of the field of view (FOV) to the size of the imaging matrix. When the field of view is constant, the more and smaller the number of pixels, the higher the resolution. Pixel size = field of view / matrix size [57], [101].

2.1.14.4. Artifacts

Artifacts refer to false images that are not present in the body examined but are visible. There are two reasons for producing artifacts [71], [102]:

- Incorrect operation of equipment.
- Physiological characteristics of the body examined.

Table 2.2: The relationship between scanning parameters and image quality [103].

Target	Scanning parameters	Results
SNR increases	↑ NEX	↑ Scanning time
	↓ Matrix	↓ Scanning time ↓ Resolution
	↑ Thickness of layer	↓ Resolution
	↓ Bandwidth of RF	↑ Min TE ↑ Chemical shift
	↑ FOV	↓ Resolution
	↑ TR	↓ T1 weighted ↑ Number of layers
	↓ TE	↓ T2 weighted
Spatial resolution increases	↓ Thickness of layer	↓ SNR
	↑ Matrix	↓ SNR ↑ Scanning time
	↓ FOV	↓ SNR
Scanning time decreases	↓ TR	↑ T1 weighed ↓ SNR
	↓ Matrix	↓ Resolution
	↓ NEX	↓ SNR ↑ Motion artifact
	↓ Number of layers	↑ SNR

2.1.15. Safety in MRI

Following safety instructions and warnings in MRI units is very important, as we deal with the influence of a huge magnetic field. It is necessary to ask patients before entering the MRI room about the metal objects they have and to confirm this when in doubt by conducting an X-ray to ensure that the patient is free of metals [104].

2.1.15.1. Magnet Field Effect

The effect of the huge magnetic field present in the MRI system is to forcefully attract ferrous materials. The patient or anyone in the flight path is harmed, and oxygen tubes are not inserted in the area of the MR system [10]. There is an area drawn around the room in which the MRI machine is located. This area has a magnetic field strength

greater than or equal to 5 gaussses [104]. It is forbidden to enter this space for people who have [10], [57]:

1. Bullets.
2. An artificial joint or artificial bone stabilizer.
3. A foreign metal body in the eye or any other place.
4. Anything metal, such as hair clips, pins, belts, metal zippers, or credit cards.
5. Mobile phones, metal jewelry, pens, watches, coins, keys, and stethoscopes.

2.1.15.2. *RF Coils Effect*

The oscillating electromagnetic fields arising in RF coils affect the human body due to eddy electrical currents that are generated in the tissues and lead to an increase in the body temperature [104], and this is measured by the Specific Absorption Ratio (SAR). 1 SAR = 0.5 degree increase in body temperature. SAR (W/Kg) = RF energy deposited per mass of tissue. This affects patients with an inability to regulate temperature, such as [57]:

1. People with heart failure or fever.
2. Pregnant women.
3. Those who have lost the sensation of heat.
4. Those who take medications such as vasodilators, diuretics, and tranquilizers.

Factors that increase the specific absorption ratio, such as [57], [105]:

1. High static magnetic field strength.
2. Using a 180-degree radiofrequency pulse.
3. Rapid repetition of radiofrequency pulses.
4. Connecting tissues (blood, brain, CSF).

Techniques that contribute to reducing the specific absorption ratio, such as [57], [106]:

1. Increase the repetition time between radio frequency pulses to prevent higher energy deposition.
2. Reducing the flip angle.
3. Reducing the number of slices.
4. Reduce the room temperature and loosen clothing.

2.1.15.3. Gradient Coils Effect

Rapidly changing gradient magnetic fields cause currents in conductive tissues, and they work to [57], [104], [107]:

1. Loud noise that may reach more than 90 decibels, the limit that causes irreversible damage, and this noise is proportional to the increase in both the magnetic field strength and the amplitude of the gradient.
2. It causes stimulation of the peripheral nerve, difficulty breathing, and ventricular fibrillation.
3. Induced voltages disable the pacemaker, ECG monitors, and cochlear implant.

2.2. computational background

This section describes the concepts of DICOM (Digital Imaging and Communications in Medicine) and PACS (Picture Archiving and Communication System). Furthermore, this section delves into the process of image processing techniques, using MATLAB software to enhance MRI images for improved clarity and diagnostic accuracy. It describes methods for reducing noise, which can obscure critical details, and techniques for segmenting medical images to isolate specific regions of interest. These image enhancement and analysis techniques are essential for optimizing the quality of MRI scans, enabling more precise interpretation and aiding in accurate medical decision-making.

2.2.1. DICOM Format and PACS System

Two related terms that are circulated together are the DICOM format and the PACS system, but they are different in meaning, as the DICOM format is a standard for medical image formats and data that store, send, and transmit, while PACS is a system that uses the DICOM format to store, transmit, and manage medical images [108].

2.2.1.1. DICOM Format

DICOM is an abbreviation of the first letters of the term "Digital Imaging & Communications in Medicine" [109]. It is a set of international standards that were established in 1985 by the American College of Radiology and the National Association of Electrical Instrument Manufacturers to process the issue of vendor-independent data formats and data transfer processes for digital medical images [110]. It represents the responsible language for information transfer, storage, printing, and transmission of digital radiology images [109]. There was an urgent need to establish a unified encoded platform for exchanging radiological images and information between medical entities, so DICOM was established and is often combined with HL7 software for higher functionality in medical facilities, hospitals, and clinics [111]. DICOM is used in all types of medical imaging (X-rays, CT, MRI, US, etc.) in interaction with printers and medical management platforms such as electronic medical records and PACS software [108].

A DICOM file contains an image of the medical examination, patient identification data (name, age, date of examination, timing of examination, type of examination, part examined, and many other details), and information about the parameter values used during the examination. This helps us not to confuse patient files. The advanced DICOM standard is used all over the world and represents the preferred tool for medical imaging.

In short, it is a format that allows medical staff to safely manage medical imaging files while adhering to the privacy and confidentiality of patients' health data [112].

2.2.1.2. PACS System

PACS is an abbreviation of the first letters of the following term "Picture Archiving & Communication System". It is a medical imaging technology that provides storage space and easy access to images through multiple methods [113]. The idea of PACS was first established in 1982 in a British hospital [114]. The importance of storing medical images is increasing as the size of images grows rapidly as a result of the development in imaging devices that can now take hundreds of images in a short time [115]. The PACS technique is an effective system that allows users to acquire medical images by any medical imaging method in the radiology department, send them, store them, retrieve them, and modify them. It enables us to have digital access to medical images, which reduces the possibility of error in treatment and prevents excess testing caused by lost examinations for the patient, which leads to improved safety and saves time and money [116].

- The PACS consists of the following [108], [117]:
 - a. Medical imaging devices are necessary to acquire examination images.
 - b. System management network and servers to share and exchange medical files among medical staff members.
 - c. A workstation for displaying images, analyzing them, diagnosing them, and writing the necessary reports.
 - d. Storage media, i.e., electronic archives for storing medical images and information and retrieving them when needed.
- The most prominent advantages of PACS [116], [118]:
 - a. Saving time for patients, staff, and radiologists.

- b. Easy access to data and information at any time.
- c. There is no need to carry films from one place to another.
- d. Reducing high costs.
- e. Improving productivity.
- f. It makes it easier for patients to get a second opinion.
- g. Ease of comparison between old and modern images and reports.

2.2.2. Image Processing Using MATLAB

An image in general is a visual representation of a specific thing, taken via any device designated for this purpose. A digital image is a representation of a finite set of discrete values called image elements or pixels on the computer. The pixel is the smallest element in the image and represents the brightness value, the intensity level. Expressing a digital image as a function $f(x,y)$, where x and y are the spatial coordinates, and f is the amplitude for certain values of x and y , and the intensity at that point (pixel) is called, as it gives information about intensity and color. A digital image can be represented in three dimensions, where x , y , and z are the spatial coordinates, and the pixels are arranged in the form of a matrix, as in an RGB image [119], [120].

Each image is a matrix consisting of rows and columns of pixels. The greater the number of pixels, the greater the resolution, that is, the clarity. Digital images are files containing digital data and have different formats and sizes. First, image sampling is done: taking the image and determining the coordinates (x, y) , and second, image quantization: determining the level of intensity for each coordinate (x, y) [121]. There are types of digital images [120]:

- 1) Binary images: In this type, each pixel can be represented by either a zero or a one; that is, they are only one-bit images, often containing two colors, white and black.

- 2) Grayscale images: In this type, each pixel can be represented by a specific degree of gray between zero and 255, and the pixel becomes lighter as it moves away from zero.
- 3) Color images: In this type, each pixel is represented by one shade of color, where the primary colors are red, green, and blue, and they are combined in specific ratios to obtain a wide range of colors.

Digital image processing (DIP) means performing operations on images to extract images with the desired results. The input to the computer system is a digital image, and then processing is done using specific algorithms, and the output is a more useful image (Figure 2.1). The primary goal of digital image processing is to improve the quality and characteristics of the image and also obtain meaningful information [122].

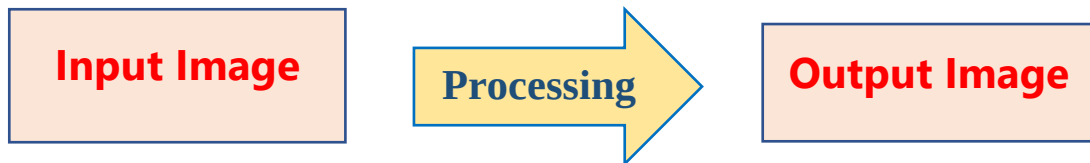


Figure 2.1: Flowchart of image processing concept.

Applying image processing techniques can avoid local defects and reach the best and most useful in the field of medicine. Image processing aims to get rid of the following problems: noise pollution, information hiding, organization of ambiguous edges, and several other things that have an impact on the accuracy of pathological diagnosis and thus on the treatment followed as well [123]. One of the advantages of obtaining medical images in digital form is allowing them to undergo procedures and steps of different types of processing by computer in order to reduce noise, increase the clarity of details, enhance the contrast characteristics of the image, and thus improve the quality of the images. The visual effects of the disease locations are manipulated, making it more visible; this makes it easier to reach maximum visibility and enables us to obtain more information about it, which leads to a more accurate diagnosis [124].

MATLAB is a powerful simulation program designed for engineers and scientists. Its high-level programming language is based on matrices and allows the expression of arithmetic mathematics [125]. It allows us to mathematically manipulate matrices and graphs and implement various algorithms. Through it, medical images can be processed and analyzed using the MATLAB toolbox [126]. In short, MATLAB is a matrix laboratory through which we can process medical images [127].

2.2.2.1. Image Processing Principle

The digital image is processed by passing the digital calculations representing the image through a series of mathematical algorithms and then obtaining the results. This leads to changing its shape and making it clearer and better suitable for specific applications, as images can be reproduced with higher quality by applying many linear and non-linear operations to them. Therefore, this digital processing is widely used in the medical field, specifically in medical diagnosis of images resulting from detection scanning with medical imaging devices [126].

2.2.2.2. Image Processing Types in MATLAB

Image processing using MATLAB can read, write, and show images using the following functions: `imread()`, `imwrite()`, and `imshow()`, respectively [120]. If the images are in dicom format, reading the image, showing it, and obtaining information related to it is done using the following functions: `dicomread()`, `imagesc()`, `dicominfo()`. It is possible to convert image formats from one form to another, and it is also possible to restore images in order to change the spatial resolution through the `imresize()` function, and the `imcrop()` function is applied to crop a specific part of the image. The `rgb2gray()` function is used to convert from color to gray scale [128].

Through the MATLAB Laboratory for matrices, we can easily obtain the data of the magnetic resonance image matrix. It is possible to reconstruct the image using certain

algorithms, and it is possible to apply the Fourier transform; in addition, there are a large number of linear and non-linear filtering functions, convolution, and other operations that are applied to the image matrix and give results in improving vision [129]. There are several things to process images after obtaining them in MATLAB, such as improving edges, softening a specific area, changing the width of windows, filtering, or calibrating. There are several functions that correspond to these functions in the MATLAB program. In short, MATLAB works to provide a more accurate diagnosis and better vision [127].

2.2.2.2.1. Image Enhancement

The basic method for processing digital images is image enhancement by contributing to the details of the image and thus improving the accuracy of the interpretation of the image and selecting or extracting the important part in the image as shown in Figure 2.89 [130]. For example, using a mask to remove the skull or to identify a tumor using the `roipoly()` function, and then multiplying the image by the mask (`Image. *mask`) to show location of interest [128]. Image enhancement represents a process through which an image is processed in a specific way for a specific purpose, such as reducing noise in the image, highlighting important details in the image by enhancing contrast, showing sharp edges in the image, improving image quality, or removing blur from the image. The results are images that are more suitable for the specific applications and a better view of the required parts [130].

It is possible to obtain several forms that include improving the image, such as a negative image using the `imcomplement()` function, setting a specific threshold using the `imbinarize()` function, highlight edges using the `edge()` function, and contrast stretching using the `imadjust()` function [128]. One of the most common methods is the

histogram correction method, using the `histeq()` function to normalize the gray point information in the form of a gray level distribution and adjust the gray scale of the image, thus achieving the goal by using the physiological characteristics of the human visual system to distinguish the details of the image [131].

2.2.2.2.2. *Filters*

The filter works to attenuate unnecessary image information [132]. There are several types of filters, and they are divided into two parts:

2.2.2.2.2.1. *Spatial Domain Filter*

Spatial field filters depend on direct manipulation and processing of image pixels [133], and are divided into two types:

- Linear filters include the operations of adding, subtracting, multiplying, and dividing the image using the following functions: `imadd()`, `imsubtract()`, `immultiply()`, and `imdivided()` respectively [128]. Also there is a smoothing filter to hide details – the Gaussian filter [132], which is used to remove Gaussian noise [130], and is applied the `imgaussfilt()` function, and the mean filter, by applying the `imfilter()` function [128]. There is also a Laplace filter to show details; the Laplace filter is a differential operator that highlights sudden changes in specific areas and attenuates small changes in the gray scale. The Laplace filter detects edges using the second derivative of the image [134], via the `locallapfilt()` function or by setting the Laplace filter $\begin{bmatrix} 0 & 1 & 0 \\ 1 & -4 & 1 \\ 0 & 1 & 0 \end{bmatrix}$ and then use the `conv2()` function [128].
- Non-linear filters: used to remove spot noise from the image, most notably the median filter, which is used to remove salt and pepper noise [130], [135], using the `medfilt2()` function [128]. There is maximum filter, minimum filter [136].

2.2.2.2.2. Frequency Domain Filter

Frequency domain filters depend on modifying the Fourier space of the image [133], and are divided into two types [131], [137]:

- High pass filter

It allows high frequencies to pass through, and attenuates low frequencies that are lower than the cutoff frequency. This makes the image clearer, and fine details appear in the image, making the image appear sharper. One of its disadvantages is that it amplifies noise, and the image becomes grainy and unnatural. Frequencies at the periphery of k-space pass through, excluding the center, using the `highpass()` function [128].

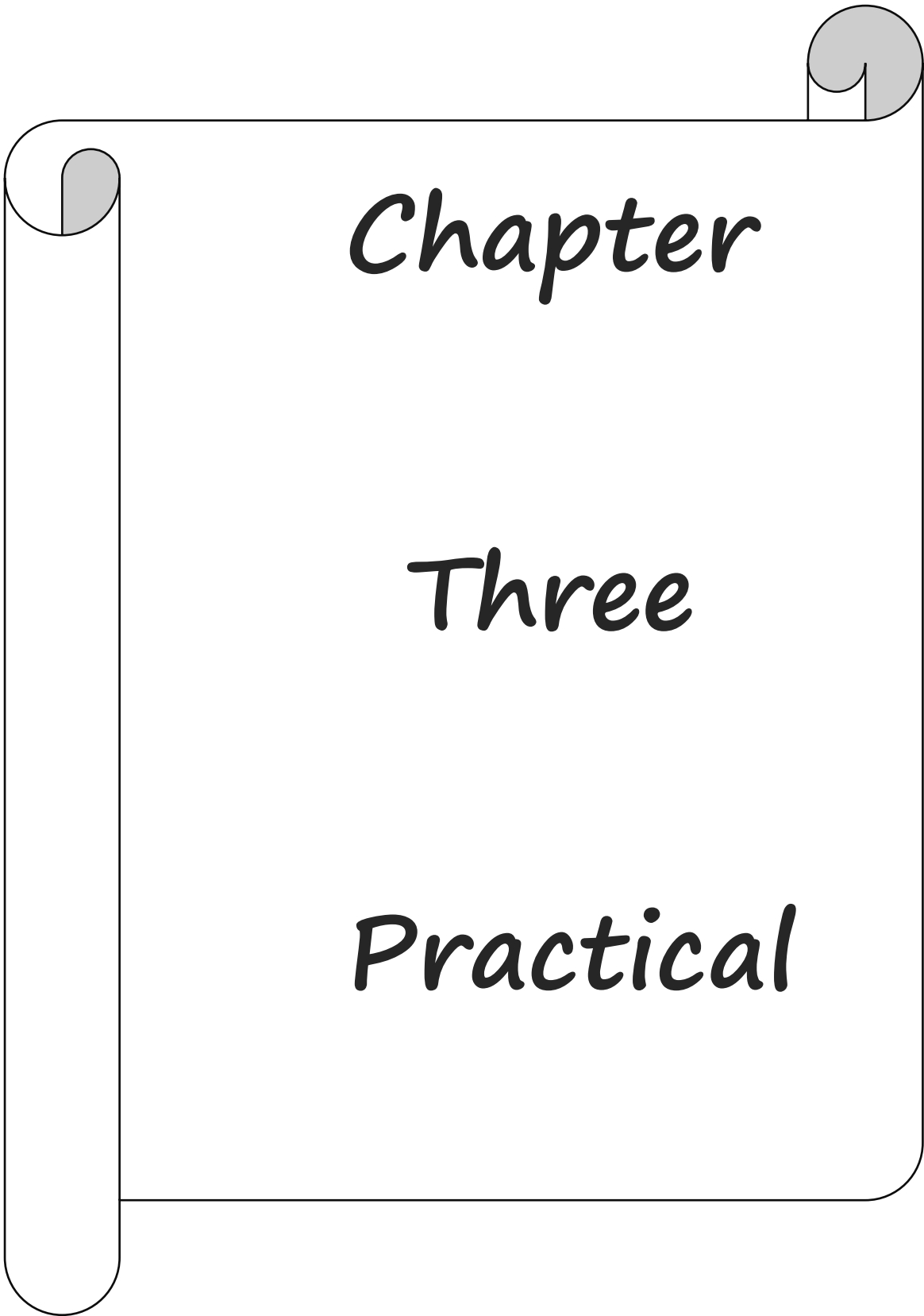
- Low pass filter

It allows low frequencies to pass, and high frequencies that are higher than the cutoff frequency are attenuated, resulting in a smoothing of the image and the absence of fine details, as the frequencies in the center of k-space pass through, using the `lowpass()` function [128].

2.2.2.2.3. Image Segmentation

Its most prominent applications are separating the image from the background [120], and image segmentation is used to determine a specific anatomical area. Brain tumors can be detected using segmentation. This method is applied so that the image is represented in another way that is easy to understand, and converting the image into different parts, by dividing the image and separating the foreground pixels from the background [130]. By using the threshold limit using the `graythresh()` function, this leads to calculating the threshold using the Otsu method, pixels with an intensity higher than the threshold limit are white and pixels with an intensity lower than the threshold limit are black. Then use the `imbinarize()` function to convert the image to a binary

image using the calculated threshold. The image also can be segmented through the binary image using the `im2bw ()` function [128].



Chapter

Three

Practical

Practical Part

This section offers an in-depth and comprehensive description of equipment, agents, and cases that were utilized in this research (Figure 3.1). It outlines the specific technologies, and materials that were employed, along with the characteristics and roles of each element in supporting the study's objectives. Furthermore, this section provides a clear and detailed explanation of the methodology that was followed throughout the study. This includes the step-by-step procedures that used in data collection, analysis, and interpretation, ensuring transparency and reproducibility in the research. The section aims to give readers a thorough understanding of the research framework, allowing them to grasp the logical progression and rationale behind the methods applied in this study.

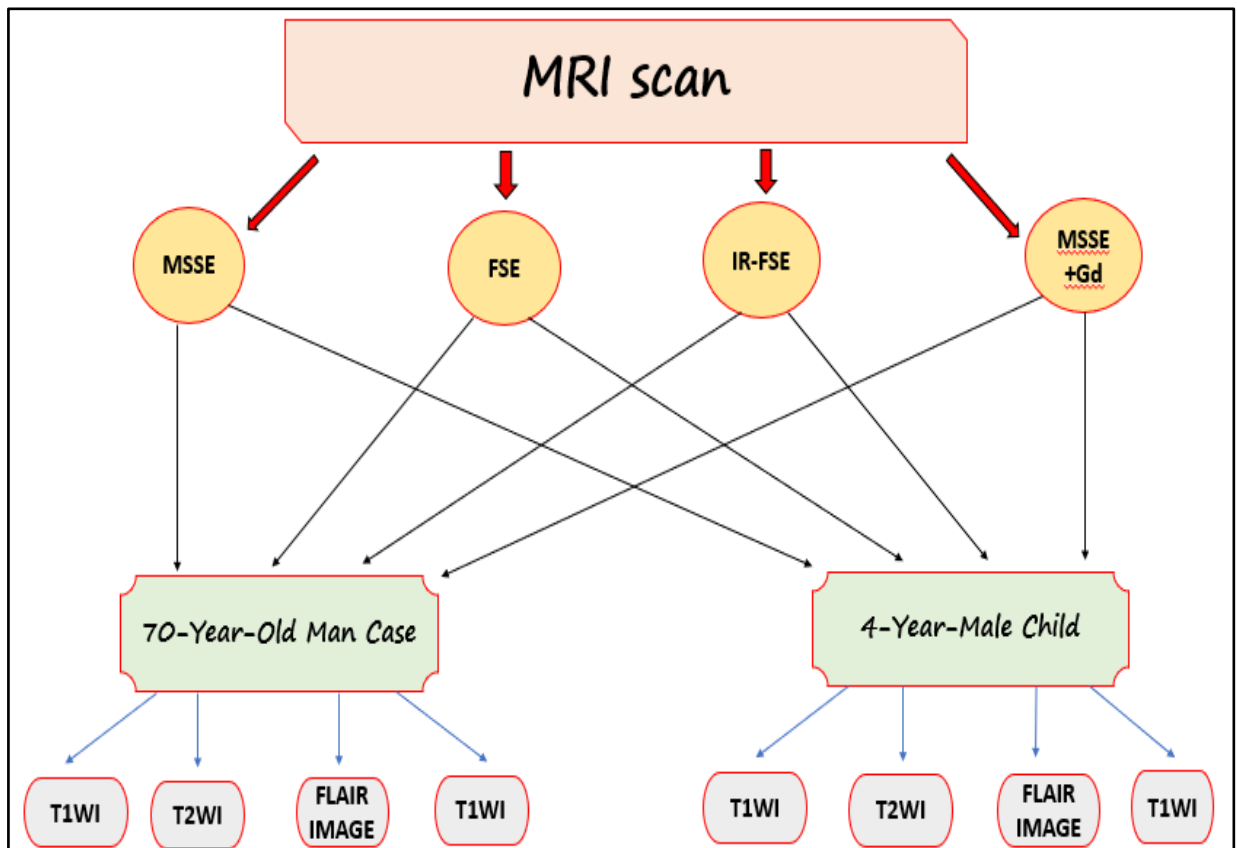


Figure (3.1): Schematic description of practical work.

3.1. Equipment

The present study was conducted at the Department of Radiology, Aden German International Hospital, located in the city of Aden. The MRI scans were performed using an open-type 0.5 Tesla MRI unit, which was manufactured by China Resources Wandong Medical Equipment Co. LTD (Figure 3.2). Further details on the components and composition of the MRI system can be found in this source [138].

For this research, an open permanent magnet with a field strength of 0.5 Tesla (with a variation of $\pm 5\%$) was utilized. The study primarily focused on comparing the signal intensity and imaging characteristics across three different types of sequences: T1-weighted images (T1WI), T2-weighted images (T2WI), and Fluid-Attenuated Inversion Recovery (FLAIR). To achieve comprehensive results, the experiment employed a variety of imaging techniques, including multi-slice spin echo, fast spin echo, and Inversion Recovery Fast Spin Echo (IR-FSE) sequences. These sequences were applied in different planes, namely sagittal, transversal, and coronal, to obtain detailed and diverse views of the anatomical structures being studied. Following the acquisition of the images, detailed analyses were conducted to evaluate and interpret the signal characteristics across the various imaging modalities and planes.

The MRI system utilized in this study operates with a vertical magnetic field direction. The magnet has a net weight of 21,000 kg and is constructed from Neodymium-Iron-Boron (Nd-Fe-B) materials, known for their strong magnetic properties. The patient table is designed to support a maximum weight of 150 kg. Additionally, a head RF coil was employed during the scanning process to ensure optimal image quality for head and brain imaging. The system functions under specific operational conditions, including a

temperature range of $23^{\circ}\text{C} \pm 2^{\circ}\text{C}$, relative humidity between 30% and 70%, and an atmospheric pressure range from 70 kPa to 106 kPa. These environmental parameters are critical to maintaining the optimal performance of the MRI system.



Figure 3.2: i-open permanent MRI system.

3.1.1. Hardware of MRI that was used during the research (i-open 0.5T permanent MRI system)

The MRI machine used in this procedure comprises several hardware components, which are outlined as described in table 3.1. The MRI machine used in this procedure consists of a range of critical hardware components that work in tandem to produce high-resolution images for diagnostic purposes. Each of these hardware elements plays a specific role in ensuring the effective operation of the MRI machine, from generating the magnetic fields to capturing the signals emitted by the body and converting them into detailed images.

Table 3.1: i-open permanent MRI system hardware [138].

NO.	HARDWARE		Quantity
1	Two column super open permanent magnet		1
2	Magnet constant temperature system		1
3	X、Y、Z gradient coil set		1
4	Gradient amplifier modules for X, Y and Z axes Gradient amplifier power module		1
5	QD Flat transmitting coil set		1
6	RF power amplifier		1
7	RF power splitter		1
8	Four channel digital spectrometer	Network Communication Module (NCM)	1
		Four Channel Data Acquisition Module (MC-DAM)	1
		Digital Control Module (DCM)	1
		Gradient Calculation Module (GCM)	1
		Gradient Output Module (GOM)	1
		RF Output Module (ROM)	1
		Gain-Adjust RF Amplifier Module (GAAM)	4
		Power Module	1
9	Receiving coil	Four channel phase array head coil	1
		Four channel phase array neck coil	1
		Four channel phase array knee coil	1
		Four channel phase array body coil (Large)	1
		Four channel phase array body coil (Medium)	1
		Two channel phase array shoulder coil	1
		Two channel phase array small joint coil (wrist)	1
		Two channel phase array small joint coil (ankle)	
		Pre amplifier	4

10	Workstation	CPU: Intel® Core™ ≥3.2GHz Memory: ≥4GB Hard disk: ≥500GB	1
		16 X DVD+/-RW	
		USB Keyboard, USB optical mouse	
		High performance graphic card	
		Windows 10 Business	
11	TFT-LCD monitor ≥ 24 inches		1
12	10/100M Switch		1
13	15KVA Stabilizer		1
14	Electric cabinet		1
15	Magnet cover		1
16	Six-direction motorized patient table		1
17	Touch-screen laser position system		1
18	Dual-way communication system		1
19	Patient cushion		1
20	Operating table		1
21	Receiving coils interface		1
22	Respiration gating & pulse gating system		1
23	Phantom	Cube (Large)	1
		Cube (Small)	1
		Cylinder	1
		Hemisphere (Optional with breast coil)	1
24	Supporting system	Filter panel	1
		Filter set	1
		Connecting cable set	1
		Other auxiliary parts set	1
25	RF cage within inner decoration		1
26	User manual		1

3.1.2. Software of MRI that was used during the research (i-open 0.5T permanent MRI system)

The software utilized in the MRI machine for this procedure, which plays a crucial role in acquiring and processing the imaging data, is outlined in detail in Table 3.2. This software is responsible for controlling various functions of the MRI machine,

Table 3.2: i-open 0.5T permanent MRI system software [138].

NO.	SOFTWARE		Quantity
1	System software	Patient register software	1
		Scanning control software	
		Sequence management software	
		Image reconstruction software	
		Image viewing software	
		Image processing software	
		Image analyzing software	
		Image format conversion software (BMP, JPG)	
		Multi-function image searching software	
		Image DVD archive software	
		DICOM 3.0 printing software	
		System quality assurance software	
2	Pulse sequence		
Spin Echo And Fast Spin Echo	Scout image with Spin Echo		1
	T1 weighted image with Spin Echo		
	Proton density weighted image with Spin Echo		
	T2 weighted image with Fast Spin Echo		
	Heavy T2 weighted image with Fast Spin Echo		
	Proton density weighted image with Fast Spin Echo		
	3D Spin echo		
	3D Fast Spin Echo		

Inversion Recovery	Fat suppress with Inversion Recovery		1
	Water suppress with Inversion Recovery		
	Fat suppress with Fast Inversion Recovery (STIR)		
	Water suppress with Fast Inversion Recovery (FLAIR)		
	Heave T1 weighted image with Inversion Recovery		
Gradient Echo	Scout image with Gradient Echo		1
	T1 weighted image with Gradient Echo		
	T2* weighted image with Gradient Echo		
	Fast dephase Gradient Echo		
	Fast rephase Gradient Echo		
	Fast dephase Gradient Echo with breath hold		
	3D Fast dephase Gradient Echo		
	3D Fast rephase Gradient Echo		
3	TOF for Angiography	2D TOF	1
		3D TOF	1
		MRCP	1
4	Water imaging	MRU	1
		MRM	1
5	LSDWI		1
6	Dixon		1
7	Movement compensation		1
8	Flow compensation		1
9	Multi slice multi angle imaging technology		1
10	Track pre-saturation band		1
11	Image filter software		1
12	Advanced 3D MRI	MIP	1
		MPR	1
13	DICOM 3.0	DICOM Storage	1
		DICOM Transfer to PACS	1

3.2. Agents & Cases

Pathological brain cases were evaluated in two patients: a 70-year-old man weighing 75 kg and a 4-year-old male child weighing 40 kg. Both individuals underwent MRI scans, which were performed both before and after the administration of contrast material to enhance the visibility of any abnormalities. The contrast agent used was a gadolinium-based solution (Gd), which was injected intravenously during the scans to improve image quality and highlight areas of concern within the brain. The gadolinium contrast agent was administered at a concentration of 0.5 mmol/ml. The dosage of the contrast medium varied between the two patients, based on their weight and age: the 70-year-old man received 15 ml of the contrast agent, while the 4-year-old child was administered a reduced dosage of 10 ml. These MRI sequences allowed for detailed imaging, facilitating a more accurate assessment of the pathological brain conditions in both cases.

3.3. Methods

Prior to MRI scanning, patients were carefully screened to exclude any ferromagnetic objects or implanted medical devices, such as pacemakers or cochlear implants, that could pose risks within the magnetic field. All MRI safety protocols were strictly observed. The patient must remain completely still so that there is no interference that would affect the quality of the imaging. Since they are unable, they were anesthetized. The patient is inserted into the middle of the magnetic field to ensure the uniformity and homogeneity of the primary magnetic field, using the laser identification located within the device.

First, when the human body is placed in the primary magnetic field, the hydrogen nuclei align in two ways: most of them are parallel to the direction of the primary magnetic field, and some of them are anti-parallel in the opposite direction of the field.

According to the Zeeman effect, this creates a net magnetization vector being in the direction of the primary magnetic field along the direction of the axes Z and precession around the direction of the primary field with Larmor frequency. The net magnetization is called a longitudinal magnetization. In order to measure this longitudinal magnetization vector, it is flipped to the transverse plane XY by using a 90-degree RF pulse generated from RF coils; furthermore, the frequency of these 90-degree RF pulses must be equal to the Larmor frequency of the proton's precession, which is known as magnetic resonance. After it is flipping the longitudinal magnetization completely to the transverse plane XY, the 90-degree RF pulse is stopped, and relaxation occurs through two separate processes.

- a. The first process is transverse relaxation, in which the dephase is occurs in the transverse plane due to a spin-spin interaction, and the parameter of time T2 (the time when tissues preserve 37% of their transverse magnetism) varies from one tissue to another according to its molecular composition and also between normal and pathological tissues.
- b. The second process is longitudinal relaxation, where the time factor T1 that it takes for tissues to recover 63% of their longitudinal magnetization changes due to the spin-lattice interaction according to the type of tissue and its molecular structure, and this gives a difference in T1 between different types of tissues and also between normal tissues and pathological tissues.

Next, the RF coils measured the free induction decay signal resulting from the decay of transverse magnetization due to the interaction of the transverse magnetization with the inhomogeneous local field, which accelerates phase removal. This leads to the damp and rapid disappearance of the free induction decay signal that we cannot measure.

Then, a second rephase 180-degree RF pulse is applied. This is allowed us to measure an analog complex echo signal that induces an electrical current in the radio frequency coils installed in the head, according to Faraday's law.

After that, the mathematical Fourier transform is used to separate the analog complex signal recorded by the radio frequency coils into its primary frequencies and then encode them to fill k-space through the following stages:

- **The first stage-slice selection:**

Gradient coils are activated in the Z direction simultaneously with the release of the 90-degree RF pulse in order to select and flip a specific slice.

- **The second stage-phase encoding:**

Phase encoding gradients are activated in the Y direction in order to determine the line that will be filled in k-space.

- **The third stage-frequency encoding:**

Frequency coding gradients are activated in the X direction in order to move to another new line of K space.

Finally, after filling the K space and processing the image, the inverse Fourier transform is applied to obtain different types of MRI brain images. Healthy normal images are compared with images taken from the patient in order to detect and diagnose the pathological lesion that the patient suffers from.

3.3.1. Sequences Used in this Research

This research utilized three different imaging sequences to analyze and compare four distinct types of diagnostic tests. Each sequence was provided a comprehensive evaluation of the tests, producing a total of four unique contrast images. These images

were carefully examined and analyzed to assess their effectiveness in diagnosing various conditions. The results of the analysis are detailed in the following sections, highlighting the strengths and potential applications of each imaging sequence and contrast type in clinical diagnostics.

3.3.1.1. *The First Sequence - MSSE Sequence,*

This sequence is the first imaging sequence employed in this procedure, and the steps involved are as follows:

1. Activate the slice selection gradient, and then we apply a 90-degree radio frequency pulse in order to flip the net magnetization vector to the transverse plane.
2. Apply the radio frequency pulse 180 degrees to re-phase in the transverse plane.
3. Run the phase encoding gradient with a certain degree, where the degree of phase encoding determines the acquired row data in K space.
4. Turn on the frequency encoding gradient or the reading gradient.
5. Use the dead time by selecting another slice and starting a cycle etc. send another 90 pulse and 180 pulse that makes the rephase and obtains a signal to fill another line of K space. We can scan many lines of different k spaces. We just need to repeat these enough times to get every line of every k-space (i.e., multiply by the number of phase encoding steps).

3.3.1.2. *The Second Sequence - FSE Sequence*

The FSE Sequence is the second imaging sequence utilized in this procedure, and the steps involved are outlined below:

1. Activate the slice selection gradient, and then we apply a 90-degree radio frequency pulse in order to flip the net magnetization vector to the transverse plane. However,

the net transverse magnetization vector decays very quickly, and this decay is called free induction decay at a rate of T_2^* due to the inhomogeneity of the local field.

2. Apply the radio frequency pulse 180 degrees to re-phase in the transverse plane, thus re-accumulating the signal back up to the level of the decay constant T_2 , which occurs due to the spin-spin interaction.
3. Run the phase encoding gradient with a certain degree, where the degree of phase encoding determines the acquired row data in K space.
4. Turn on the frequency encoding gradient or the reading gradient.
5. Don't wait for another TR to fully recover the longitudinal magnetization, but rather we send another 180 pulse that makes the rephase and obtains a signal to fill another line of K space.
6. During the next echo, we apply the rewinder phase encoding gradient to cancel the effects of the initial phase encoding gradient.
7. Repeating these echoes several times before repeating the sequence at the next TR, and in the end, we obtained multiple echoes and thus multiple lines of K space, reducing the acquisition time by a factor of the echo train length (ETL), which is the number of echoes generated during one 90-degree pulse.

3.3.1.3. The Third Sequence - IR-FSE Sequence

This sequence is the first imaging sequence employed in this procedure. The steps involved in this sequence are as follows:

1. Apply the 180-inversion pulse simultaneously with the operation of the slice selection gradient, which leads to the magnetic moment of the desired slice being flipped in the direction of the negative z axis.

2. Tissue recovery of the longitudinal magnetization M_z occurs, varying according to the relaxation time T_1 for each tissue, and after partial recovery of balance with a time of TI at the zeroing point of a specific tissue signal, the 90-radio frequency pulse is also applied at the same time with the selection of the slice to invert the magnetization to the transverse plane M_{xy} of the slice.
3. The phase encoding gradient is operated between 90 and 180 pulses, and the number of steps is usually from 128 to 256 steps. The phase encoding gradient is not applied after the first 180 pulse as there is no transverse magnetization, and also between 90 and 180 pulse the elimination gradient is used in order to determine the beginning of signal acquisition at the edge of K space.
4. Apply a pulse of 180 simultaneously with the slice selection gradient to refocus and obtain a pulse with its maximum value, where the free decay after the 90 pulse is not used.
5. The frequency encoding gradient is triggered after the 180-second pulse during the echo measurement time.
6. The cycle is repeated again.

3.3.2. MATLAB Software Used for Processing Medical Images

The MRI images were sent to PACS system and read using the software Micro dicom DICOM viewer and processed using MATLAB. In this program, the DICOM MRI images, were transformed to matrices and each pixel for various slices was processed in order to the visual effects of the disease locations are enhanced, increasing their visibility. This improvement allows for better detection, providing more information that contributes to a more accurate diagnosis.

After acquiring primary data from the MRI machine and reconstructing the images, they were processed using specific filters or segmented to identify particular anatomical regions.

3.3.2.1. *Isolation of Abnormal Tissues with MATLAB*

The medical images were processed by employing a series of image processing techniques in the MATLAB software program. These techniques involved several steps, including image enhancement, edge detection, and segmentation, to effectively highlight the areas of interest that correspond to the pathological region. Specific MATLAB functions and commands were used to manipulate the image data, allowing for the accurate identification and isolation of abnormal tissues or structures. The following commands and operations were applied to achieve this goal, which isolates the pathological region, such as a tumor or cyst, in steps as follows:

1. **Read the DICOM image:** The first step involved loading the DICOM image using the `dicomread` function. This function reads the medical image data stored in a DICOM file and makes it available for processing.

```
Img = dicomread('image.dcm');
```

2. **Normalize intensity values from 0 to 1:** To standardize the image intensity values and prepare it for further processing, the `mat2gray` function was applied. This function rescales the intensity values of the image to a range between 0 and 1, which is often necessary for image manipulation tasks like segmentation.

```
Inorm = mat2gray(Img);
```

3. **Display the normalized image:** The normalized image was then displayed using the `imshow` function. This step allows visual inspection of the image and helps confirm that the normalization was applied correctly.

```
imshow(Inorm);
```

4. **Select the pathological region (Tumor/Cyst):** The next step involved selecting the specific pathological region (e.g., a tumor or cyst) using the `roipoly` function. This function enables interactive selection of a polygonal region of interest (ROI) in the image. The user can click around the area of interest with the mouse, and the program will define this area as the ROI mask. After selecting the desired region, the user can right-click and choose "Create mask" to generate the mask.

```
mask = roipoly();
```

The interactive process involves pressing **Ctrl + Enter**, using the cursor to draw around the pathological region, and then right-clicking and selecting "Create mask" to generate the mask.

5. **Display the selected mask:** After the mask is created, it can be displayed to visually verify that the correct area has been selected. The mask shows a binary image where the selected region is represented by 1 (white), and the rest of the image is represented by 0 (black).

```
imshow(mask);
```

6. **Apply the mask to the normalized image:** Once the mask has been created, it was applied to the normalized image by performing an element-wise multiplication. This operation isolates the pixels corresponding to the pathological region (tumor/cyst) by keeping only the values within the mask and setting all other pixels to zero.

```
Inorm = Inorm .* mask;
```

7. **Display the image with the isolated pathological region:** The final step involved displaying the image with the isolated pathological region. The result shows only the selected region (tumor/cyst), allowing for further analysis of that specific area.

```
imshow(Inorm);
```

By applying these operations in MATLAB, the pathological region (such as a tumor or cyst) was successfully isolated from the original DICOM image. This process is useful for medical image analysis, as it highlights the areas of interest for further diagnostic purposes.

3.3.2.2. *Negative Image with MATLAB*

To obtain the negative image, the specific medical image was processed by applying the following commands in the MATLAB software:

1. **Read the image:** The medical image was first imported into the workspace using the `imread` function, which loads the image data into MATLAB for further processing.

```
img = imread('medical_image.png');
```

2. **Convert to grayscale:** The image was in color (RGB), it was converted to grayscale to simplify the negative transformation process. This was achieved using the `rgb2gray` function, which converts the color image into a single-channel intensity image.

```
gray_img = rgb2gray(img);
```

3. **Normalize the image:** The grayscale image was then normalized to a 0-1 scale using `im2double`, which ensures that pixel values are in the appropriate range for image processing tasks.

```
norm_img = im2double(gray_img);
```

4. **Generate the negative:** The negative of the image was created by applying the `imcomplement` function or subtracting the pixel values from 1 (for an image scaled between 0 and 1). This operation reverses the intensity values, converting bright regions to dark ones and vice versa.

```
negative_image = imcomplement(norm_img); or  
negative_img = 1 - norm_img;
```

5. **Display the original and negative images:** The original and the negative image were displayed side by side for visual comparison using the `imshow` function.

```
subplot(1, 2, 1), imshow(gray_img), title('Original Image');  
subplot(1, 2, 2), imshow(negative_img), title('Negative Image');
```

6. **Save the negative image:** Finally, the negative image was saved to the disk for further analysis or use, using the `imwrite` function.

```
imwrite(negative_img, 'negative_medical_image.png');
```

By applying these steps in MATLAB, the negative of the medical image was successfully obtained, facilitating further analysis or diagnostic tasks.

3.3.2.3. *Detected Edges with MATLAB*

To obtain the detected edges, the specific medical image was processed by applying the following commands in the MATLAB software:

1. **Read the image:** The first step was to load the medical image into MATLAB using the `imread` function. This function imports the image from a specified file path and stores it in the workspace for further processing.

```
img = imread('medical_image.png');
```

2. **Convert to grayscale:** Since edge detection typically works on grayscale images, the image was originally in color (RGB), it was converted to grayscale using the `rgb2gray` function. This step reduces the complexity of the image by focusing on intensity information.

```
gray_img = rgb2gray(img);
```

3. **Apply a Gaussian filter (optional):** To reduce noise and improve the accuracy of edge detection, a Gaussian blur was applied to the grayscale image. This was achieved using the `imgaussfilt` function, which smooths the image and makes the edges more distinct. The standard deviation of the Gaussian filter (sigma) can be adjusted for more or less smoothing.

```
smoothed_img = imgaussfilt(gray_img, 2); % 2 is the standard deviation
```

4. **Edge detection using the Canny method:** The edges of the image were detected using the Canny edge detection algorithm, which is one of the most widely used methods for edge detection. The `edge` function in MATLAB implements the

Canny algorithm, which identifies areas in the image where intensity changes sharply.

```
edges_img = edge(smoothed_img, 'Canny');
```

5. **Display the original and edge-detected images:** To visualize the result, the original grayscale image and the detected edges were displayed side by side using the `subplot` and `imshow` functions. This allows for a quick comparison between the original image and the extracted edges.

```
subplot(1, 2, 1), imshow(gray_img), title('Original Grayscale Image');  
subplot(1, 2, 2), imshow(edges_img), title('Detected Edges');
```

6. **Save the edge-detected image:** Finally, the edge-detected image was saved to the disk for further analysis or use in diagnostics using the `imwrite` function. This step ensures that the result can be stored for future reference if needed.

```
imwrite(edges_img, 'edge_detected_image.png');
```

By applying these steps in MATLAB, the edges in the medical image were successfully detected, providing a clearer understanding of structural boundaries and important features within the image. The edge-detected image could be used for further analysis, such as segmentation, measurements, or enhanced visualization of important areas within the medical image.

3.3.3. Case of 70-Year-Old Man

In the first case, tests were applied to obtain types of images using the general parameters: flip angle of 90° , pixel bandwidth of 108.5069 HZ, and imaging frequency of 20.70 HZ.

3.3.3.1. First Test for 70-Year-Old Man

In this test, a T1-weighted image was obtained to evaluate the image using a multi-slice spin echo (MSSE) sequence. The sequence included in the protocol was as follows: TR= 630 ms, TE= 16 ms, slice thickness 6 mm, spacing between slices 7.2 mm, echo number 131, number of phase encoding steps 131, and echo train length 1.

3.3.3.2. Second Test for 70-Year-Old Man

This second test was obtained a T2-weighted image contrast using a fast spin echo (FSE) sequence. The sequence included in the protocol was as follows: TR= 5610 ms, TE= 138 ms, slice thickness 6 mm, spacing between slices 7.2 mm, echo number 14, number of phase encoding steps 149, and echo train length 11.

3.3.3.3. Third Test for 70-Year-Old Man

The third test was obtained a FLAIR image using an inversion recovery fast spin echo (IR-FSE) sequence. The sequence included in the protocol was as follows: TR= 9382 ms, TE= 115 ms, TI= 2334 ms, slice thickness 6 mm, spacing between slices 7.2 mm, echo number 18, number of phase encoding steps 152, and echo train length 9.

3.3.3.4. Forth Test for 70-Year-Old Man

In this test, a T1-weighted image was obtained after injection of 15 ml of gadolinium contrast material (Gd-ATPA) at a concentration of 0.5 mmol/ml to evaluate the image using a multi-slice spin echo (MSSE) sequence. The sequence included in the protocol was as follows: TR= 923 ms, TE= 16 ms, slice thickness 4 mm, spacing between slices 4.6 mm, echo number 131, number of phase encoding steps 131, and echo train length 1.

3.3.4. Case of 4-Year-Old Male

In this case, four tests were performed to obtain four types of images using the general parameters: slice thickness of 6 mm, spacing between slices of 7.2 mm, flip angle of 90° , frequency imaging of 20.70 HZ, and pixel bandwidth of 108.5069 HZ.

3.3.4.1. First Test for 4-Year-Old Male

In this test, a T1-weighted image was obtained to evaluate the image using a multi-slice spin echo (MSSE) sequence. The sequence included in the protocol was as follows: TR = 630 ms, TE = 16 ms, echo number 131, number of phase encoding steps 131, and echo train length 1.

3.3.4.2. Second Test for 4-Year-Old Male

This second test was applied to obtain a T2-weighted image to be able to evaluate the image by using a fast spin echo sequence. The sequence included in the protocol was as follows: TR= 5610 ms, TE= 138 ms, echo number 14, number of phase encoding steps 149, and echo train length 11.

3.3.4.3. Third Test for 4-Year-Old Male

In the third test, an inversion recovery fast spin echo (IR-FSE) sequence was used to obtain the FLAIR image when the previous two tests fail to evaluate the images. As the fluid signal is attenuated. The sequence included in the protocol was as follows: TR = 9382 ms, TE= 115 ms, TI= 2334 ms, echo number 18, number of phase encoding steps 152, and echo train length 9.

3.3.4.4. *Fourth Test for 4-Year-Old Male*

In this test, a T1-weighted image was obtained after injection of 10 ml of gadolinium contrast material (Gd-ATPA) at a concentration of 0.5 mmol/ml to evaluate the image using a multi-slice spin echo (MSSE) sequence. The sequence included in the protocol was as follows: TR= 630 ms, TE= 16 ms, echo number 131, number of phase encoding steps 131, and echo train length 1.



Chapter Four

Results

&

Discussions

Results and Discussions

This section presents the detailed results obtained from the research, which involved the application of i-open permanent MRI technology on two distinct cases, with both procedures incorporating the use of a gadolinium (Gd) contrast agent. The results are carefully analyzed and discussed, focusing on key differences and similarities when compared to normal, healthy MRI images, to aid in achieving a more accurate and comprehensive diagnosis. The comparison between the abnormal and normal imaging patterns serves as a critical step in identifying and understanding the underlying pathologies. Furthermore, advanced digital image processing techniques are employed to enhance and highlight the pathological regions within the MRI scans. These techniques provide a clearer visualization of abnormalities, which aids clinicians in pinpointing the precise areas of concern, ensuring more efficient diagnostic accuracy and potentially improving patient outcomes.

4.1. Results:

The results (images) from the two cases will be thoroughly examined and discussed in detail in the following sections. Each case will be analyzed individually, with a focus on the key features, observations, and significant findings observed in the images. The descriptions will provide a comprehensive understanding of the outcomes and highlight any notable patterns or abnormalities. These detailed explanations will offer a clear interpretation of the results and their implications.

4.1.1. Results of 70-Year-Old Man's Case

Four distinct types of images were captured in the transverse plane, each providing valuable insights into the precise location of the lesion and its impact on the surrounding

structures. These images effectively illustrate the spatial relationship between the lesion and nearby tissues, highlighting the extent of its influence on the affected area. The following visual representations demonstrate these findings in detail:

4.1.1.1. MSSE Sequence

Longitudinal relaxation time weighted image (T1WI); this type was obtained using a spin echo sequence, but with a short TR= 630 ms and a short TE= 16 ms as well, and the fluids appear black and the fats are white (Figure 4.1).

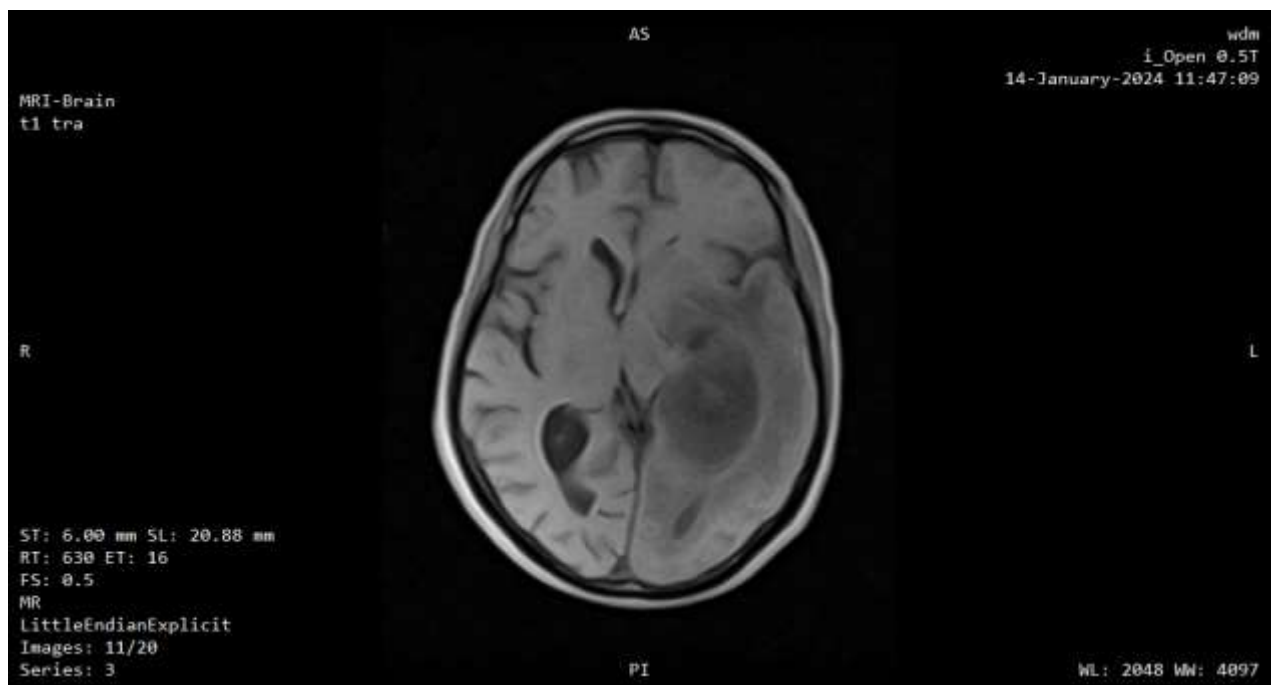


Figure 4.1: T1-Weighted Image of the brain on transverse plane for 70-Year-Old Man's Case (Test1).

4.1.1.2. FSE Sequence

Transverse relaxation time weighted image (T2WI): This type was obtained using a spin echo sequence with a long TR= 5610 ms and long TE = 138 ms, and the fluids appear white and the fats are dark (Figure 4.2).

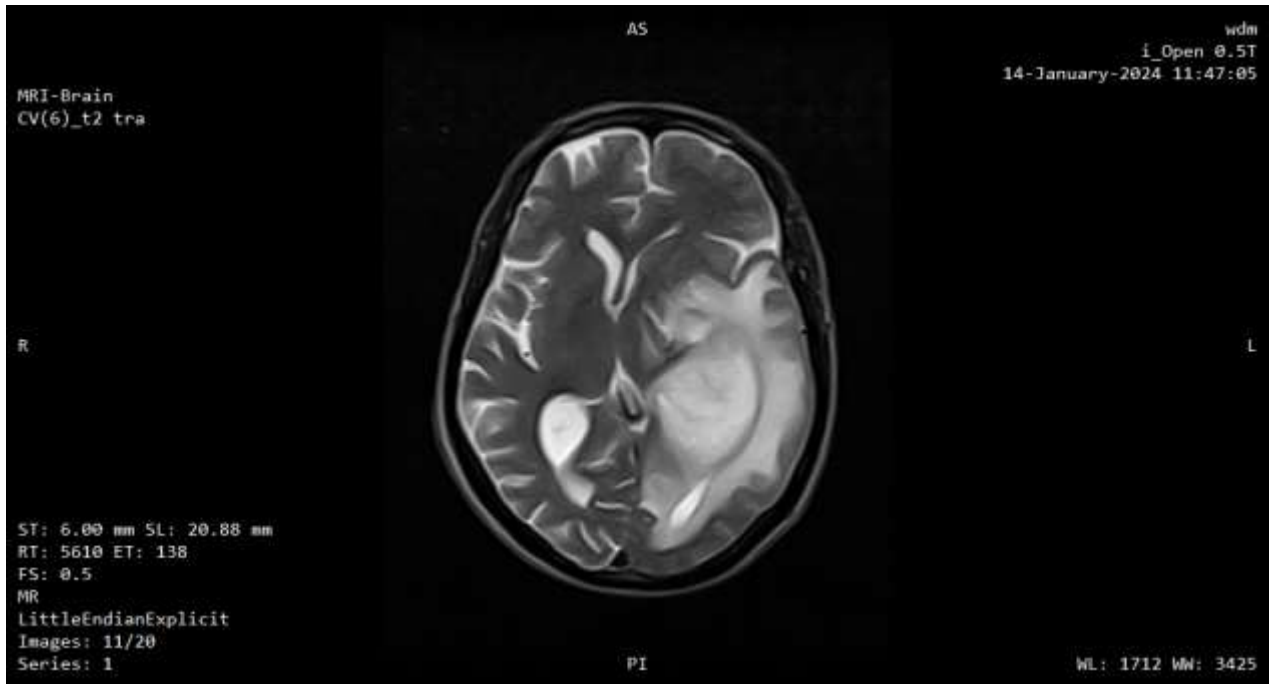


Figure 4.2: T2-Weighted Image of the brain on transverse plane for 70-Year-Old Man's Case (Test2).

4.1.1.3. IR-FSE Sequence

Fluid attenuated inversion recovery (FLAIR) image, using a spin echo sequence preceded by a 180 pulse (i.e., inversion recovery fast spin echo) with a very long time of repetition ($TR = 9382$ ms), long $TE = 115$ ms and long $TI = 2334$ ms, shows a similar T2 image with the cancellation of the fluid signal (Figure 4.3).

4.1.1.4. MSSE After Gadolinium Contrast Agent

The weighted image of the longitudinal relaxation time after injecting the contrast medium. The image appears as in T1 if it is normal, but an accumulation of gadolinium occurs in the disrupted blood-brain barrier. It is considered one of the magnetic materials that work to create a local magnetic field that works to shorten T1. If there is a tumor, it appears brightly (Figure 4.4).

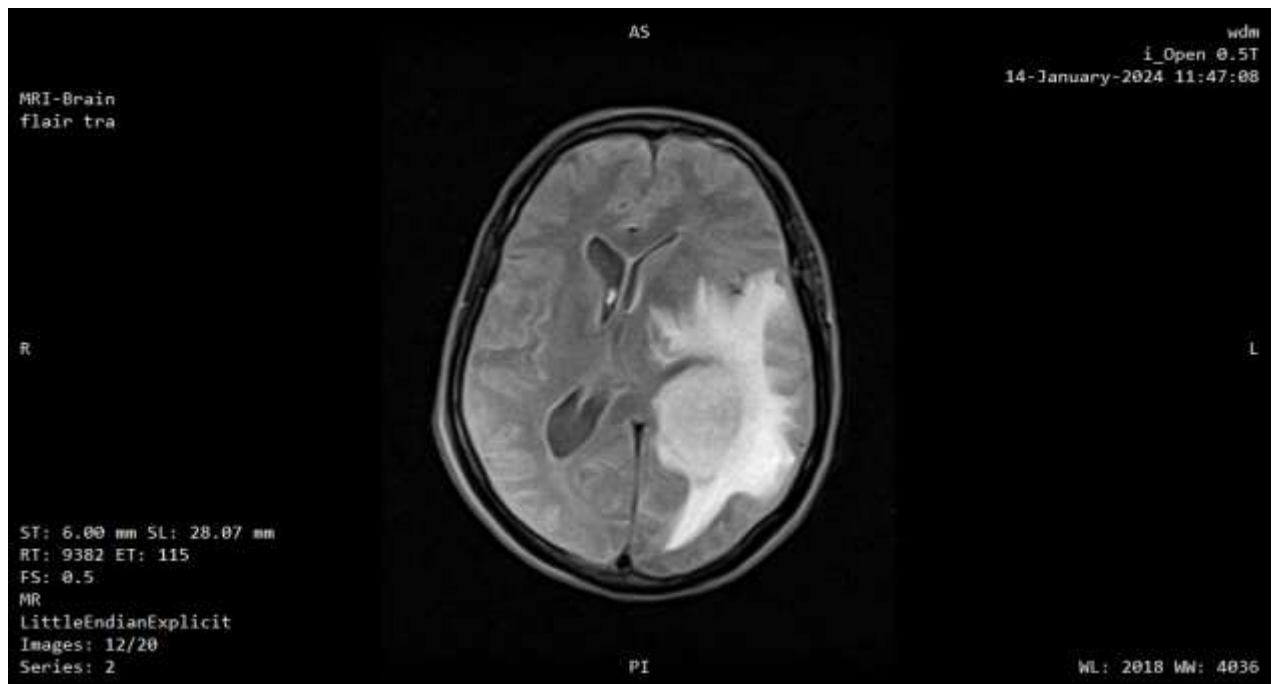


Figure 4.3: FLAIR Image of the brain on transverse plane for 70-Year-Old Man's Case (Test3).

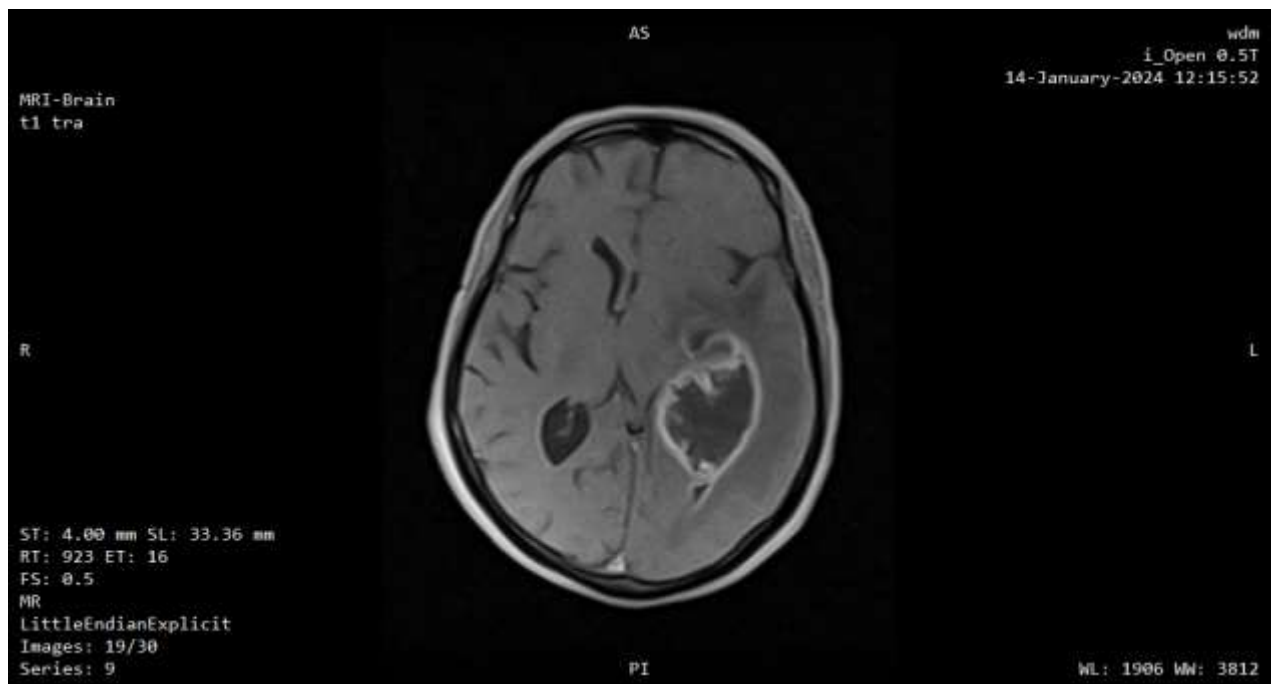


Figure 4.4: T1-Weighted Image + Gd contrast agent of the brain on transverse plane for 70-Year-Old Man's Case (Test4).

4.1.2. Results of 4-Year-Old Male's Case

Three distinct MRI sequences were utilized to acquire detailed images that accurately define the precise location of the lesion and its effects on the adjacent tissues. Each

sequence was selected for its unique ability to highlight specific characteristics of the lesion and its surrounding structures, providing a comprehensive view of the area of interest. These sequences allowed for a clearer understanding of the lesion's size, shape, and the extent of its influence on neighboring anatomical structures. The images generated by these sequences are presented below, offering a thorough visualization of the findings.

4.1.2.1. MSSE Sequence

In this sequence was obtained a T1- weighted image (T1WI) by using a short TR = 630 ms and a short TE = 16 ms spin echo sequence. Fluids in this sequence appear dark and the fats appear bright, as shown in Figure 4.5.

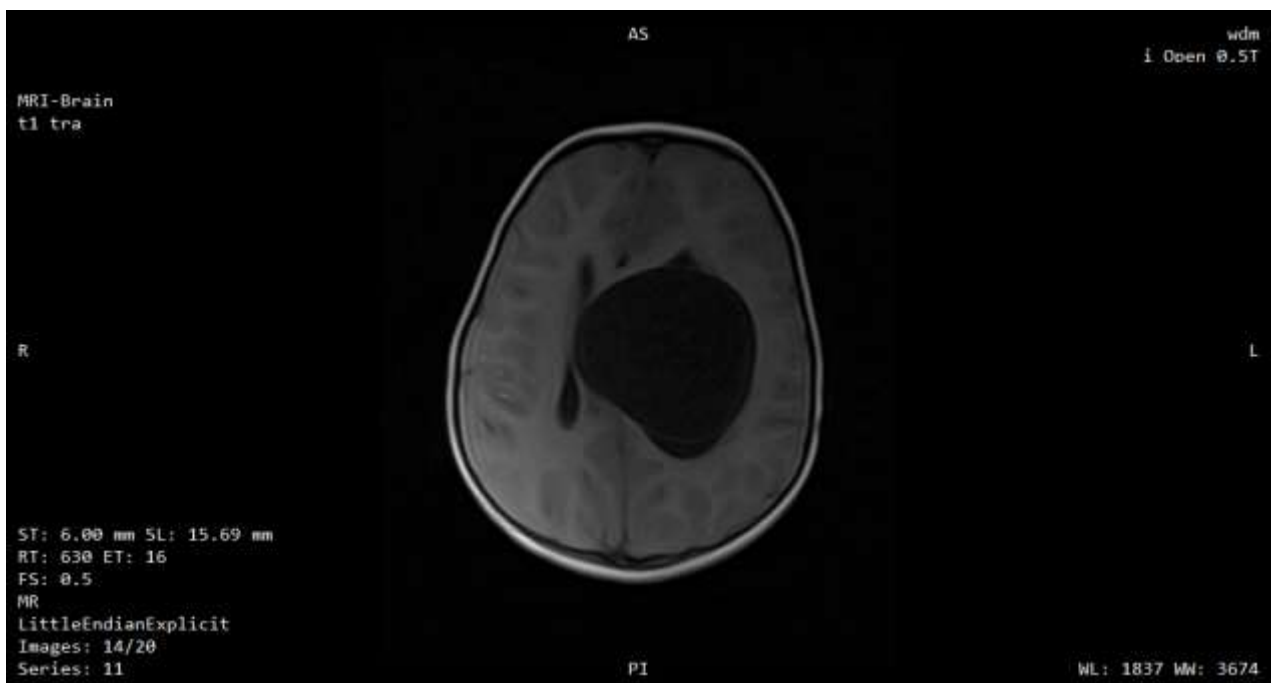


Figure 4.5: T1-Weighted Image of the brain on transverse plane for 4-Year-Old Male's Case (Test1).

4.1.2.2. FSE sequence

This sequence produced a T2-weighted image (T2WI) using a spin echo sequence with a long TR and long TE. In this sequence, fluids and lesions appear bright because they

have long TR = 5610 ms and long TE = 138 ms, while fats and proteins appear dark because they have short TR and short TE, as shown in Figure 4.6.

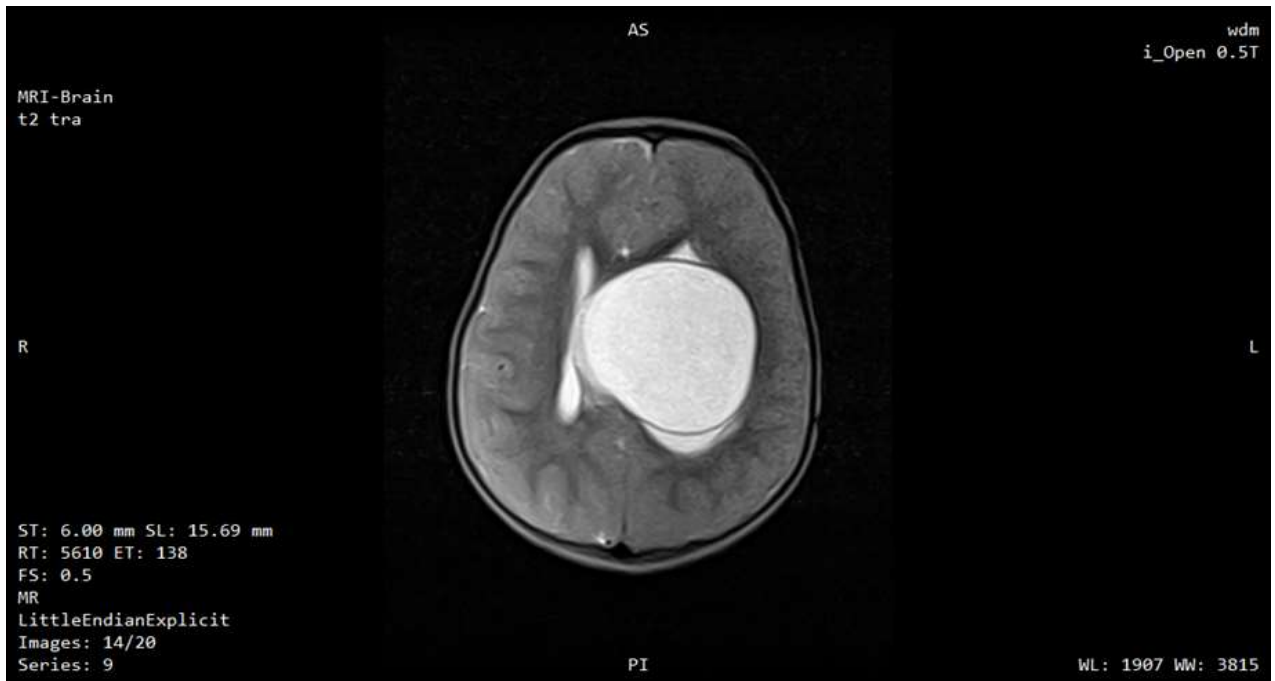


Figure 4.6: T2-Weighted Image of the brain on transverse plane for 4-Year-Old Male's Case (Test 2).

4.1.2.3. IR-FSE Sequence

To distinguish the lesion from cyst fluid, a FLAIR sequence was implemented. This fast spin-echo technique employed a 180° pulse with extended TR of 9382 ms, TE of 115 ms, and TI of 2334 ms to effectively suppress the cyst fluid signal. The resulting image exhibited T2-like characteristics but free fluids are dark, as depicted in Figure 4.7.

4.1.2.4. MSSE After Gadolinium Contrast Agent

The weighted image of the longitudinal relaxation time after injecting the contrast medium. The image appears as in T1WI if it is normal, but an accumulation of gadolinium occurs in the disrupted blood-brain barrier (BBB). It is considered one of the magnetic

materials that work to create a local magnetic field that works to shorten T1. If there is a tumor, it appears brightly (Figure 4.8).

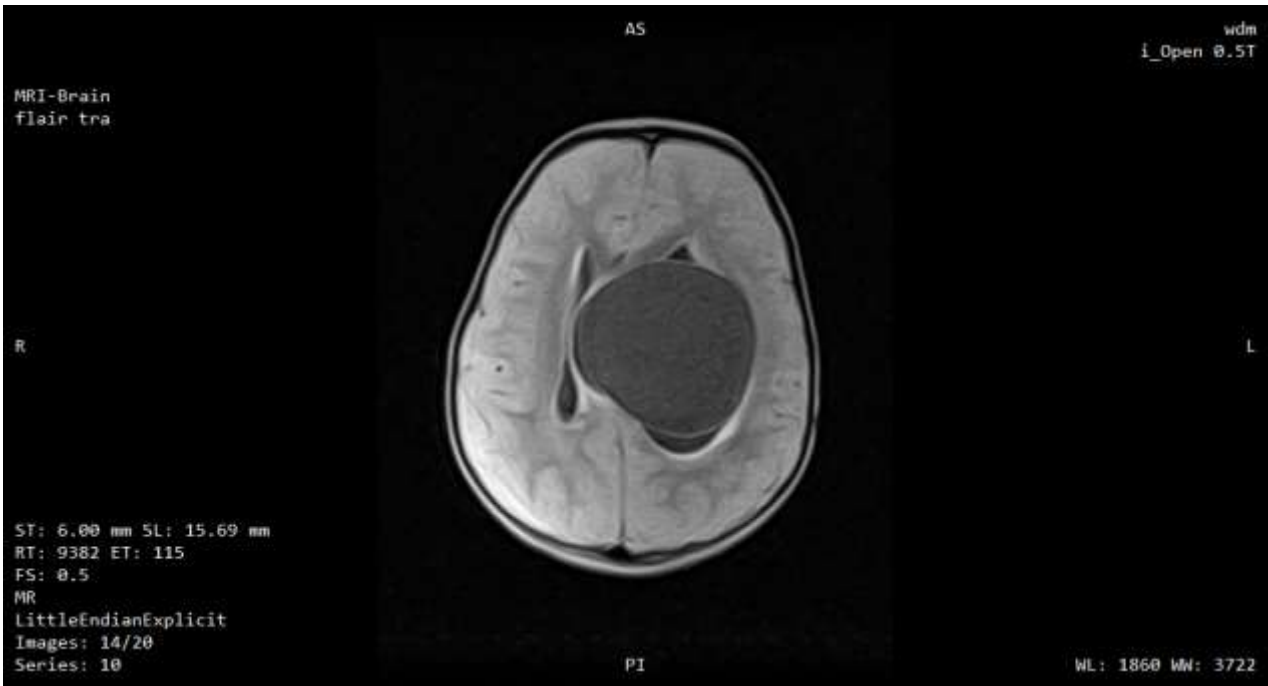


Figure 4.7: FLAIR-image of the brain on transverse plane for 4-Year-Old Male's Case (Test3).

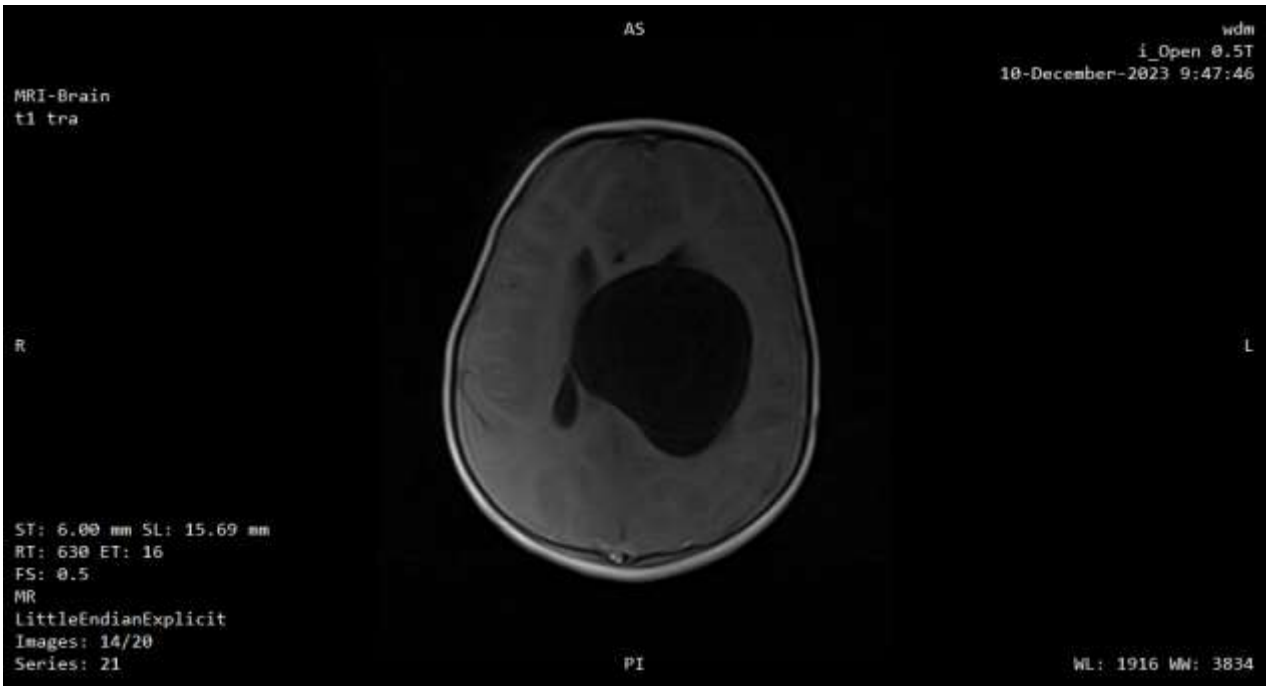


Figure 4.8: T1-Weighted Image + Gd contrast agent of the brain on transverse plane for 4-Year-Old Male's Case (Test4).

4.2. Discussion and Analysis

In this section, the MRI images obtained from two distinct cases were discussed and analyzed. Both cases underwent different MRI sequences. These images will be compared to normal reference images to aid in further diagnosis and provide insights into the abnormalities detected. By examining the differences and similarities between the case images and the normal scans, we can assess the nature and extent of the lesions or anomalies present in each case.

Additionally, the analysis will include images processed using MATLAB program software. These images were generated through advanced image processing techniques, offering enhanced visualization and clarity of the lesions. The use of MATLAB software allows for more precise measurement, contrast adjustment, and visualization of subtle features that might not be immediately apparent in the raw MRI scans. This comparison of the processed images with the original MRI scans will contribute to a more accurate diagnosis and a deeper understanding of the underlying conditions in each case. The following figures illustrate the MRI images from both cases, along with the enhanced images produced by the MATLAB software, providing a comprehensive overview of the diagnostic process.

4.2.1. Discussion & Analysis of 70-Year-Old Man's Case

After obtaining the various types of images from the MRI sequences mentioned earlier, the next step is to conduct a thorough comparison of these images with normal reference images corresponding to each specific sequence type. This comparison will allow us to identify any deviations or abnormalities present in the images from the cases under study. By juxtaposing the images from the MRI sequences with the normal images,

we can assess the characteristics of the lesion, such as its size, shape, and location, as well as its effect on surrounding tissues.

Each MRI sequence has unique imaging properties that reveal different aspects of the tissue, such as structural details, fluid accumulation, or potential changes in tissue composition. Therefore, it is crucial to compare the case images against the normal images for each sequence type to understand how the pathological features stand out from typical healthy anatomy. This comparative analysis will provide valuable insights into the presence and extent of any abnormalities, supporting a more accurate and informed diagnosis.

4.2.1.1. *T1WI*

Fluid signal: In the normal image, fluids have a long T1 (to recover their longitudinal magnetization). But the T1-weighted image uses a short TR, so the fluids in the image appear dark. As in cerebrospinal fluid (CSF). Signal of fats and proteins: In the normal image, fats and proteins have a short T1 to restore their longitudinal magnetization, so although the TR is short in the T1WI-weighted image, the signal is still high and the fats and proteins appear bright (Figure 4.9).

In abnormal image, the fat signal and the water signal appear as in the normal condition, except for the area marked with a red circle that appears in a darker color than normal, and this indicates the presence of abundant fluids in this place (Figure 4.10). This indicates the presence of a pathological lesion. Usually, the pathological tissues have a higher water content than normal. There was also a mass effect that led to the displacement of the midline and pressure on the ventricles.

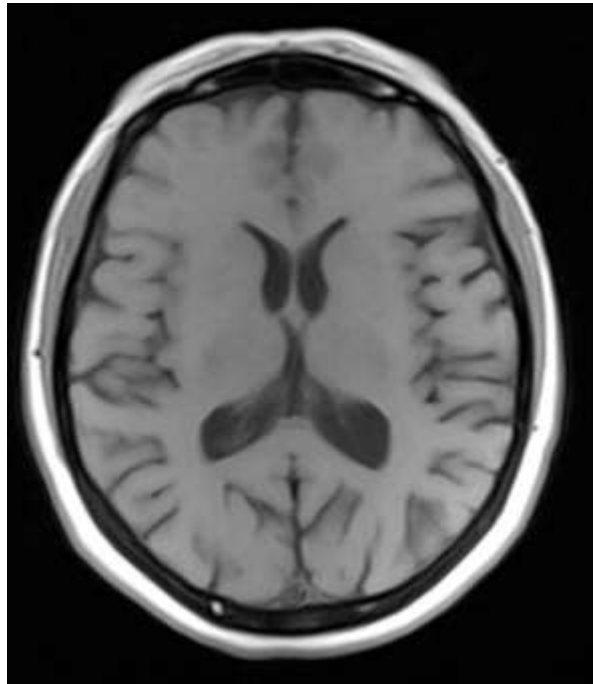


Figure 4.9: Normal T1WI on transverse plane with short TR & short TE.

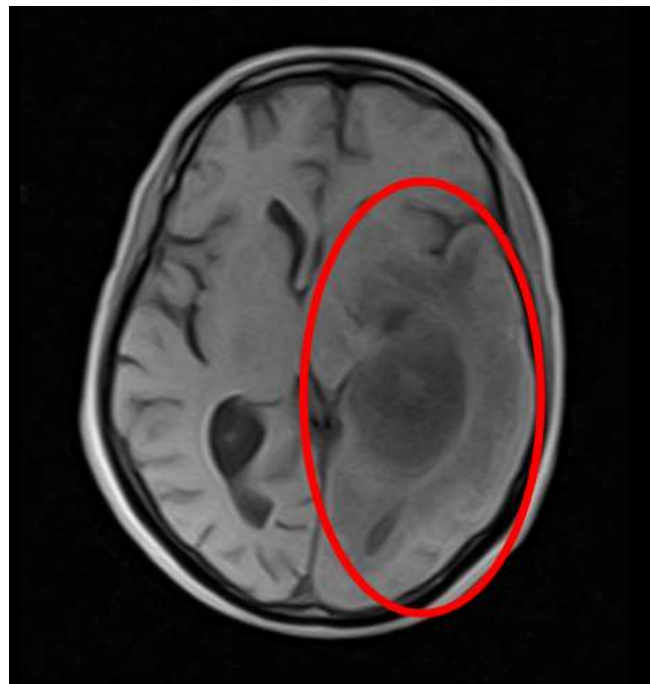


Figure 4.10: T1-Weighted Image of abnormal brain, where the image highlights the abnormal tissues (red circle) of 70-Year-Old Man's Case.

4.2.1.2. T2WI

Fluid signal: In the normal image, fluids have a long T2 (when their transverse magnetization is lost). They retain their magnetism for a long time, and the recorded signal is strong, so the fluids in the image appear bright. As in cerebrospinal fluid (CSF).
Fats and proteins: In the normal image, fats and proteins contain a short T2 when their transverse magnetization decays. Therefore, they lose their magnetization quickly, and the recorded signal is weak and appears dark in the image (Figure 4.11).

In abnormal image, fat and water appear as in the normal case, except for the area marked with a red circle that appears brighter and lighter in color than normal (Figure 4.12). This indicates the presence of water in abundance in this place, and this indicates the presence of a pathological lesion. Usually, pathological tissues have a higher water content than normal, but in this type of image, no distinction is made between lesions and ventricles.

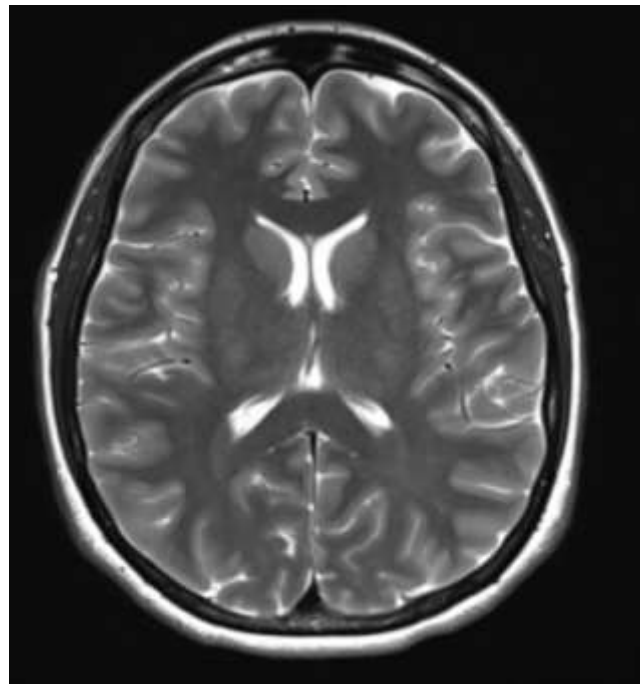


Figure 4.11: Normal T2WI on transverse plane with TR long & long TE.

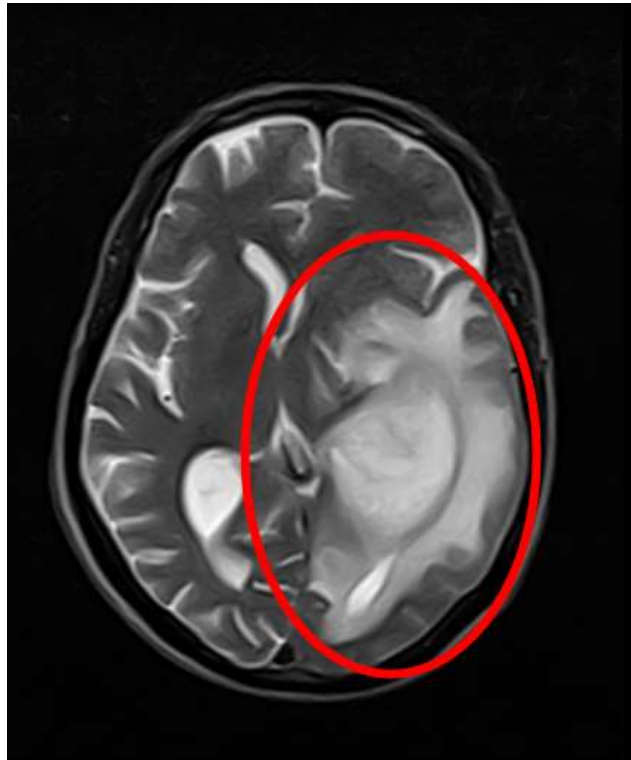


Figure 4.12: T2-Weighted Image of abnormal brain, where the image highlights the abnormal tissues (red circle) of 70-Year-Old Man's Case.

4.2.1.3. FLAIR Image

In the normal image, when the FLAIR sequence is used to suppress the fluid signal, it is similar to the weighted image transverse relaxation (T2WI) time in the spin echo sequence, except that the free fluid appears black and the restricted fluid appears white (Figure 2.13).

In abnormal image, after suppressing the signal of free fluid, the specific place appeared in a bright color, and this is evidence that it is a tumor and not fluid (Figure 4.14). This tumor is surrounded by edema, whose signal appears brighter than the tumor because it contains a lot of restricted fluid. The tumor, by its nature, causes a significant multiplication of cells, so the fluid increases. Therefore, the protons increase, and the amount of magnetization increases, so the signal is strong here. Suppression of the free fluid signal allowed us to distinguish between the CSF in the ventricles and the tumor and

demarcate the boundaries between them. The pathological anatomical region was extracted using MATLAB programe, as shown in Figure 4.15.

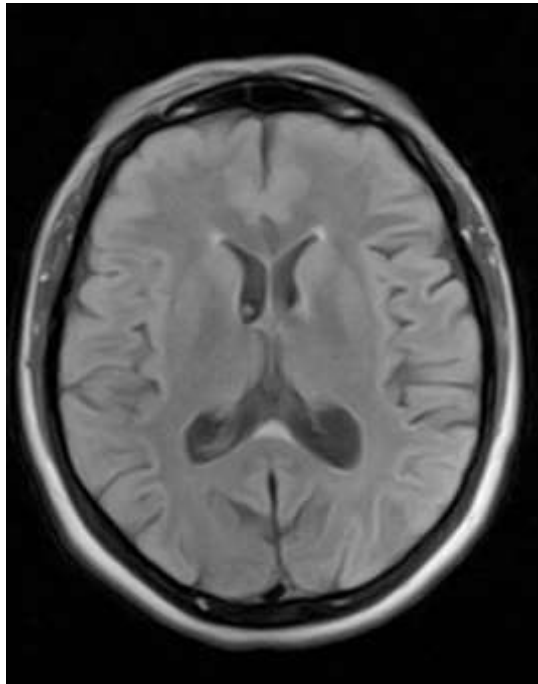


Figure 4.13: Normal FLAIR image on transverse plane.

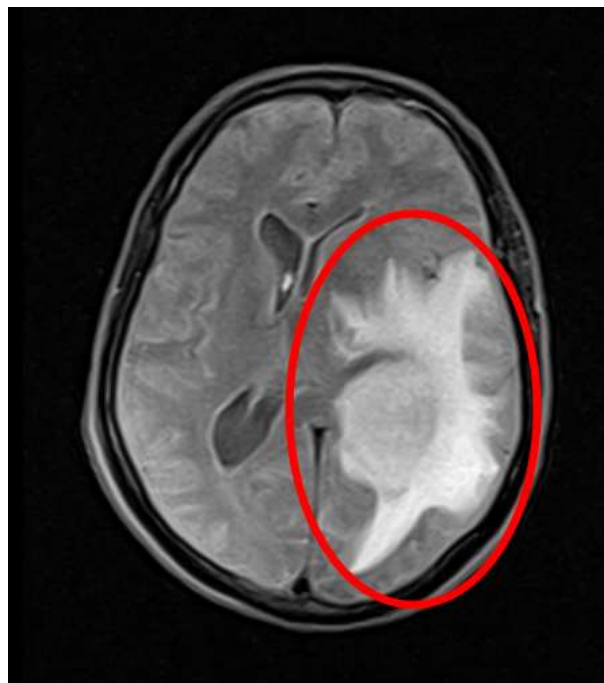


Figure 4.14: FLAIR Image of abnormal brain, where the image highlights the abnormal tissues (red circle) of 70-Year-Old Man's Case.

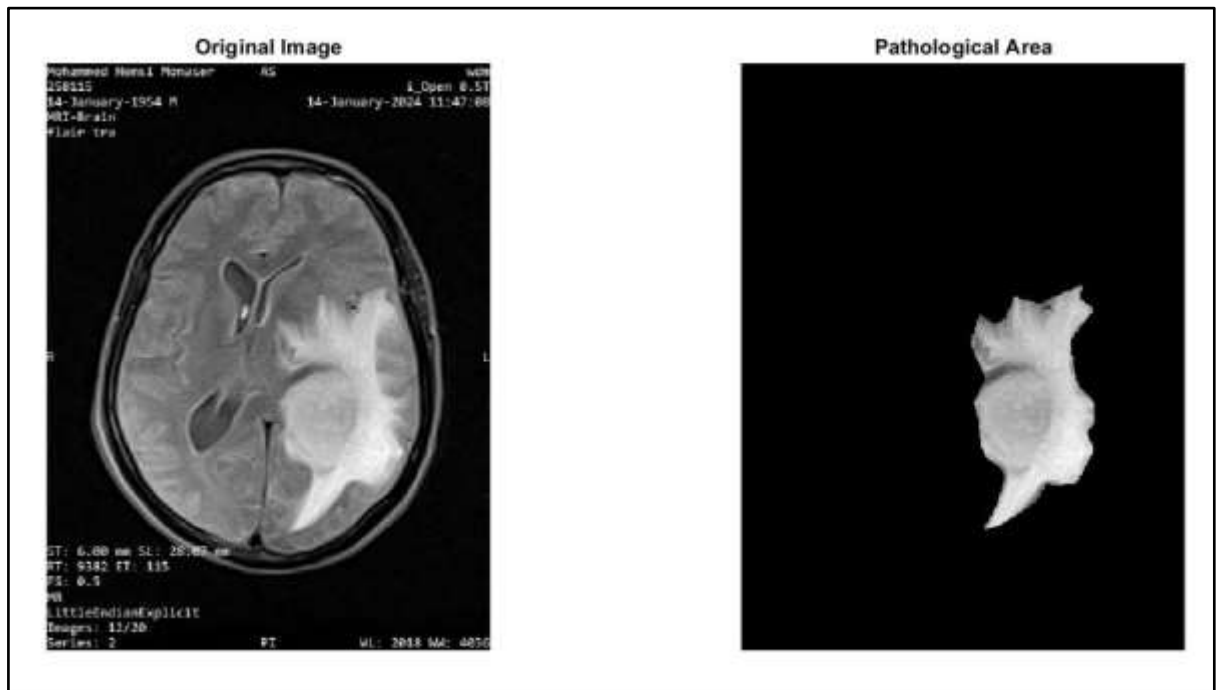


Figure 4.15: Processing FLAIR image to identify and extract the pathological anatomical region using MATLAB.

4.2.1.4. $T1WI + Gd$

Magnetic materials such as gadolinium possess small local magnetic fields that shorten the relaxation times of the surrounding protons, noting that gadolinium does not pass through the normal blood-brain barrier (BBB) but rather through the BBB (Figure 4.16), and this leads to shortening of T1, which enhances the vascularization of the tumor tissue. It makes the tumor bright and the surrounding edema is not affected, Interestingly, the surrounding edema (swelling due to fluid accumulation) is typically not affected by the gadolinium in the same way. This does not happen normally in the case of a normal BBB. Thus, the ability of gadolinium to cross a disrupted BBB provides valuable diagnostic information, as it helps to differentiate between normal tissue, tumor tissue, and areas of edema, allowing for more accurate tumor localization and assessment of the vascularization within the affected area.

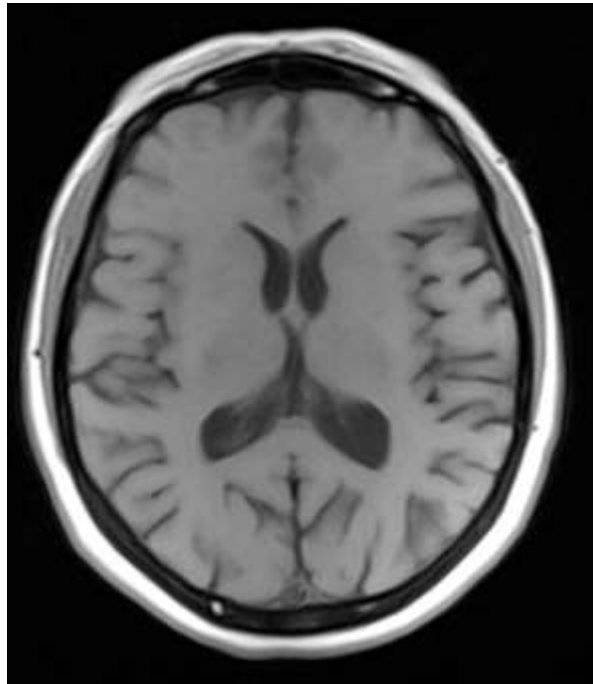


Figure 4.16: Normal T1WI + Gd on transverse plane.

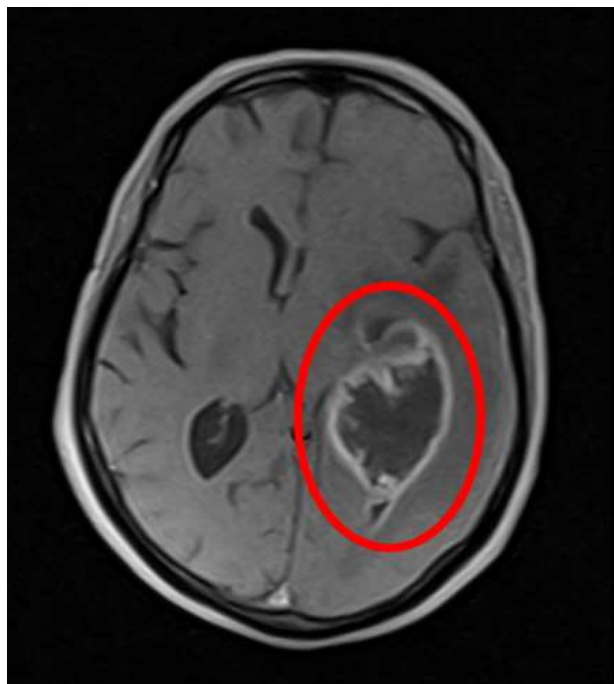


Figure 4.17: T1-Weighted Image + contrast medium Gd of abnormal brain, where the image highlights the abnormal tissues (red circle) of 70-Year-Old Man's Case.

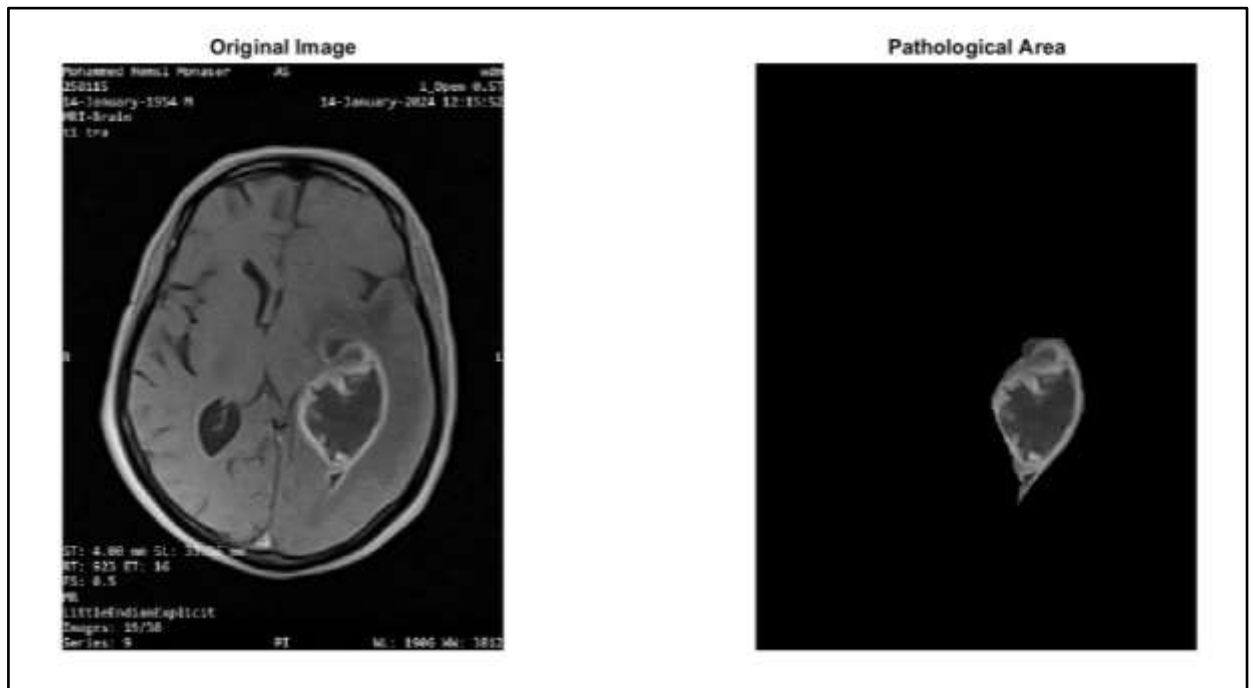


Figure 4.18: Processing T1-Weighted Image after gadolinium injection to identify and extract the pathological anatomical region using MATLAB.

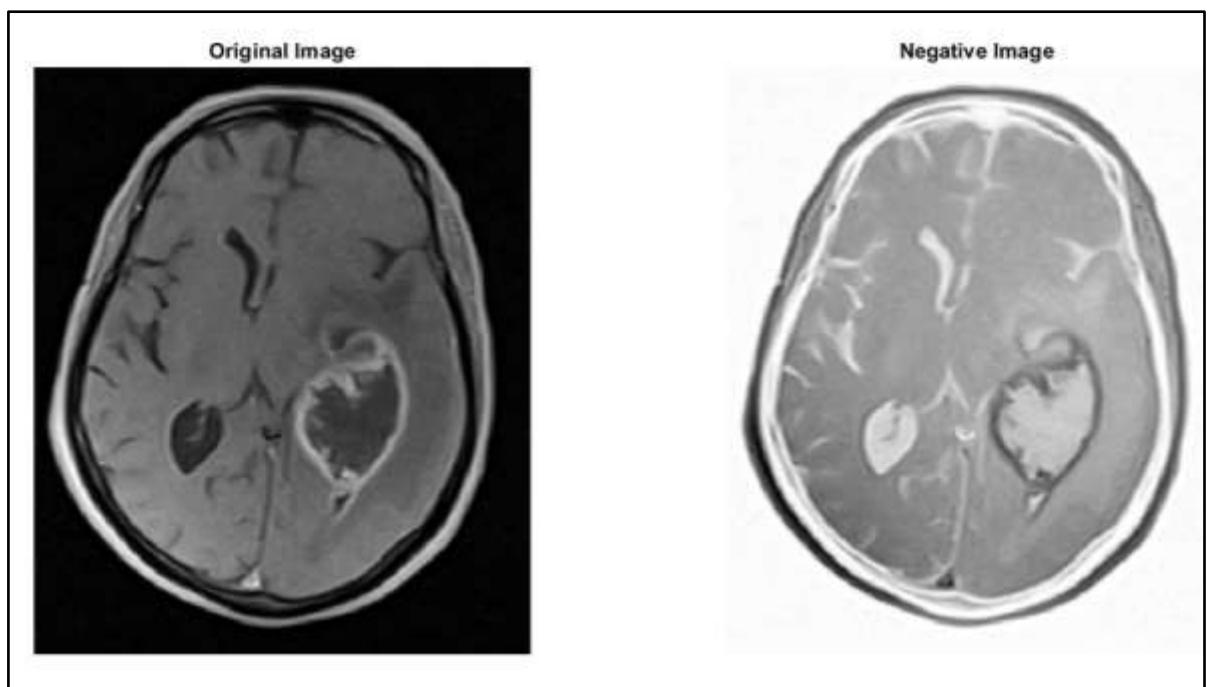


Figure 4.19: Processing T1-Weighted Image after gadolinium injection to extract negative image by using MATLAB.

In abnormal image, after injecting the patient with the contrast agent (gadolinium), there was absorption and accumulation of the contrast agent in the patient's blood-brain

barrier, which led to a shortening of T1 and an irregular enhancement of the edges of the tumor, and no effect of edema around the tumor (Figure 4.17). The pathological anatomical region was extracted using MATLAB programe, as shown in Figure 4.18. The negative image was created using MATLAB software to improve the clarity and contrast of the original image, making certain features more visible and easier to analyzeas. The negative image was created using MATLAB software to improve the clarity and contrast of the original image, making certain features more visible and easier to analyze (Figure 4.19).

4.2.2. Discussion & Analysis of 4-Year-Old Male's Case

Subsequent to acquiring the aforementioned MRI sequences, a thorough comparative analysis was performed by carefully examining the acquired images in relation to corresponding normal reference images for each sequence. This comparison aimed to identify any deviations or abnormalities in the patient's imaging data by evaluating various structural and characteristics. The analysis sought to highlight any discrepancies, anomalies, or variations in tissue composition, signal intensity, or anatomical structures. This step was crucial in determining the presence of any pathological conditions and in assessing the overall quality and reliability of the MRI data in comparison to a typical baseline.

4.2.2.1. T1WI

In the normal image, fluids have a long T1 (to recover their longitudinal magnetization). But the T1-weighted image uses a short TR, so the fluids in the image appear dark. As in CSF. Signal of fats and proteins: In the normal image, fats and proteins have a short T1 to restore their longitudinal magnetization, so although the TR is short in the T1WI-weighted image, therefore the signal is still high and the fats and proteins appear bright, as shown in Figure 4.20.

In the abnormal image, the fat signal and the water signal appear as in the normal condition, except for the area marked with a red circle (Figure 4.21). The water signal and the fat signal appear as in the normal case, except for the area marked with a red circle that appears dark in black and gives the same water signal (the CSF signal).

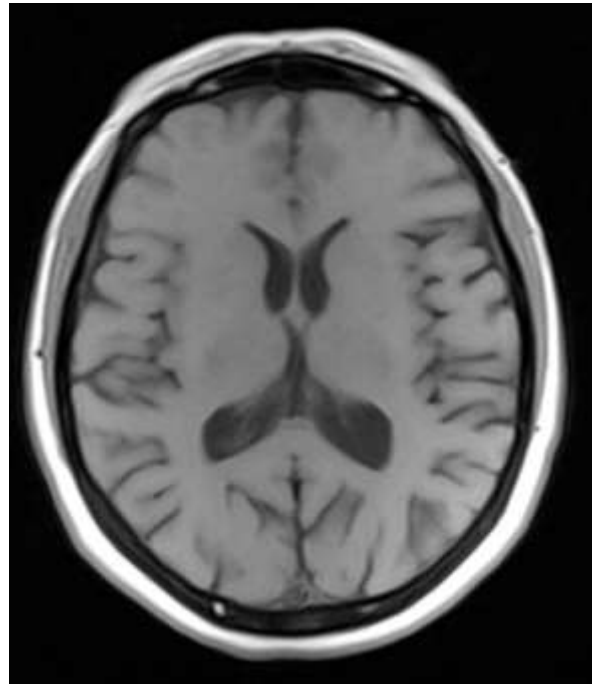


Figure 4.20: T1-Weighted Image of normal brain on transverse plane.

4.2.2.2. T2WI

In the normal image, fluids have a long T2 (when their transverse magnetization is lost). They retain their magnetism for a long time, and the recorded signal is strong, so the fluids in the image appear bright. As in CSF. Signal of fats and proteins: In the normal image, fats and proteins contain a short T2 when their transverse magnetization decays. Therefore, they lose their magnetization quickly, and the recorded signal is weak and appears dark in the image, as shown in Figure 4.22.

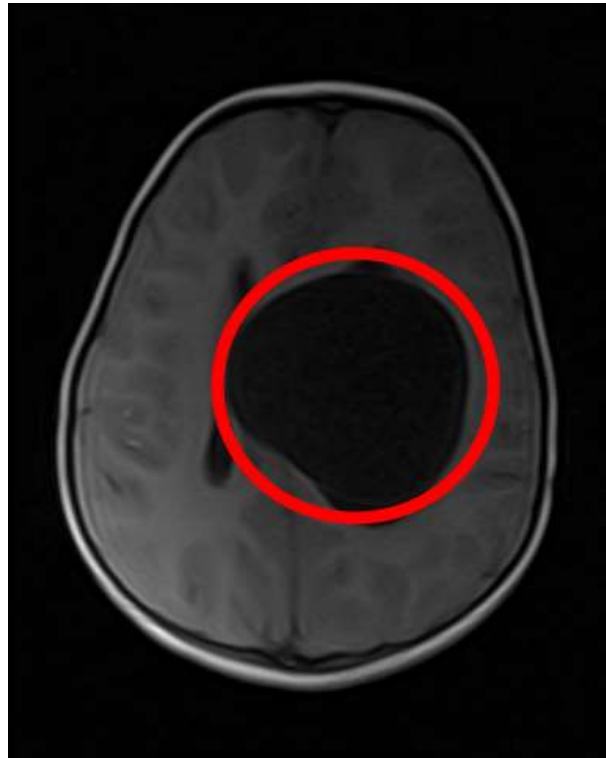


Figure 4.21: T1-Weighted Image of abnormal brain, where the image highlights the abnormal tissues (red circle) of 4-Year-Old Male's Case.

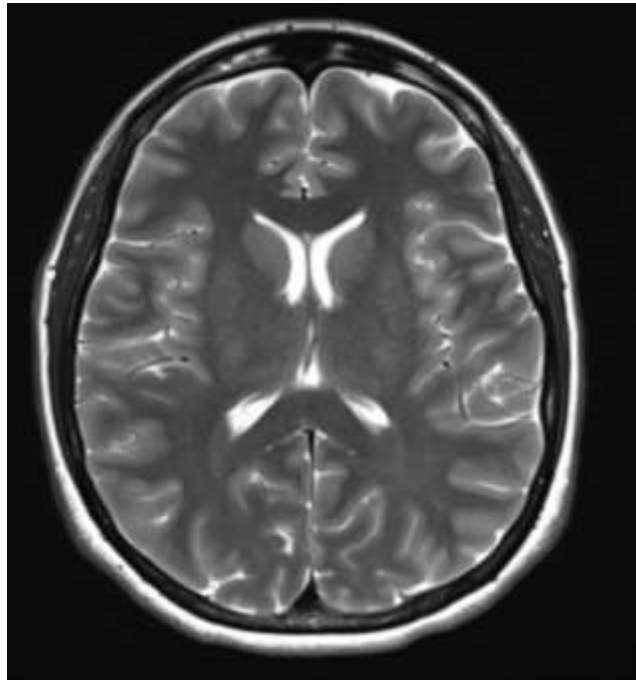


Figure 4.22: T2-Weighted Image of Normal brain on transverse plane.

In the abnormal image, the fat signal and the water signal appear as in the normal case, except for the area marked with a red circle (Figure 4.23). The water signal and the fat

signal appear as in the normal case, except for the area marked with a red circle that appears bright and gives the same water signal (the CSF signal). The pathological anatomical region was extracted using MATLAB programme, as shown in Figure 4.24.

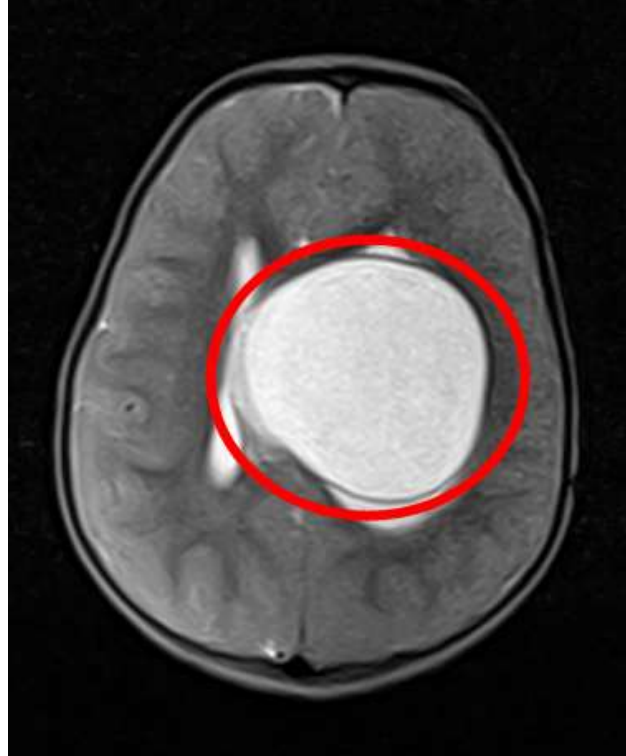


Figure 4.23: T2-Weighted Image of abnormal brain, where the image highlights the abnormal tissues (red circle) of 4-Year-Old Male's Case.

4.2.2.3. FLAIR Image

In the normal image, when the FLAIR sequence is used to suppress the fluid signal, by using FSE-IR, the sequence is under the condition of keeping T2 contrast, restrain the signal of free water. This is done by applying a 180-RF pulse to flip the longitudinal magnetization (M_z) to the negative pole (M_{-z}); by recovering the magnetization during relaxation, it will reach zero before becoming positive again. At this point, when a 90-RF pulse is applied, it will not create magnetization in the transverse plane of the free liquid. This in turn leads to the cancellation of the free fluid signal as in CSF. FLAIR image is similar to the weighted image transverse relaxation time (T2WI) in the spin echo

sequence, except that the free fluid appears black and the restricted fluid appears white, as shown in Figure 4.25.

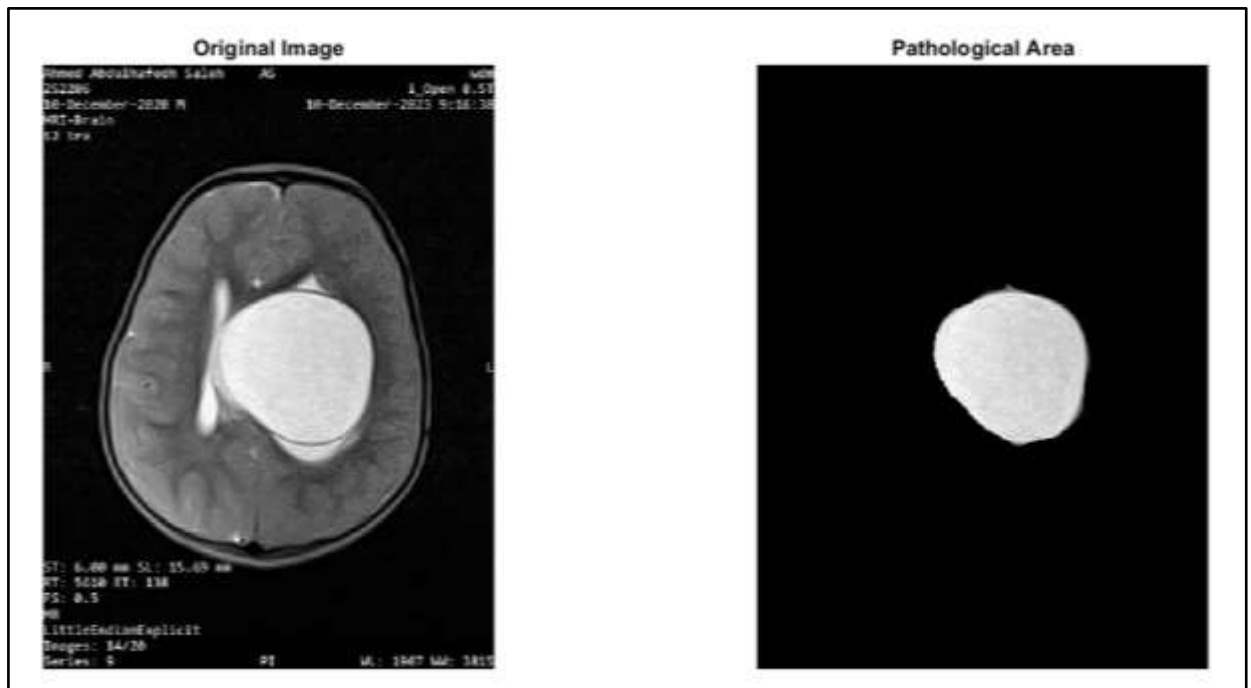


Figure 4.24: Processing T2-Weighted Image to identify and extract the pathological anatomical region using MATLAB.

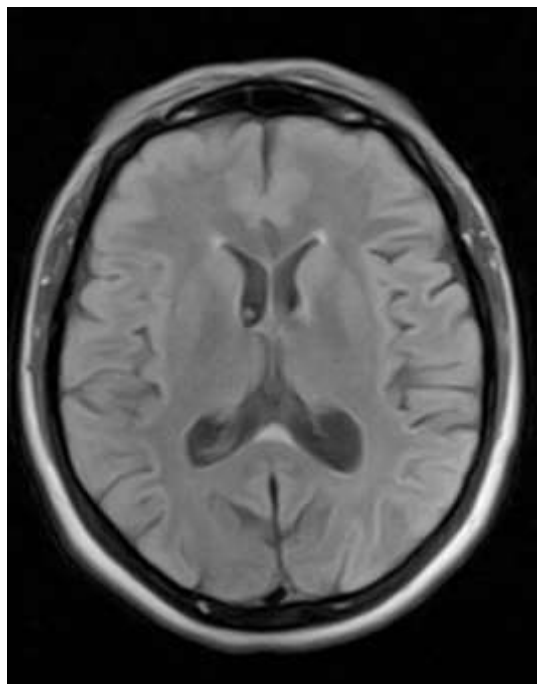


Figure 4.25: FLAIR-image of normal brain on transverse plane.

A third test was used in order to differentiate whether the brightness in T2-weighted image was a tumor or a fluid. Following signal attenuation of free fluid, the lesion also exhibited signal suppression, appearing dark on the image (Figure 4.26).

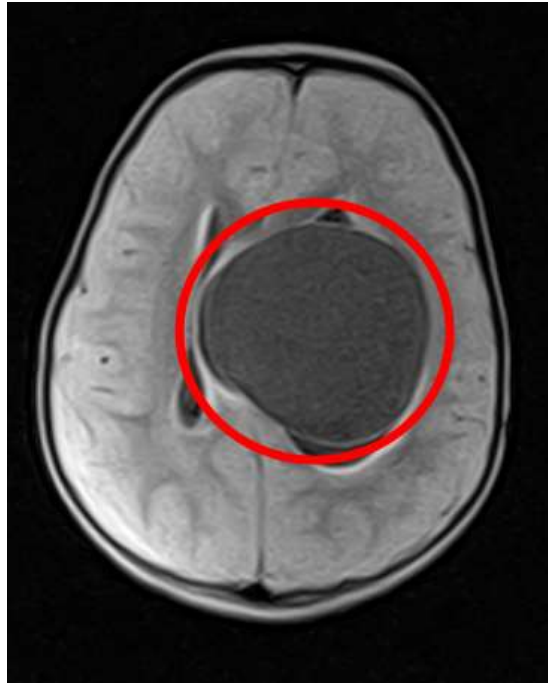


Figure 4.26: FLAIR-Image of abnormal brain, where the image highlights the abnormal tissues (red circle) of 4-Year-Old Male's Case.

This dark signal intensity resembled that of cerebrospinal fluid, indicating a fluid-filled lesion rather than a tumor. The effective attenuation of the free fluid signal was crucial in distinguishing between a tumor and a cystic lesion. These study results are consistent with the study of Sahu, et al. [28], in the effectiveness of the FLAIR sequence over the T2-sequence. The results also agreed with the study of Aprile, et al.[26], that the FLAIR sequence is useful for detecting intracranial lesions. In contrast, in our previous case, the lesion signal was not attenuated in the FLAIR image, which made the abnormal brain tissue appear bright, and on this basis, the tumor was diagnosed. The pathological anatomical region was extracted using MATLAB programme, as shown in Figure 4.27. The edges of the image were detected using MATLAB software to highlight the boundaries

and contours of the relevant structures. The resulting edge-detected image, shown in Figure 4.28, clearly outlines the important structures.

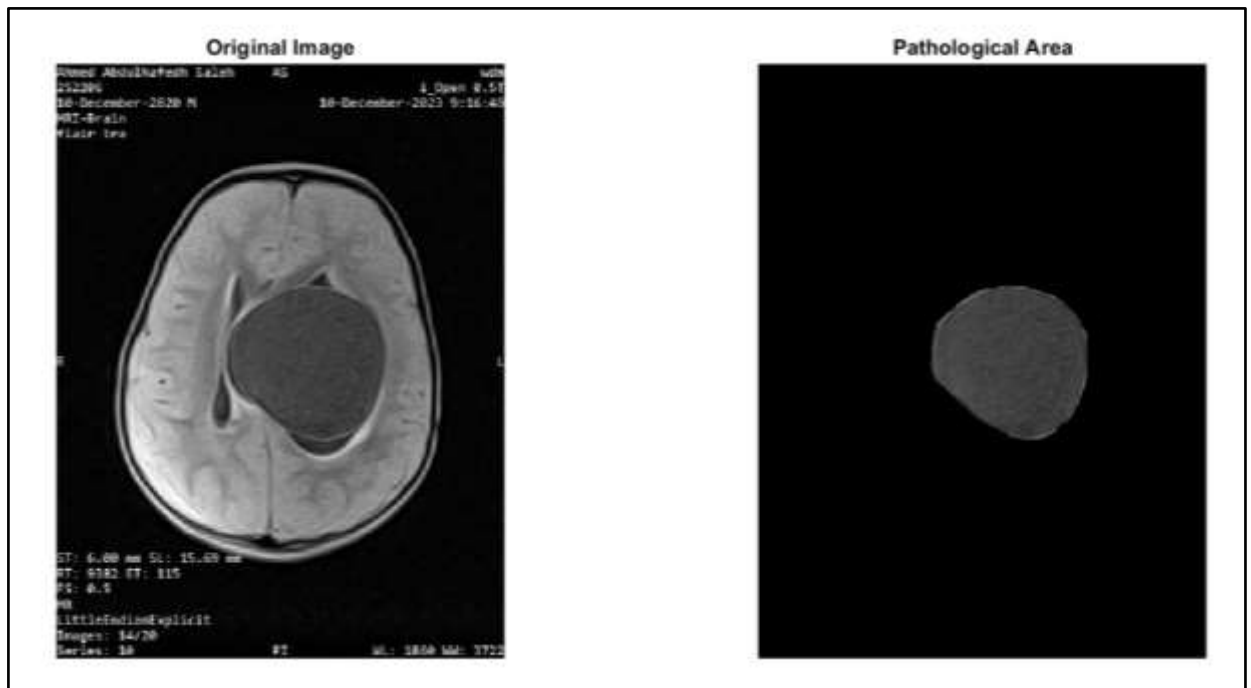


Figure 4.27: Using MATLAB on FLAIR image to identify and extract the pathological anatomical region.

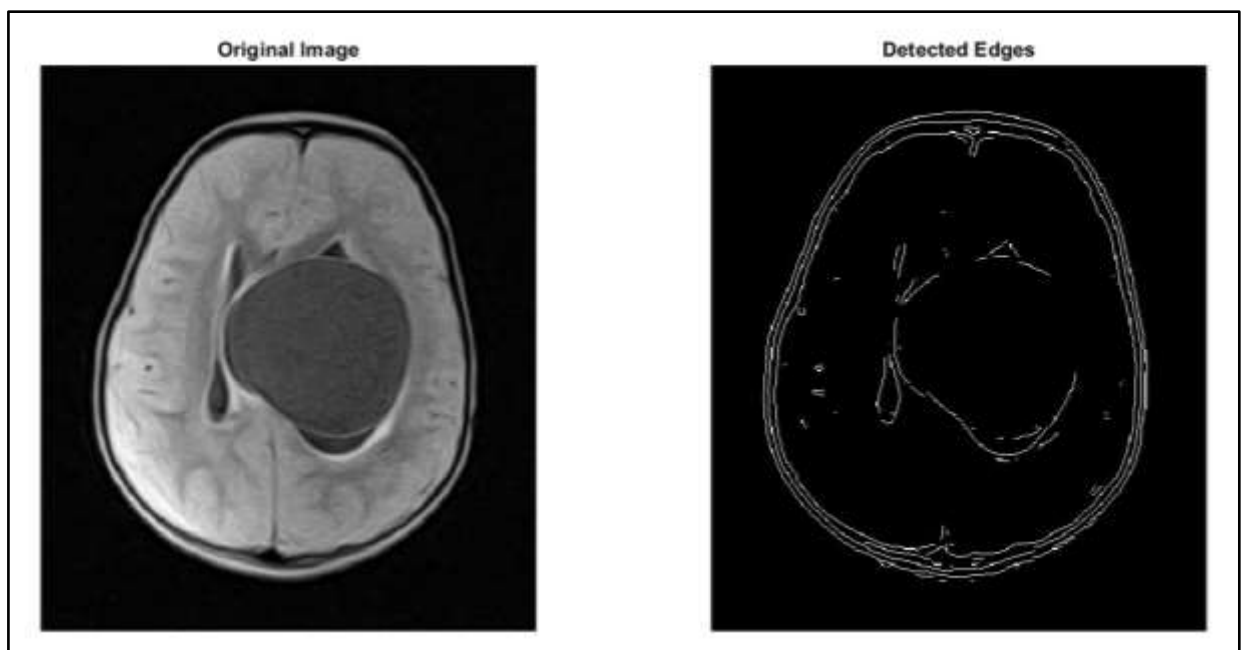


Figure 4.28: Using MATLAB on FLAIR image to process and detect edges of medical image.

4.2.2.4. T1WI + Gd

Magnetic materials, such as gadolinium, are commonly used as contrast agents in MRI due to their ability to generate small, localized magnetic fields. These fields have the effect of shortening the relaxation times of surrounding protons, particularly the longitudinal relaxation time (T1). When gadolinium is introduced into the body, it interacts with the protons in the tissue, influencing their magnetic properties and leading to a faster return to equilibrium after excitation by the MRI machine. This shortening of T1 enhances the imaging contrast, allowing for more detailed and clearer images of the tissue being studied.

A key characteristic of gadolinium, however, is that it cannot cross the normal BBB, which acts as a protective filter between the blood and the brain. The BBB prevents many substances, including gadolinium, from entering the brain tissue under normal conditions. However, when there is disruption or damage to the blood-brain barrier, as can occur in certain pathological conditions such as tumors or inflammation, gadolinium can cross into the brain tissue. This breach in the BBB allows gadolinium to accumulate in the affected area, leading to a noticeable shortening of T1 relaxation times in the surrounding protons, which is detectable through MRI.

The disruption of the BBB is particularly relevant in the context of tumors. When the BBB is compromised in or around tumor tissue, gadolinium can enhance the vascularization (blood supply) of the tumor, making the tumor more visible on MRI scans. This effect is particularly useful for identifying and delineating tumors, as the contrast difference between the tumor and normal brain tissue is more pronounced. In contrast, when the BBB is intact, gadolinium does not enter the brain tissue in significant amounts, and this enhancement effect does not occur. As a result, cysts or surrounding

edema (swelling) in areas where the BBB remains intact are typically not affected by gadolinium, and the MRI scan may not show the same level of contrast enhancement in these areas.

In summary, the ability of gadolinium to cross a disrupted BBB enhances the imaging contrast in regions with compromised vascular integrity, particularly in tumor tissues. However, this effect is absent when the BBB is intact, limiting gadolinium's utility in imaging unaffected areas like normal brain tissue or regions with intact BBB, such as cysts or non-tumorous edema, as shown in Figure 4.29.

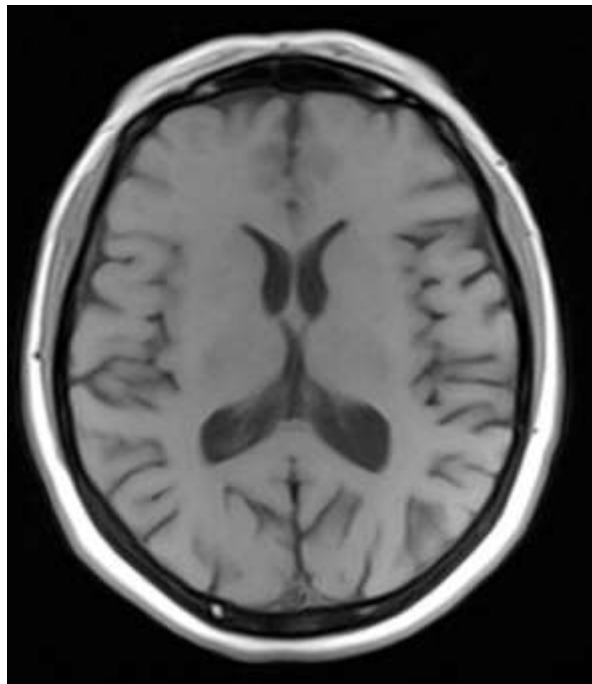


Figure 4.29: T1WI + Gd on transverse plane (normal).

In the abnormal image, when the patient was injected with the contrast agent (gadolinium), there was no absorption or accumulation of the contrast agent in the blood-brain barrier. Under normal circumstances, gadolinium enhances the visualization of abnormal structures by crossing the blood-brain barrier, particularly in areas with disrupted or abnormal tissue, such as tumors. However, in this case, the injection of the contrast agent, there was no shortening of T1 relaxation times, a key indicator of

gadolinium's effect. Consequently, no irregular enhancement was observed in the region of interest. This lack of enhancement suggests that the lesion is not a tumor, but rather a cyst, as cystic structures typically do not show the same contrast enhancement characteristics due to the absence of abnormal vascularization or disruption of the blood-brain barrier. Therefore, the imaging findings support the diagnosis of a cyst rather than a tumor, as shown in Figure 4.30.

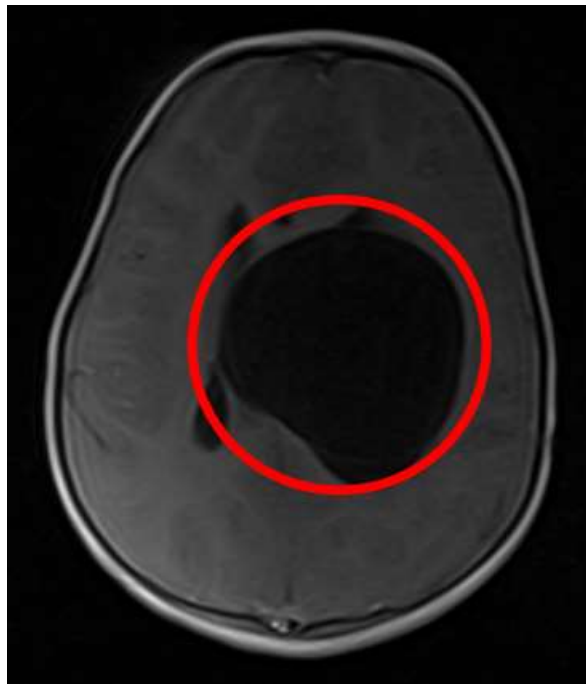
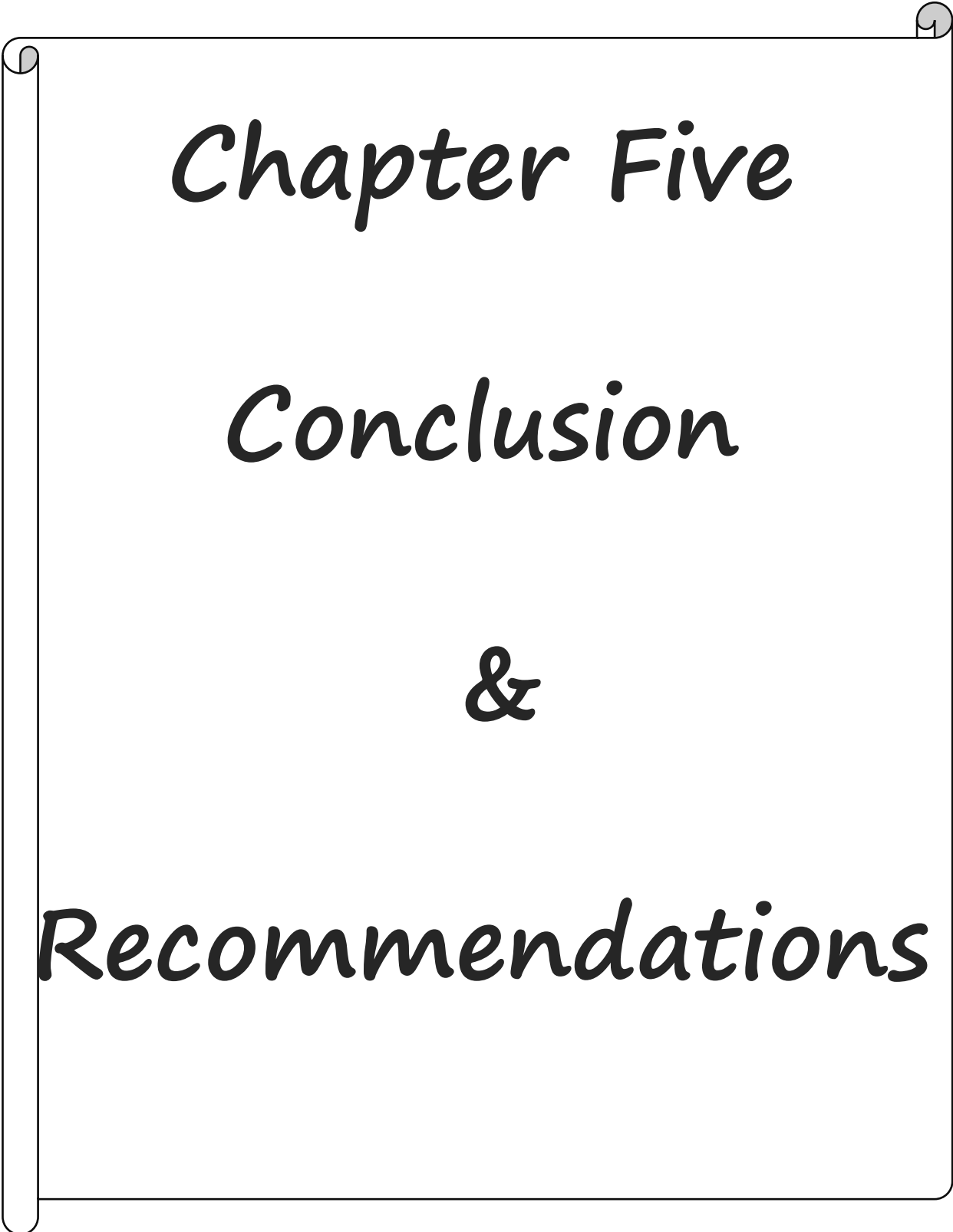


Figure 4.30: T1-Weighted Image + contrast medium
Gd of abnormal brain, where the image highlights
the abnormal tissues (red circle) of 4-Year-Old
Male's Case.



Chapter Five

Conclusion

&

Recommendations

Conclusion and Recommendations

This section presents the conclusion of the research, summarizing the key findings and insights gained throughout the study. It also includes several recommendations for future research directions, highlighting areas where further investigation could enhance understanding or improve existing methods. The recommendations aim to guide future work that could build upon the current study, addressing limitations and exploring new opportunities for advancement in the field.

5.1. Conclusion

This research utilized advanced MRI sequences, including FSE, MSSE, and FLAIR, to image brain lesions within an i-open MRI system. The findings of the study demonstrate the significant effectiveness of MRI, particularly the FLAIR sequence, in accurately diagnosing brain lesions. Notably, FLAIR imaging played a crucial role in diagnosing a brain tumor in a 70-year-old male patient and a brain fluid cyst in a 4-year-old male patient, highlighting the versatility of this imaging modality across diverse age groups. The use of Inverse Fourier Transform for reconstructing the MRI-processed brain images proved essential for accurate cyst and tumor localization, further enhancing the diagnostic capabilities of the MRI system.

In this study, T1-weighted, T2-weighted, and FLAIR images were acquired and compared with corresponding normal brain images to aid in the analysis of these lesions. While T1-weighted and T2-weighted sequences provided useful preliminary insights into the brain's structure, they were found to be inadequate in definitively characterizing the lesions. In contrast, FLAIR imaging significantly improved lesion detection by selectively enhancing the tumor's signal while suppressing the signal from the cystic

fluid, allowing for a clearer distinction between the lesions and their surrounding brain tissues. Clearly, differentiating them from surrounding tissues. This contrast was particularly evident in the brain tumor case, where FLAIR imaging effectively highlighted the abnormal tissue without interference from surrounding other normal brain structures.

Additionally, the study demonstrated that the administration of a gadolinium-based contrast agent in T1-weighted imaging further refined lesion identification. The post-gadolinium enhancement in T1-weighted images not only accentuated the boundaries of the tumor but also helped differentiate it from surrounding edema, providing critical diagnostic information. The ability of gadolinium to shorten T1 relaxation time in abnormal tissues—such as tumors—further underscores its value in distinguishing malignant or abnormal tissue from normal brain tissue.

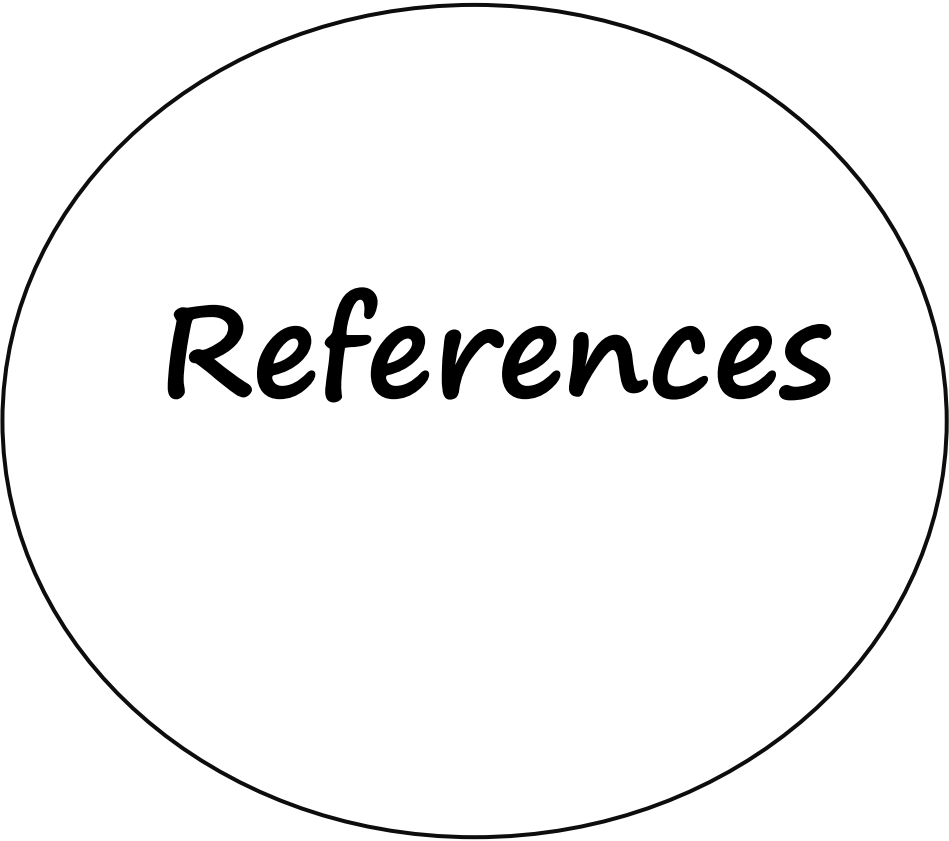
These results collectively emphasize the essential role of the FLAIR sequence as a highly effective diagnostic tool for identifying brain fluid cysts and tumors, with the added benefit of gadolinium contrast agents enhancing the accuracy of tumor detection. The study highlights the complementary nature of different MRI sequences, reinforcing the importance of employing a multi-modal approach to fully assess brain lesions and improve clinical outcomes. Therefore, integrating FLAIR imaging, alongside other sequences like T1WI with gadolinium, can significantly improve the early detection and characterization of brain lesions, leading to better-informed treatment plans for patients.

The results of our study serve to facilitate and enhance the clarity of the image by using some techniques such as attenuation of the free fluid signal and injection of contrast agent in addition to processing the obtained images with a MATLAB program in order to compensate for the low magnetic field instead of high digitization, and this in turn provides better diagnostic information while optimizing the imaging process itself.

5.2. Recommendations and Future Scope

Through our study on the MRI (i-open permanent MRI system) and its different sequences in detecting different lesions, we recommend the following:

- 1- Investigating the potential of advanced MRI techniques, such as diffusion-weighted imaging (DWI), in conjunction with FLAIR could provide additional insights into the nature and characteristics of lesions.
- 2- Optimize the protocols of the MRI machine used in research in order to improve image quality or reduce imaging time.
- 3- Study another MRI machine with a higher magnetic field strength in order to compare the results.
- 4- Research for more effective contrast agents and use them in all domains of MRI.
- 5- Develop special programs to enhance image quality, such as MATLAB and Python programs.



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الملخص

الرنين المغناطيسي النووي (NMR) هو ظاهرة فيزيائية تستخدم لدراسة الخصائص المغناطيسية للنوى الذرية. يعبر عن امتصاص وانبعاث الطاقة بواسطة النوى في مجال مغناطيسي، والذي يمكن أن يوفر معلومات واسعة النطاق حول بنية وديناميكيات وحالة التفاعل والبيئة الكيميائية للجزيئات. التصوير بالرنين المغناطيسي هو الطريقة الأكثر أمانًا للتصوير، وهو يعطي تمثيلًا مرئيًا للأنسجة البشرية دون تدخل جراحي للتشخيص السريري. في هذا البحث، تم استخدام تسلسلات صدى الدوران السريع (FSE)، وصدى الدوران متعدد الشرائح (MSSE)، وتوهين السائل باستعادة الانقلاب (FLAIR). يتم وضع الحالات قيد الدراسة في نظام التصوير بالرنين المغناطيسي الدائم من النوع المفتوح من أجل تصوير الآفات المرضية للدماغ. يتم ترميز الإشارات التي تم الحصول عليها لملء فضاء k من خلال ثلاث مراحل؛ اختيار الشريحة وترميز الطور وترميز التردد. تتم معالجة الإشارات التناظرية التي تم الحصول عليها إلى إشارة منفصلة باستخدام تحويل فورييه، ثم رقمتها وإعادة بنائها باستخدام تحويل فورييه العكسي للحصول على صورة للدماغ. تم إجراء اختبارات ترجيح زمن الاسترخاء الطولي (T1-weighted) قبل وبعد حقن عامل التباين القائم على الجادولينيوم وترجيح زمن الاسترخاء العرضي (T2-weighted) وتوهين السائل باستعادة الانقلاب (FLAIR) للحصول على أنواع الصور ومقارنة الصور الناتجة بالصور الطبيعية في كل نوع من التسلسل المطبق. في صورة توهين السائل باستعادة الانقلاب (FLAIR)، الحالة الأولى، تم تعزيز الإشارة في الموقع المستهدف، وتحولت إلى إشارة ساطعة، وظهرت كماء محصور، مما يشير إلى أنه ورم، بينما في الحالة الثانية، يتم إخماد إشارة السائل على وجه التحديد في موقع الهدف، وتحولت إلى داكنة، وظهرت كماء حر، مثل السائل الدماغي الشوكي، مما يشير إلى أنه سائل وليس ورمًا. كل هذا بدوره يساعد على التمييز بين الورم والسائل في الصورة المرجحة لزمن الاسترخاء العرضي (T2WI). كما أن عامل التباين (الجادولينيوم) في صورة ترجيح زمن الاسترخاء الطولي (T1WI) قلل من وقت الاسترخاء الطولي (T1) وكشف بدقة عن حواف الورم وساعد على تمييزه عن السائل. وبالتالي، تم الكشف عن الآفات المرضية. تضع دراستنا أساسًا لاستخدام تقنية التصوير بالرنين المغناطيسي مع تسلسلات التباين مثل توهين السائل باستعادة الانقلاب (FLAIR) وتسلسلات أخرى للتمييز بين علامات المرض المختلفة مثل الأكياس السائلة والأورام عندما تفشل التقنيات الأخرى في القيام بنفس الشيء.

تتكون هذه الأطروحة من خمسة فصول؛ وقد تم تلخيصها بإيجاز على النحو التالي:

١. المقدمة: يقدم هذا القسم مفهوم الرنين المغناطيسي النووي وتطبيقاته في المجال الطبي. كما يغطي تاريخ الرنين المغناطيسي النووي والتصوير بالرنين المغناطيسي، علاوة على ذلك فهو يشمل أهمية ودافع وأهداف البحث، بالإضافة إلى مراجعة الأدبيات للدراسات السابقة في تكنولوجيا التصوير بالرنين المغناطيسي.

٢. الجزء النظري: يناقش هذا الفصل المبادئ الفيزيائية وراء تكنولوجيا التصوير بالرنين المغناطيسي، ومكونات الأجهزة التي تعمل معًا لتوليد صور مقطعية، وأنظمة الكمبيوتر المتعلقة بالتصوير بالرنين المغناطيسي.

٣. الجزء العملي: يقدم هذا القسم نظرة عامة على المعدات والعوامل والحالات التي استعملت، بالإضافة إلى المنهجية المستخدمة.

٤. النتائج والمناقشة: يقدم هذا الفصل نتائج دراسة حالتين خضعتا للتصوير بالرنين المغناطيسي، إلى جانب مناقشة وتحليل للنتائج.

٥. الخاتمة والنطاق المستقبلي: يلخص هذا القسم الأخير الاستنتاجات المستخلصة من البحث ويقدم توصيات للتطورات المستقبلية في هذا المجال.



الجمهورية اليمنية

جامعة عدن

كلية التربية - عدن

قسم الفيزياء

استخدام تقنية التصوير بالرنين المغناطيسي في التطبيقات السريية لإظهار وتشخيص علامات أمراض الدماغ

رسالة مقدمة الى قسم الفيزياء - كلية التربية عدن
جامعة عدن

لاستكمال المتطلبات الجزئية للحصول على درجة الماجستير في الفيزياء
التخصص: الفيزياء الطبية

مقدمة من

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تحت إشراف

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